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Macquarie Coal Complex

Package E: Economics, Social, and Market Assessment Report

Lake Macquarie City Council

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Abbreviations

Acronym	Definition
ABS	Australian Bureau of Statistics
BESS	Battery Energy Storage System
CBD	Central Business District
C&D	Construction and Demolition
Council	Lake Macquarie City Council
DPHI	Department of Planning, Housing and Infrastructure
EPA	Environmental Protection Authority
ESA	Economic Study Area
EbD	Enquiry by Design
EIA-N05	Environmental Impact Assessment Practice Note – Socio-economic Assessment (EIA-N05) (Transport, 2020).
EPR	Extended Producer Responsibility
FY	Financial Year
GRP	Gross Regional Product
HVHH	Hunter Valley Hydrogen Project
LEP	Local Environmental Plan
LGA	Local Government Area
NZEA	Net Zero Economy Authority
PMLU	Post-Mining Land Use
PTAL	Public Transport Accessibility Level
R&D	Research and Development
REIP	Renewable Energy Industrial Precincts
REF	Review of Environmental Factors
REZ	Renewable Energy Zone
SEARs	Secretary's environmental assessment requirements
SIA	Social Impact Assessment
SSA	Social Study Area
SEIFA	Socio-Economic Indexes for Areas
SHR	State Heritage Register
SSD	State Significant Development
SOHI	Statement of Heritage Impact assessment
SA1s	Statistical Areas Level 1
SA2s	Statistical Areas Level 2
SA3s	Statistical Areas Level 3
The Precinct	Macquarie Coal Complex Precinct
The Site	Macquarie Coal Complex Precinct and Teralba Southgate
Transport	Transport for NSW

Executive summary

Lake Macquarie City Council engaged Aurecon to undertake an economic, social and market assessment of redevelopment opportunities for the Macquarie Coal Complex (Precinct) and Teralba Southgate, collectively referred to as the "Site". The assessment has been undertaken to provide an evidence base for future land use decision-making as the Site transitions through closure and rehabilitation towards productive post-mining uses. This report ("Package E") brings together baseline demographic, labour market, economic, social and market evidence to support the Master Plan process. It tests the land use directions emerging from the Enquiry by Design process and considers their feasibility, economic contribution, social implications, market readiness, risks and dependencies. The assessment is prepared for an early planning and pre-investment context. It provides a structured evidence base to support land use decision-making, future business case development and further technical testing.

Market assessment and land use opportunities for employment

The Site's scale, industrial legacy, proximity to established communities, access to regional labour markets and potential infrastructure links favour employment-generating uses. The land use opportunities are grouped into three categories. Table 0-1 summarises land use opportunities by priority, time horizon and employment.

Table 0-1: Market assessment summary of potential post-mining land uses[1]

Land use	Priority	Time horizon	Employment generation (ongoing)	Key opportunities
Circular economy, industrial and advanced manufacturing	High	Near term (0–5 years)	Up to 40 jobs per facility[2]	Integrated circular economy; precision fabrication; advanced industrial and manufacturing uses.
Waste and resource recovery	High	Near term (0–5 years)	Up to 100 jobs per facility	Accumulating feedstock for circular economy and manufacturing, e.g. Recycling and e-waste facilities.
Energy transition and renewable infrastructure	High	Near-long term (0–5 years)	Up to 20 jobs per facility	Battery energy storage systems (BESS) and related energy infrastructure.
Freight, logistics, distribution and supply chain hub	Conditional	Long term (15+ years)	Up to 400 jobs per facility	Expansion of logistics and transport operations; e-commerce hub; cold chain and perishable goods logistics.
Data centres and digital infrastructure	Conditional	Medium-long term (10–15 years)	Up to 50 jobs per facility	Data storage centres supporting cloud and enterprise clients; small-scale facilities for regional data processing.
Community, tourism and agriculture	Conditional	Near-medium term (5–10 years)	Up to 130 jobs per facility	Open spaces; visitor economy and eco-tourism; cultural experiences; and controlled environment agriculture.
Residential	Complemen-tary	Near-medium term (5–10 years)	Up to 50 jobs per facility[3]	Residential development and senior living.
Education, research and innovation	Optional	Long term (15+ years)	Not assessed for optional land use	Clean technology innovation hub; energy transition training and workforce development centres.

[1] Note: Excludes construction jobs.

[2] Up to 40 jobs per facility, noting that these will use more robotics than traditional manufacturing.

[3] Note: Applies to Senior Living facilities.

The categories above reflect the relative readiness and strategic role of each land use within the Site, having regard to market demand, alignment with site attributes, infrastructure and servicing requirements, and the extent of further investigation and delivery planning required. These are defined below:

- **Highly feasible**: circular economy, industrial and advanced manufacturing, waste and resource recovery, and energy transition infrastructure and renewables. These uses have near-term feasibility, alignment with the Site's scale, industrial legacy and workforce. They can proceed earliest and anchor investment.
- **Conditional:** freight and logistics, data centres, and community, tourism and agriculture. These uses have strategic merit but depend on conditions being met first, principally infrastructure capacity, market maturity, anchor tenants and policy support. They are sequenced to later horizons as those conditions are resolved.
- **Complementary**: residential uses support the Site's function, liveability and public benefit but are not the primary economic drivers.

The Site's key opportunity lies in its flexibility to accommodate a range of employment-generating, infrastructure-led and complementary uses as market, infrastructure and policy conditions evolve. **In total, these uses have the potential to support around 1,130 ongoing jobs.**

Social implications and consideration for land use

- Minimising social impacts and sustaining social outcomes will depend on how effectively economic benefits are accessed, shared and supported over time.
- **Transport access is the key consideration** across all land uses. Limited public transport accessibility will reduce future access to jobs and services, particularly for lower-income households, older residents, people with disabilities and those without private vehicles.
- **Employment-generating uses** (freight, logistics, data centres) offer strong economic potential but introduce trade-offs such as increased traffic, amenity impacts and pressure on local services. Industrial and circular economy uses align with regional transition objectives, but their social outcomes depend on how well impacts are managed and local workforce pathways are enabled.
- **Complementary uses** (community infrastructure, housing, recreation, health services) are likely to deliver more direct social benefits, supporting liveability, wellbeing and place identity.
- **Cumulative impacts** from workforce growth, concurrent regional activity, and construction and operational phases will increase demand for housing and services and require coordinated planning.
- **Social outcomes** can be strengthened through improved transport access, local employment pathways, Aboriginal participation and community benefit sharing.

Key concluding considerations and next steps

- The Site has **credible potential to contribute** to Lake Macquarie's economic transition and employment land supply and is **not immediately market-ready** for all uses. Success will depend on staged de-risking, clear planning pathways, infrastructure sequencing, targeted market testing and land use flexibility.
- **The social findings highlight key considerations which require careful planning, further investigation and engagement with the community**. Future decisions should not weigh only on the number of jobs or scale of investment, but whether benefits are accessible, locally captured and supported by housing, transport, social infrastructure and biodiversity outcomes.
- This assessment recommends a pragmatic approach to **prioritise feasible employment-generating uses** and **protect longer-term optionality.**
- **Advance to Business Case development for high priority near term uses**, building on this assessment through refined technical studies, commercial analysis, and stakeholder validation to support an informed investment decision.

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Part A: Land use assessment

1 Introduction and background

This Section focuses on project overview, Site, the purpose of the report, strategic alignment and relationship to other packages.

1.1 Project overview

The Macquarie Coal Complex Pilot Project was initiated as part of the NSW Government's wider response to the Beneficial and productive post-mining land use Parliamentary Inquiry.[4] In July 2025 the NSW Government committed to undertake two pilot projects to demonstrate how alternative post-mining land uses can be delivered under the existing planning and regulatory framework.

The Macquarie Coal Complex Master Plan was prepared by the Department of Planning, Housing and Infrastructure (DPHI) in partnership with Lake Macquarie City Council (Council), Net Zero Economy Authority (NZEA) and relevant NSW Government agencies involved in post-mining land use, regional development and investment delivery. The Department and Council have undertaken master planning informed by technical studies, to support the coordinated transition of a former mine site to alternative employment generating land uses in the Precinct

The Rezoning Proposal applies to approximately 1,160 hectares of land which includes the Macquarie Coal Complex Precinct and Teralba Southgate, collectively referred to as the "Site".

The **Precinct** sits adjacent to the Lake Macquarie Northwest Catalyst Area. It is bounded by the suburb of Barnsley to the north, Cockle Creek and the Central-Coast Newcastle main passenger and freight train line to the east, Teralba and Wakefield to the south and Killingworth to the west. The M1 Pacific Motorway and Sugarloaf State Conservation Area adjoin the western boundary of the Site.

Teralba Southgate is located at 91 York Street, Teralba, approximately 1.5 km southeast of the Precinct, 400 m from Teralba Train Station and 250 m from Booragul Train Station.

Strategic road and rail connections position the Site to support established and emerging industries and contribute to a diverse post-mining economy. The Site also provides opportunities to protect and enhance natural systems, cultural and historic assets, and support long-term community and environmental benefit.

The Site is located in a region with a strong and transferable skills base in mining, heavy industry, logistics, construction and land management. As regional mining operations transition toward closure, these skills represent a significant asset that can be redirected to support new and emerging industries.

The Master Plan identifies the arrangement of developable areas, environmentally constrained areas, recreational areas, movement networks and key constraints across the Site and outlines the spatial logic for future development at a precinct scale, providing a framework within which detailed design, subdivision and development assessment will occur.

The Site is a non-operational mining site located within the Lake Macquarie Local Government Area (LGA) undergoing rehabilitation. The Site forms part of a broader regional landscape that is undergoing economic, environmental and land use transition as coal related activity declines and mining and energy assets are considered for alternative productive land uses.

The Site has historically supported mining related activity and associated infrastructure. The Site now presents an opportunity to consider future post-mining land uses that can deliver economic, social and community benefit while responding to rehabilitation, environmental, infrastructure and planning requirements.

[4] Cooperative Research Centre for Transformations in Mining Economies (CRC TiME) (2024) Inquiry into Beneficial and Productive Post-Mining Land Use, Submission No. 67.

This assessment tests how former mining land can be planned, enabled and transitioned to support employment-generating land uses, regional economic resilience and broader community benefit.

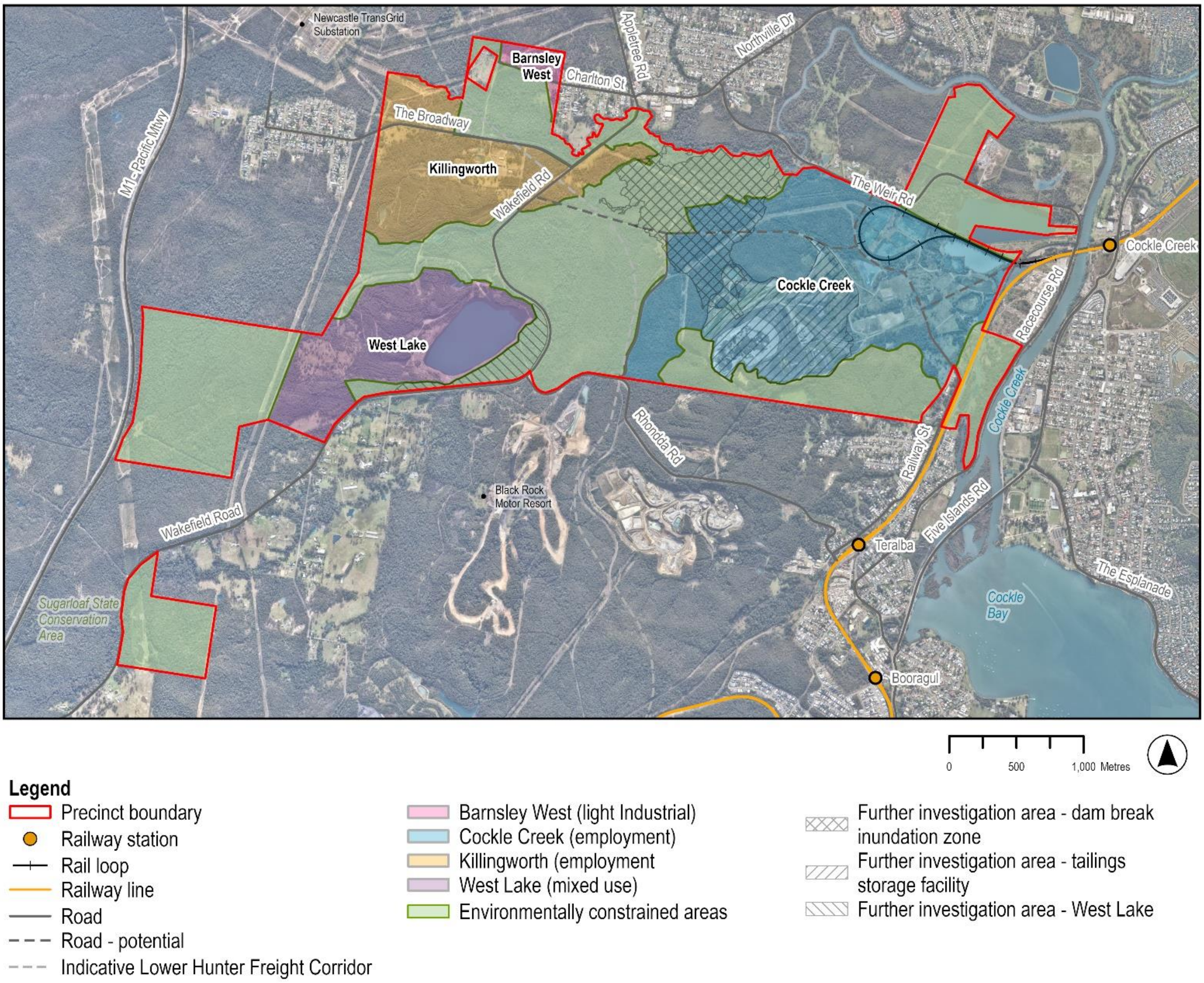

Figure 1-1: Site context and Master Plan land use areas

Figure 1-1 shows the Site context and Master Plan land use areas, including developable areas, environmentally constrained areas and further investigation areas.

1.2 Lake Macquarie's strategic role in regional transition economy

Lake Macquarie is strategically positioned within the Greater Newcastle and Hunter regional economy, with a large employment base and strong connectivity to the Port of Newcastle, established freight networks, major transport corridors, the Central Coast and Sydney. [5] These locational advantages support the area's role in the Lower Hunter economic system and provide a basis for future employment-generating land uses linked to regional transition.

The Lake Macquarie community strongly supports the economic transition away from coal alongside a clear expectation that redevelopment delivers new jobs, affordable housing, and improved infrastructure.[6] Residents place high value on environmental quality, public spaces, and overall liveability, while expressing concerns about housing affordability, transport connectivity, and involvement in decision-making.[7]

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[5] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

[6] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[7] Place Score (2023) 2023 Australian Liveability Census Lake Macquarie City Council - Priorities, Values and Performance Report

Forecasts indicate that Lake Macquarie may require around 113 hectares of additional employment land under the most likely growth scenario, with current vacant serviced employment land falling short of projected demand by around 54 hectares by 2046.[8] This highlights a key constraint on economic development, particularly where land is not appropriately located, serviced, or zoned to meet industry needs.

Lake Macquarie's key employment land use planning goals are:

- Maintaining a long-term pipeline of serviced employment
- Aligning infrastructure provision with growth
- Enabling adaptive reuse of post-mining and energy land
- Supporting emerging industries such as the circular economy.

Figure 1-2 provides a snapshot of the Lake Macquarie LGA.

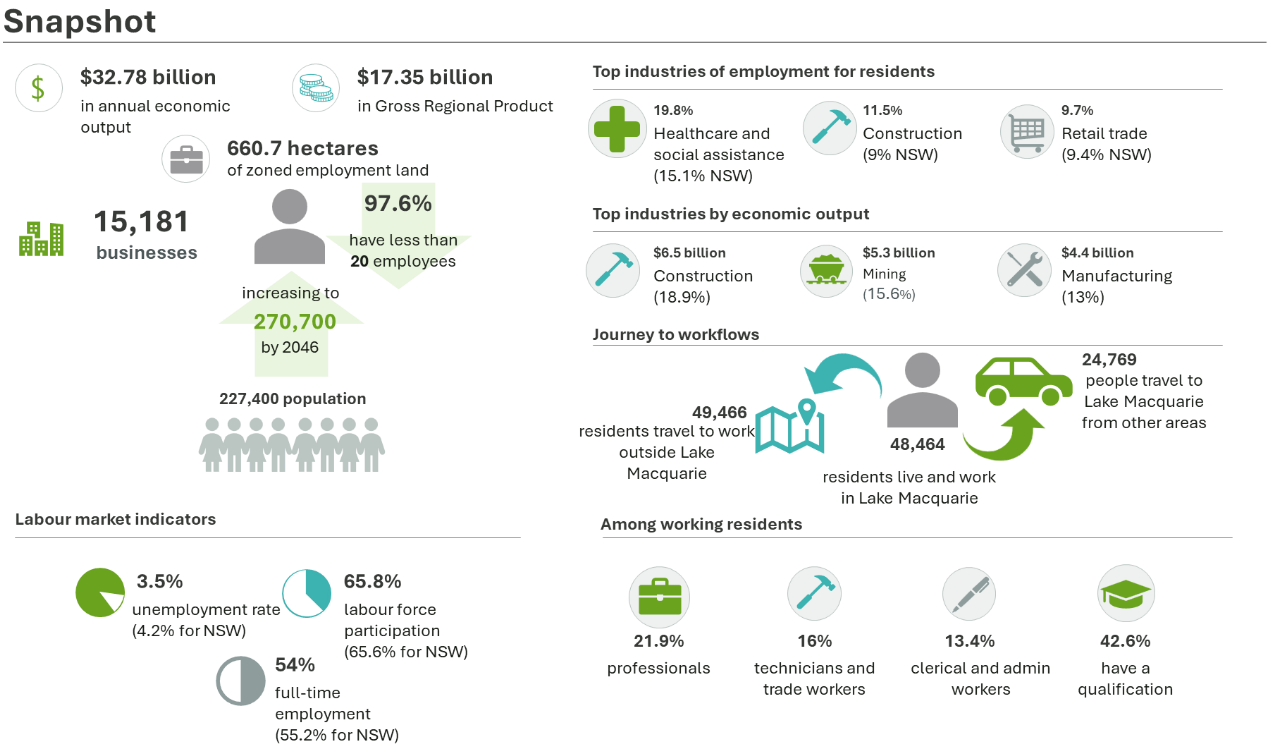

Figure 1-2: Lake Macquarie LGA economic snapshot[9]

Lake Macquarie LGA reflects the Hunter's broader regional economic transition. The economy is shifting from a goods-producing base toward service-oriented industries, with manufacturing and mining declining from 37% to 29% of total output over time.[10] Major assets such as Eraring Power Station and associated mining operations have historically underpinned employment and supply chains.

Mining contraction has had cascading impacts across freight, logistics, engineering, manufacturing, and port operations with a loss of 1,200 coal mining reliant jobs between 2011 and 2021 in the Lake Macquarie LGA.[11] Future employment growth is expected to be concentrated in healthcare, education, retail, construction and hospitality, with up to 15,000 additional jobs by 2046 within the Lake Macquarie LGA.[12] This growth is expected to drive increasing demand for employment land, particularly for larger-scale commercial construction, industrial services and business support sectors which rely on well-serviced, appropriately zoned employment land to operate efficiently.

[8]Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

[9]Ibid.

[10]Ibid.

[11]Australian Bureau of Statistics (ABS) (2021) Industry of employment (INDP).

[12]Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

The Precinct is therefore strategically relevant because it seeks to address constrained employment land supply while supporting broader economic transition objectives. The Site should therefore be considered as part of a broader transition landscape, not as an isolated land parcel.

The strongest strategic fit is likely to be for uses that:

- Leverage **existing infrastructure** and access advantages
- Support **employment generating activity**
- Align with **circular economy, industrial, energy transition and logistics opportunities**
- Avoid areas affected by significant environmental, flooding, contamination, subsidence or rehabilitation constraints
- **Delivered through a staged** and infrastructure aligned pathway.

1.3 Purpose and scope of this report

The purpose of this report is to establish the strategic case for the Site by testing whether the land use directions in the draft Master Plan are market-grounded, socially equitable and aligned with regional development goals.

This report consolidates the findings of the staged scope of work, emerging through the Enquiry by Design (EbD) workshop, [13] to identify the opportunities, constraints, risks and dependencies associated with future post-mining land use. The report specifically seeks to:

- **Establish the demographic, labour market, economic and social context** relevant to the Site and its surrounding communities.
- **Identify the current and potential future economic role** of the Site within Lake Macquarie, the broader Hunter and Central Coast context, and the NSW economy.
- **Assess the alignment between existing economic conditions**, workforce capability, infrastructure context and potential land use directions.
- **Consider the social implications** of potential land use directions, including community benefit, social infrastructure, access to jobs, equity, inclusion and Aboriginal participation.
- **Test the market plausibility**, likely timing and feasibility of potential land uses emerging from the Enquiry by Design process.
- **Identify risks, dependencies** and post-mining impediments that may affect feasibility, sequencing, scalability and delivery.
- **Provide recommendations and evidence-based inputs** to support future master planning, business case development, market testing and implementation planning.

The report is structured in two parts.

- **Part A** provides the main findings and planning implications. It supports the Master Plan by explaining the land use opportunities, their relative supportability, key dependencies, post-mining delivery considerations and implications for future planning, rezoning and business case development.
- **Part B** provides the supporting evidence base that was used to inform the findings in Part A and the Master Plan. It includes the methodology, strategic context, demographic, labour market, economic, social and market evidence. It provides the technical basis for the assessment and can be used to test the assumptions, data sources and analysis that underpin the report's conclusions.

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[13] The Enquiry by Design workshop was held on 7-8 May 2026 with considerable involvement across all packages subject matter experts, key interested stakeholders and industry representatives.

1.4 Relationship to other technical packages

This report forms one component of the broader technical workstream informing the future planning of the Site. It should be read alongside the Master Plan and the other technical packages summarised in Table 1-1.

Table 1-1: Macquarie Coal Complex Land Transformation Precinct technical packages

Package	Lead Agency	Lead Consultant	Focus and Key Deliverables
Package A – Urban Design	DPHI	GHD	Overall Master Plan (includes a single page Structure Plan)
Package B – Engineering	DPHI	Aurecon	Flooding and water cycle management, Geotechnical, Transport and traffic, Contamination, Utilities and Servicing, Noise and vibration, Air quality and odour
Package C – Environment	DPHI	SLR / Blackash	Biodiversity; bushfire; Indigenous and non-Indigenous heritage.
Package D – Statutory planning	DPHI	Patch Planning	Explanation of Intended Effect (and statutory mapping), planning pathways for resolving legislative barriers under EP&A Act and Mining Act to enable land use change.
Package E – Economics, Social and Market Assessment (this report)	Council	Aurecon	Economic assessment, social scoping report, and market assessment.

2 Preliminary land use feasibility assessment

This Section presents a preliminary land use feasibility assessment of the Site to identify key opportunities and constraints that may influence future development. The assessment provides a high-level review of the existing land use context and site characteristics to support the identification of suitable land use and employment outcomes.

2.1 Land use feasibility assessment framework

Figure 2-1 shows the land use feasibility assessment framework. The assessment identified land uses based on market assessment aligned to the Master Plan, these are defined as high feasibility, conditional, complementary, and optional land uses. For example, a land use is considered highly feasible when it demonstrates strong market fit, is appropriate for the Site, and can be delivered within a five-year timeframe based on the Site's readiness.

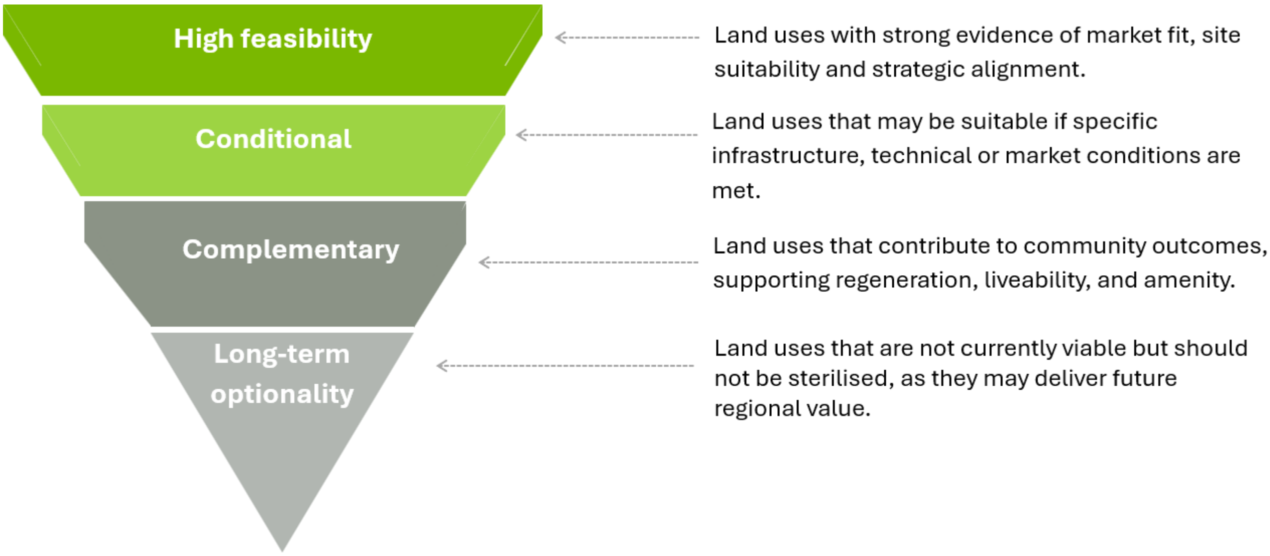

Figure 2-1: Land use feasibility assessment framework

The feasibility assessment for each land use and associated opportunity theme is based on Aurecon's understanding of the market, informed by:

- Research
- Benchmarking analysis
- Market soundings and insights from comparable projects

Several land use opportunities have been identified using the assessment framework outlined below.

2.2 Land uses identified for the Site

The identified land uses for the Site discussed in this Section are mapped in Figure 2-2 below. Time horizons are based on five-year planning period, taking into account the rapid pace of technological change.

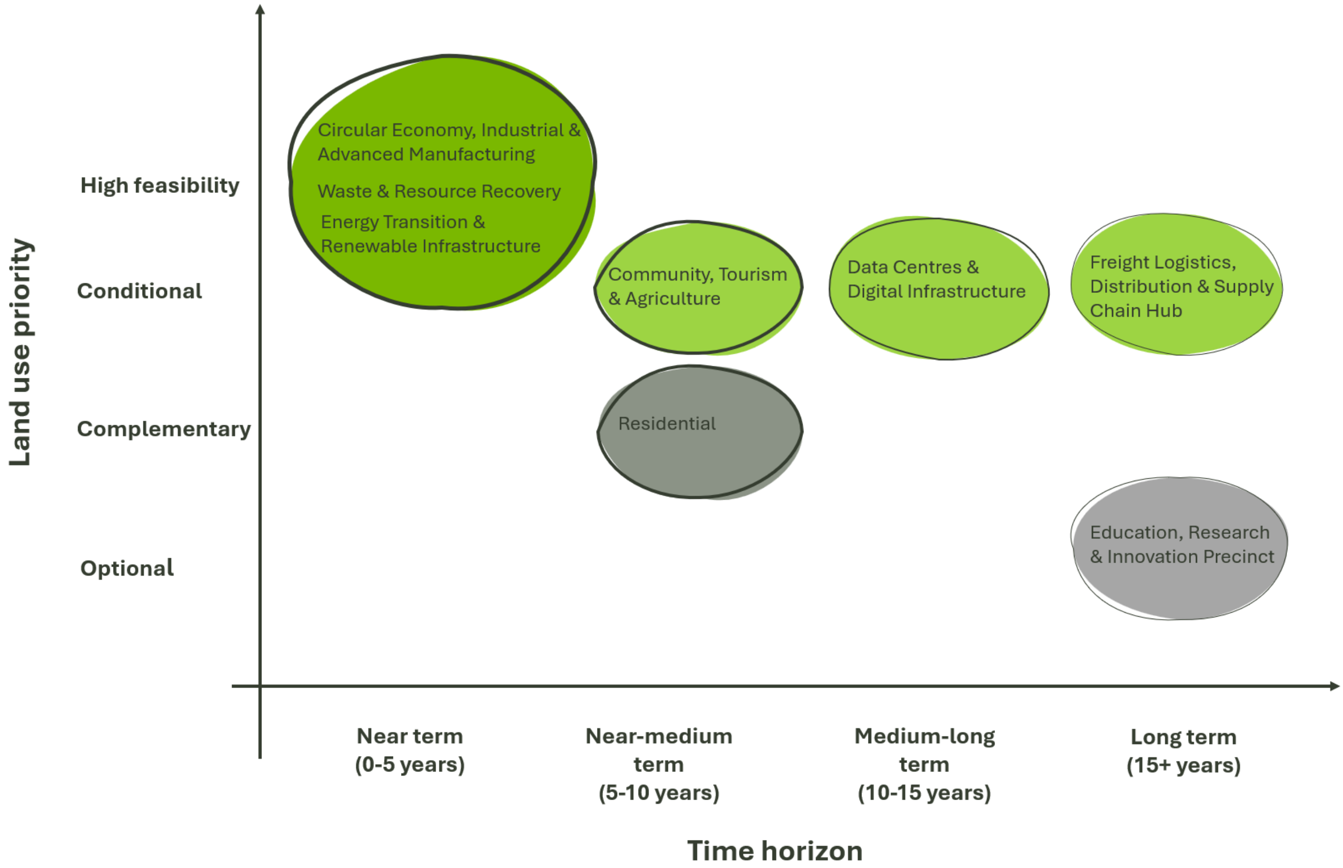

Figure 2-2: Land use priority mapping

Strategic land use categories have been assigned to sub-precincts alongside their associated land use areas including road contingency and allowance for open space on the revised structure plan post EbD workshop as presented in Figure 2-3.

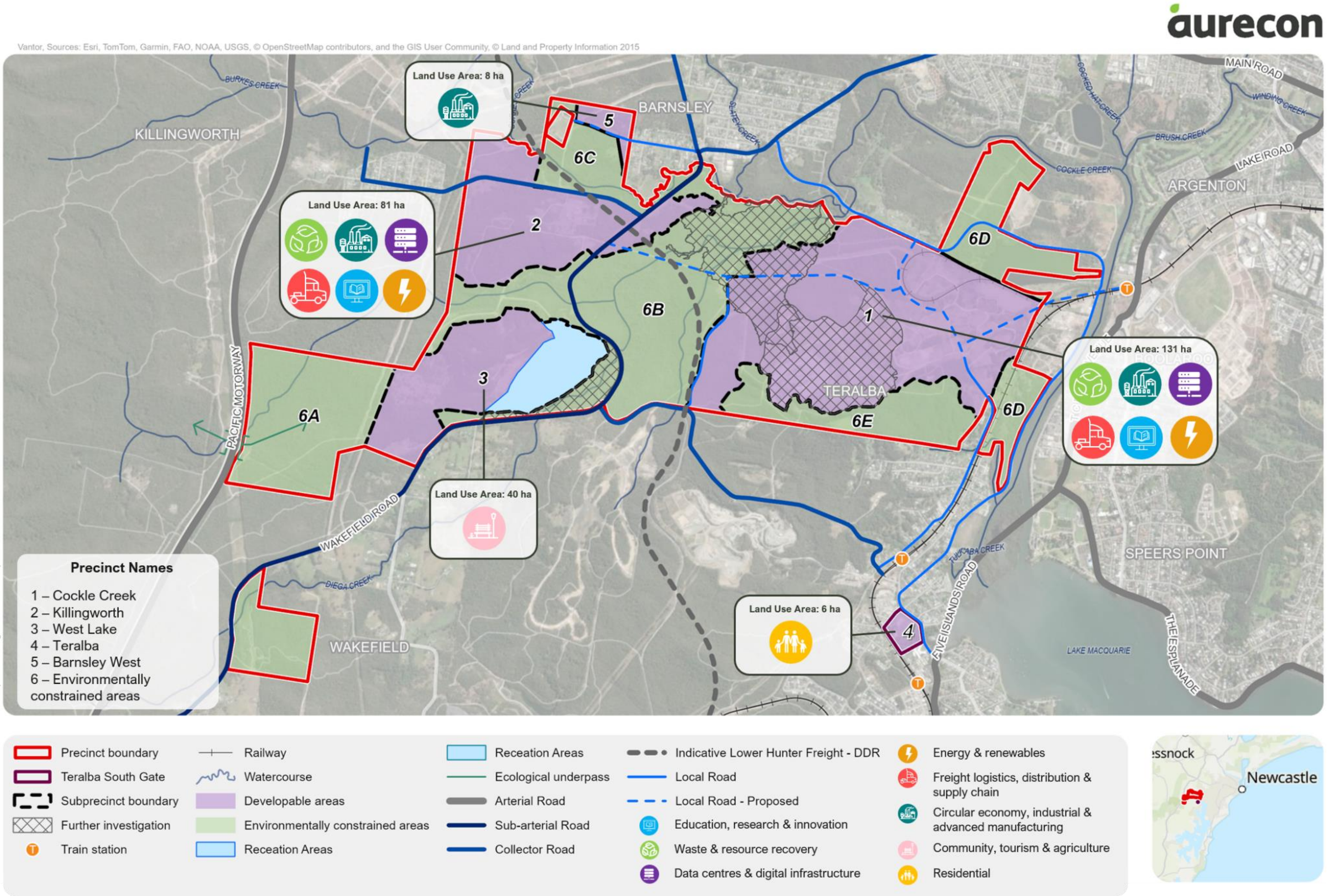

Figure 2-3: Indicative categorisation of sub-precincts for the Site

2.3 Market opportunities for the Site

The opportunities outlined in Table 2-1 sit within a competitive Hunter landscape, where sites such as Eraring Power Station, Tomago Industrial Estate and Williamtown Royal Airforce Base offer established advantages in grid access, tenants and industry clustering. The Site is best positioned for uses that leverage its scale, buffers and regional access, particularly energy transition, industrial and logistics uses. It is less competitive for port-dependent logistics, defence/aerospace, and high-amenity uses, which are better suited to more established precincts. Overall, the Site's advantage lies in delivering a large-scale and flexible industrial precinct that complements the broader Hunter economy.

Table 2-1: Market opportunities for the Site

Land use types	Target industries	Market drivers	Infrastructure requirements	Site advantages	Time horizon
Circular Economy, Industrial and Advanced Manufacturing	▪ Circular economy precinct ▪ Advanced materials ▪ Precision engineering ▪ Renewable energy component manufacturing ▪ Equipment manufacturing ▪ Defence-related manufacturing	▪ Circular economy schemes encouraging manufacturers to produce sustainable products. ▪ Skilled workforce availability in engineering and manufacturing ▪ Regionalisation of supply chain	▪ Large land parcels ▪ Reliable water supply ▪ Power supply (high-capacity, renewable-ready) ▪ Road, rail, and port access for inputs and outputs	▪ Access to Port of Newcastle for export ▪ Rail access to Sydney and regional markets ▪ Potential for co-location with complementary industries ▪ Opportunity to position the Site as circular economy leader	Near term (0-5 years)
Waste and Resource Recovery	▪ Recycling, resource and materials recovery ▪ Industrial symbiosis ▪ Water recycling	▪ Landfill diversion targets ▪ Accumulation of feedstocks for circular economy and manufacturing ▪ Water scarcity and recycled water demand ▪ Industrial symbiosis opportunities (waste from one industry as input to another)	▪ Water and wastewater treatment capacity ▪ Waste handling and processing facilities ▪ Environmental monitoring and compliance systems ▪ Interconnected utility networks	▪ Distance from residential areas and lower sensitivity to contamination ▪ Existing water and wastewater infrastructure ▪ Large land parcels for integrated facilities ▪ Potential for sustainable water integration	Near term (0-5 years)
Energy Transition and Renewable Infrastructure	▪ Battery storage	▪ NSW Government 12 GW renewable capacity target by 2030 ▪ Declining coal-fired generation creating grid stability demand ▪ Federal renewable energy investment programs	▪ High-voltage transmission connectivity ▪ Water supply for cooling ▪ Grid connection infrastructure	▪ Strong proximity to existing Newcastle Substation, 33kV / 132kV / 330kV transmission significantly reduce connection costs	Near term (0-5 years)

Land use types	Target industries	Market drivers	Infrastructure requirements	Site advantages	Time horizon
		▪ National Electricity Market modernisation requirements		▪ Established power generation workforce transition pathway ▪ Proximity to grid infrastructure and market demand ▪ Transition linked to Renewable Energy Zones (REZs), transmission upgrades, Awaba Battery, Eraring BESS and broader clean energy investment. ▪ Closure of major assets (e.g. Eraring) driving new investment	
Community, Tourism and Agriculture	▪ Regional recreation and open space ▪ Intensive horticulture ▪ Farmer's markets and agritourism	▪ Wellness tourism and experiential travel ▪ Regional food system resilience and local production demand ▪ Community trail and open space undersupply relative to Sydney metropolitan alternatives	▪ Accessible transport connectivity ▪ Utility services for community facilities ▪ Environmental restoration for recreational use	▪ Established recreational infrastructure ▪ Aging workforce transition support opportunity ▪ Large, remediated land parcels suitable for diverse community uses	Near-to-medium term (5-10 years)
Residential	▪ Residential precincts ▪ Aged care and retirement living	▪ Current/future large older cohort[14] (with aging population 65+ growing from 18% to 24% by 2036, driving demand and growing pressure for health and care services) ▪ Broader community demand for liveability and social outcomes[15]	▪ Utility services ▪ Integration with open space and social facilities ▪ Buffers from heavy industry to create safe and suitable environments.	▪ Edge-of-site locations with strong accessibility may support targeted community outcomes	Near-to-medium term (5-10 years)
Data Centres and Digital Infrastructure	▪ Cloud computing ▪ Data storage ▪ Digital services hubs	▪ Rapid growth in data consumption and cloud services ▪ Demand for regional data centre capacity outside Sydney CBD ▪ Renewable energy availability in Hunter region	▪ Reliable, high-capacity power supply ▪ Significant water supply from mine dam for cooling systems ▪ Fibre optic connectivity ▪ Secure, controlled access	▪ Strong 330kV on-site transmission and established fibre connectivity ▪ Large, contiguous land parcels ▪ Proximity to Newcastle and Sydney markets	Medium-long term (10-15 years)

[14] Lake Macquarie City Council (2022) Lake Mac - Ageing Population Strategy 2022–2026.

[15] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

Land use types	Target industries	Market drivers	Infrastructure requirements	Site advantages	Time horizon
		▪ Latency requirements for Asia-Pacific markets		▪ Potential for renewable energy integration to reduce operating costs relative to Sydney.	
Freight Logistics, Distribution and Supply Chain Hub	▪ Road freight and logistics ▪ Distribution centres ▪ Warehousing ▪ Freight consolidation ▪ Supply chain management	▪ E-commerce growth driving demand for regional distribution ▪ Port-centric logistics optimisation ▪ Rail freight capacity underutilisation ▪ Supply chain resilience post-COVID	▪ Road and rail access ▪ Large, flexible warehouse space ▪ Truck parking and driver facilities	▪ Proximity to road and rail network (if upgraded) ▪ M1 Motorway ▪ Proposed freight corridor ▪ Existing freight transport employment base ▪ Potential for rail freight expansion ▪ Proximity to substations	Long term (15+ years)
Education, Research and Innovation	▪ Research centres ▪ Innovation hubs ▪ Training and apprenticeship facilities	▪ Mining workforce reskilling ▪ University and TAFE partnerships to develop workforce capabilities for emerging industries. ▪ Government transition funding & policy support ▪ Regional economic diversification – decrease heavy industry dependence	▪ Accessible transport connectivity ▪ Utility services (power, water, sewer, digital) ▪ Supporting urban amenities	▪ Learning environments for environmental and engineering research. ▪ University of Newcastle, RMIT Advanced Manufacturing Precinct, and TAFE NSW partnerships can deliver workforce development aligned with emerging industries.	Long term (15+ years)

2.4 Opportunities for high feasibility land uses

High feasibility land uses represent priority opportunities for the Site, characterised by strong alignment with market demand, site suitability, and strategic objectives. These uses demonstrate a high likelihood of near-term delivery, with limited constraints or preconditions required to support implementation.

The following land uses have been identified as highly feasible:

- Circular Economy, Industrial and Advanced Manufacturing
- Waste and Resource Recovery
- Energy Transition and Renewable Infrastructure.

2.4.1 Circular Economy, Industrial and Advanced Manufacturing

Overview of land use

Lake Macquarie sits within the Hunter's established industrial ecosystem, which already supports advanced manufacturing and energy-intensive industry. Nearby precincts such as Tomago Industrial Estate (anchored by Tomago Aluminium and Ampcontrol) and the Kurri Kurri-Loxford precinct (anchored by Orica and major energy infrastructure) demonstrate the region's capacity for large-scale, technically complex industrial activity.

Within Lake Macquarie, areas such as Morisset and Cardiff accommodate manufacturing, fabrication and logistics, providing a practical foundation for developing a new post-mining industrial and advanced manufacturing precinct based on proven regional capabilities. The Cardiff Advanced Industry Precinct is a significant hub, supporting thousands of jobs and playing a key role in advanced manufacturing.[16]

Strategic rationale

Circular economy growth is a priority for the Council. This opportunity also aligns with the Hunter and Central Coast Circular Economy Roadmap, which positions the region around keeping materials in use for longer, improving resource recovery and building local circular economy capability. Circular economy activities tend to create up to 25 times more jobs than in recycling and 80 times more than in landfill.[17]

A Hunter Joint Organisation's business case demonstrates that implementing industrial circular economy practices across the region generates significant economic returns of $1.1 billion in gross regional product and 1,020 new jobs,[18] while simultaneously delivering environmental benefits through low-carbon industry development and contamination remediation. However sufficient resource recovery is needed to provide feedstock to circular economy manufacturing.

In addition, the NZEA Regional Economic Forecasts highlight a structural shift away from mining, while growth is expected in construction, professional services and advanced industries.[19] While these sectors may not fully replace mining's economic contribution, they can support employment growth and diversification. This transition creates demand for new industrial land uses capable of absorbing displaced labour and supporting emerging sectors.

The Lower Hunter's established strengths in structural steel fabrication, mining equipment and engineering services provide a strong platform for industrial diversification. The Site offers a compelling location for manufacturers targeting regional and export markets, supported by large land parcels, existing infrastructure and access to the Port of Newcastle. The Site offers logistics connectivity and workforce availability compared to established industrial zones such as Botany, Wetherill Park, Penrith.

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[16] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

[17] Lake Macquarie City Council (2026) Draft Waste and Circular Materials Strategy 2026-2030.

[18] Hunter Joint Organisation (2024) Inquiry into Beneficial and Productive Post-Mining Land Use (Submission 25)

[19] Oxford Economics Australia (2025) Regional Economic Transition Analysis - Economic Outlook for Hunter

Similarly, advanced manufacturing, clean energy and supply chain industries have been identified as priority investment sectors for the Hunter region, noting strong alignment with existing infrastructure and export capability.[20]

Further, Federal Government initiatives are positioning NSW as an advanced manufacturing hub, creating significant opportunities for defence-related industrial clustering, although Defence institutions may prefer established locations to minimise clearance and coordination costs.

Alignment with government policy

The NSW Industry Policy prioritises advanced manufacturing as part of its local manufacturing mission. Regional manufacturing development also aligns with the Hunter Regional Plan 2041's emphasis on economic resilience and employment diversity. Hunter and Central Coast Circular Economy Roadmap, positions the region around keeping materials in use for longer, improving resource recovery and building local circular economy capability.

Market and employment opportunities

Industrial and advanced manufacturing opportunities at the Site could focus on sectors aligned with regional strengths, workforce capability and infrastructure access. Industrial opportunities could potentially be accommodated within the Cockle Creek and Killingworth sub-precincts where larger development parcels, internal road access and buffers to ecological corridors can be planned. Also, light industrial use could be supported in the Barnsley West sub-precinct.

Large-format industrial uses requiring flexible staging and lower land costs have been noted as most viable in the near term, particularly where infrastructure delivery can be aligned with demand.

The key opportunities are detailed in Table 2-2. Details on how job numbers have been estimated are outlined in Appendix A.

[20] Oxford Economics December (2025) Regional Economic Transition Analysis - Regional Investment in the Hunter.

Table 2-2: Circular Economy, Industrial and Advanced Manufacturing market opportunities for the Site

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Modular construction & prefabrication	Off-site construction of residential and infrastructure components to meet strong regional housing / infrastructure demand and reducing labour costs. Firms such as Blokable and Modular Space, currently expanding from Sydney, represent potential anchor tenants for regional hub establishment at the Site.	Existing precast manufacturing operations in Teralba (e.g. Structural Concrete Industries) demonstrate established capability in the area. Demand from residential development across the Central Coast, Hunter, and inland NSW provides sustained market opportunity. Large-format industrial land and proximity to the M1 Pacific Motorway and Newcastle-Sydney rail corridor support logistics-heavy operations.	Highly feasible	~40
Structural steel & metal fabrication	Fabrication of steel and metal components for infrastructure, transport and energy projects, supporting ongoing regional infrastructure investment and renewable energy rollout. Structural steel fabrication operations currently concentrated in Sydney can establish regional production facilities targeting infrastructure projects across NSW and Queensland.	The Hunter has demonstrated capability in this sector, with existing firms such as Hunter Structural Steel and regional fabricators maintaining strong market positions. Relocation or expansion to the Site would provide: ▪ Access to bulk material inputs via Port of Newcastle and regional suppliers ▪ Lower land and operational costs compared to Sydney precincts ▪ Proximity to major infrastructure projects (Pacific Highway upgrades, rail infrastructure, renewable energy installations)	Highly feasible	~40
Precision engineering[21] & advanced manufacturing	CNC machining, additive manufacturing and precision fabrication leverage transferable skills from mining and heavy industry workforce. Firms such as Bradken Limited and other regional precision machining and advanced manufacturing firms have demonstrated successful operations in the Hunter, validating market viability.	The Hunter's strong mining and metallurgical manufacturing base provides a skilled workforce capable of transitioning to advanced manufacturing. Ability to cluster anchor tenants aligns with need to diversify industrial base as mining employment declines. Precision engineering firms currently concentrated in Sydney precincts could establish regional operations at significantly reduced cost.	Highly feasible	~25
Integrated circular economy precinct	A coordinated precinct combining C&D processing, battery recycling and waste transfer operations could enable industrial symbiosis where waste from one process are repurposed as inputs to another. Similar	Strong long-term opportunity if staged around anchor users and underpinned by infrastructure, regulatory approvals and market demand.	Highly feasible	~40

[21] Precision engineering refers to the manufacture of highly accurate components and products using advanced machining, fabrication and production technologies.

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
	models include BINGO Industries' Eastern Creek Recycling Ecology Park in Western Sydney.			
Critical minerals and battery material recovery	Establish a facility for reprocessing battery materials and enabling closed-loop recovery of critical mineral such as lithium, nickel, cobalt and manganese from end-of-life batteries associated with renewable energy systems and electric vehicles. Considered as a modular, longer-term opportunity, potentially co-located with other energy and advanced manufacturing activities as feedstock volumes and market conditions mature.	The Site favours industrial-scale, energy-intensive uses with strong logistics and power access. International experience, particularly in Germany, demonstrates that successful facilities are typically integrated within established industrial precincts and supported by strong policy settings, scale and downstream demand. In Australia (including Neometals, Livium Ltd, Ecocycle), activity remains at an early stage, with pilot and demonstration projects beginning to emerge as policy support strengthens.	Conditional	~40
Advanced materials & composites	Production of high-value composite materials for aerospace, defence, automotive and renewable energy applications (e.g. wind turbine components) is supported by regional R&D capability and emerging clean energy industries and represents a high-value manufacturing opportunity.	Opportunity to co-locate with research institutions and advanced manufacturing ecosystem in the Hunter. Large industrial footprints and buffer distances suit specialised, high-value manufacturing. The Hunter region has emerging capability in this sector, with RMIT's Advanced Manufacturing Precinct in Newcastle providing research and development support. The Site could host: ▪ Composite manufacturing facilities serving aerospace supply chains ▪ Automotive component production for electric vehicle manufacturers ▪ Wind turbine blade and nacelle component fabrication	Conditional	~30

2.4.2 Waste and Resource Recovery

Overview of land use

Municipal waste management in Lake Macquarie is handled by the Council, which provides recycling, organics, garbage, and bulky household goods services. This creates opportunities for integrated waste management and circular economy infrastructure development on post-mining sites, rather than limiting service provision.

Strategic rationale

Waste recycling can play a critical role in reducing waste, improving resource efficiency, and supporting the transition to more sustainable and resilient regional economies, particularly as waste volumes increase and landfill capacity becomes more constrained across NSW. The Lower Hunter currently lacks a purpose-built, large-scale circular economy hub capable of serving regional waste streams and supporting the transition to a circular economy, creating a significant market gap that the complex could address.

The Site's character as a post-mining site creates regulatory advantages over greenfield locations, with incremental contamination risk lower than undisturbed sites and more straightforward pathways for remediation and repurposing.

Waste processing operators increasingly seek to establish regional recycling and recovery hubs to serve growing demand across regional NSW, driven by landfill diversion targets, and critical mineral recovery opportunities. The Cockle Creek and Killingworth sub-precincts could be positioned as alternative locations within the Site, benefiting from access to regional waste sources, transport networks, and end markets for recovered materials.

Alignment with government policy

Lake Macquarie City Waste and Circular Materials Strategy 2026 – 2030 identifies reducing organic waste and increasing reuse, repair, trade and share economy capacity as key priorities. These objectives reinforce the need for sustained investment in infrastructure and partnerships that enable resource recovery and support the transition to a circular economy.

Market and employment opportunities

Waste and resource recovery market opportunities value land, buffers, logistics access and proximity to regional waste streams. Highly feasible opportunities detailed in Table 2-3 are best suited to resource recovery uses that can operate away from the Cockle Creek wetland and floodplain areas. Estimated employment per waste and resource recovery facility is 50-100 jobs.

Table 2-3: Waste and Resource Recovery market opportunities for the Site

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Construction and demolition (C&D) waste recovery hub	A 200,000-300,000 tonne per annum C&D waste processing facility could serve the Greater Newcastle region and northern NSW.	Strong fit due to land requirements, buffers and regional construction demand.	Highly feasible	~50[22]
Mixed waste transfer and sorting facility	A 100,000-150,000 tonne per annum mixed waste transfer and sorting facility could serve Lake Macquarie, Wyong, and regional council contracts.	Supports regional waste aggregation and could reduce pressure on existing landfill-dependent systems.	Highly feasible	~100
Battery and e-waste recycling centre	A specialised 10,000-20,000 tonne per annum battery and e-waste facility, positioned as a regional aggregation and pre-processing hub could leverage growing waste streams from renewable energy systems and consumer	Aligns with energy transition and growth in battery and electronics waste streams, while addressing gaps in regional processing capacity; would require strong environmental controls.	Highly feasible	~60

[22] Employment estimates should be interpreted with caution, as some specialised data centre roles may be performed remotely or sourced from larger metropolitan labour markets, reducing on-site local employment.

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
	electronics while addressing gaps in local processing infrastructure.			

2.4.3 Energy Transition and Renewable Infrastructure

Overview of land use

Battery energy storage systems are emerging as important land uses across the Newcastle and Lake Macquarie region as coal-fired power station retire. This transition is evidenced by projects such as the Waratah Super Battery (50 km from the Site), which will provide 850MW/1680MWh of capacity to stabilise the grid and enable increased renewable energy penetration.

Energy transition investments are increasing demand for energy-enabling infrastructure, advanced manufacturing, and associated supply chains. Key drivers include the proposed 50MW Awaba Battery, 700MW Eraring BESS, EnergyCo's Hunter Transmission project, the Site's location within the Hunter-Central Coast REZ, and proximity to the Central West Orana and New England REZs.

Strategic rationale

Renewable energy, grid infrastructure and clean energy supply chains are priority investment sectors in the Hunter, though delivery remains dependent on grid capacity, market demand and infrastructure sequencing.[23] The existing workforce brings transferable operations, maintenance and technical skills, reducing retraining requirements and supporting faster project deployment.

Whilst renewable energy manufacturing was considered due to the sector's growth potential, the foreseeable renewable energy-related land use at the Site is installation and operation of BESS, reflecting difficulty of competing with current low-cost supply of renewable energy equipment from China.

The Killingworth sub-precinct is well suited to energy-intensive uses due to its proximity to Transgrid Newcastle 330kV / 132 kV Substation. Lower energy uses within the sub-precinct may require step-down electrical infrastructure. The limited size of the Ausgrid Argenton Substation 132kV / 33kV / 11kV in close proximity to the Cockle Creek sub-precinct restricts its capacity to accommodate a large BESS relative to Killingworth. Heavy energy uses in the Cockle Creek sub-precinct such as large BESS may require the provision of step-up electrical infrastructure to accommodate demand.

BESS facilities also require thermal management to maintain battery performance and safety. While smaller systems commonly use air cooling, contemporary utility-scale BESS facilities increasingly utilise liquid-cooling systems due to their superior thermal control and energy density. Water use is generally limited, as liquid-cooling systems typically operate as closed-loop systems.

However, the number of jobs lost from closed mines are unlikely to be matched by renewable energy opportunities identified for the Site. Overall, energy transition uses represent a strategically aligned but infrastructure-dependent opportunity, best pursued in coordination with broader regional energy investment.

Alignment with government policy

Energy transition use at the Site aligns directly with the State's 12 GW of renewable energy capacity targets by 2030[24], and the Site could be positioned to contribute toward this goal. The Hunter Regional Plan 2041 explicitly prioritises energy transition as regional development priority, ensuring alignment with coordinated regional planning frameworks.

Market and employment opportunities

[23] Oxford Economics Australia (2025) Regional Economic Transition Analysis - Regional Investment in the Hunter.

[24] NSW Department of Planning, Industry and Environment (2020) NSW Electricity Infrastructure Roadmap – Building an Energy Superpower Detailed Report.

Table 2-4 lists key opportunities in the Energy Transition & Renewable Infrastructure sector for the Site, indicating approximately 20 jobs could be generated per facility.

Table 2-4: Energy Transition & Renewable Infrastructure market opportunities

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
BESS	200-850MW BESS to support peak demand and grid stability.	Potential proximity to transmission infrastructure reduces connection complexity and supports staged deployment. Large sites can accommodate cooling infrastructure required for BESS.	High	~20

2.5 Opportunities for conditional land uses

Conditional land uses represent opportunities that demonstrate strategic alignment and relevance but are dependent on specific infrastructure, technical, market, or policy conditions before they become viable. These uses typically require enabling investments, regulatory support, or stronger market signals, resulting in longer development timeframes compared to high feasibility opportunities.

The following land uses have been identified as conditional:

- Data Centres and Digital Infrastructure
- Freight Logistics, Distribution & Supply Chain Hub
- Community, Tourism and Agriculture.

2.5.1 Data Centres and Digital Infrastructure

Overview of land use

Driven by increasing data consumption, AI and digitalisation across industry, data centres and digital infrastructure are emerging employment land uses. This has led to strong expansion of data centre capacity in NSW, particularly in Sydney and emerging secondary markets. AEMO estimates Australian data centres consumed 3.9 TWh in FY25, with demand projected to grow materially by FY30 and beyond, while Clean Energy Finance Corporation / Baringa forecasts Australian data centre capacity could increase from 1.35 GW today to 4.7-7.4 GW by 2035.[25,26]

Strategic rationale

The Site's scale and industrial setting, along with alignment to the energy transition, may support data centres and energy-intensive digital infrastructure, most likely within the Killingworth or Cockle Creek sub-precincts.

However, the suitability of the Site is conditional. Data centres require high-capacity, reliable electricity supply, secure fibre connectivity, and substantial cooling solutions. While the Site has access to clean raw water, the availability may be constrained relative to demand. This limitation could potentially be mitigated through alternative cooling technologies, such as air- or fan-based systems, supported by high-capacity power supply. Larger facilities may trigger SSD pathways in NSW, with data centres above 15 MW consumption treated as SSD.

While the Hunter region offers potential advantages in lower land costs and access to renewable energy, digital infrastructure has been identified as an emerging but not yet dominant regional sector, with stronger clustering in metropolitan areas.[27] AEMO's regional diversification sensitivity indicates that data centre demand could be allocated

[25] Oxford Economics Australia (2025) Data Centre Energy Demand – Final Report.

[26] CEFC and Baringa (2025) Getting the balance right: data centre growth and the energy transition.

[27] Oxford Economics Australia (2025) Regional Economic Transition Analysis - Regional Investment in the Hunter.

more broadly across states and sub-regions if location decisions place greater weight on population growth, professional workforce, fibre connectivity and regional market factors. While Sydney remains the dominant NSW market, this sensitivity supports the potential for secondary regional locations to attract demand where they can demonstrate sufficient power capacity, fibre connectivity and development ready land.

As a result, data centres represent a longer-term or opportunistic use, rather than a primary driver of development.

Alignment with government policy

Data centres and digital infrastructure are well aligned with the NSW Industry Policy, which supports digital transformation, innovation and investment in future-focused industries, and the Hunter Regional Plan 2041, which promotes economic diversification and the growth of advanced industries. At the local level, the Lake Macquarie Local Strategic Planning Statement 2025-2045 supports employment-generating investment, industry diversification and the attraction of knowledge-intensive businesses. As a result, digital infrastructure represents a logical long-term land use that can contribute to regional economic resilience and high-value employment outcomes.

Market and employment opportunities

Key opportunities in the data centres and digital infrastructure sector for the Site are detailed in Table 2-5. Estimated employment per data centre and digital infrastructure facility is 5-50 jobs.

Table 2-5: Data centres and digital infrastructure market opportunities

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Hyper-scale centre (long-term)	Hyper-scale data centre facilities providing cloud hosting and enterprise data services.	Potentially viable where large land parcels, reliable power supply and fibre connectivity are available	Conditional	~50
Regional data centre/ edge computing facility	Smaller-scale facilities supporting regional data processing, latency reduction and service resilience.	Stronger near-term opportunity; aligned with regional population and business demand and lower infrastructure requirements.	Conditional	~5
Digital services/ technology support hub	Data storage, IT services, network operations and support functions.	Suitable for industrial precinct co-location with reduced infrastructure intensity	Conditional	~5

2.5.2 Freight Logistics, Distribution and Supply Chain Hub

Overview of land use

Freight, logistics, distribution and supply-chain activities are an established land use across the Lower Hunter. These activities typically include bulk storage, warehousing, container handling, construction materials logistics and last-mile distribution servicing regional and metropolitan markets.

Major NSW hubs including Moorebank Intermodal Terminal and Eastern Creek demonstrate the demand for large, well-located sites that integrate road, port and industrial supply chains. At a post-mining site, such uses would involve large-format logistics facilities, hardstand areas and supporting transport operations.

Strategic rationale

The Killingworth sub-precinct is particularly well suited to logistics and distribution activities given its proximity to the M1 Pacific Motorway. The Cockle Creek sub-precinct also offers strategic advantages for freight-related activities through its access to the Main Northern Railway Line, proximity to the Port of Newcastle and potential connection to the proposed Lower Hunter Freight Corridor.

Continued growth in transport, postal and warehousing, construction and service-based industries, reflects increased demand for logistics and supply chain infrastructure across the Hunter.[28] This is reinforced by population growth and increasing demand for goods and services across Greater Newcastle and the Central Coast. Similarly, shortage of large, serviced industrial land, particularly for logistics and warehouse uses requiring scale and heavy vehicle access limits the ability of existing precincts to accommodate new large-format logistics development.

While the Site is not directly competitive with port-dependent or intermodal logistics (better suited to Port of Newcastle or Sydney terminals), it is well suited to road-based distribution, regional servicing and construction supply chains, where access to the motorway network and large contiguous land parcels is critical. Also, feasibility studies highlight that specific supply chain, and industrial uses could be viable early-stage land uses, due to lower infrastructure requirements and stronger near-term demand.[29] However, significant investment may be required to make rail-related logistics viable.

Alignment with government policy

The Hunter Regional Plan 2041 prioritises supply chain efficiency and logistics infrastructure development as key economic enablers. Delivering Freight Policy Reform in NSW identifies the importance of industrial land for distribution centres. Port of Newcastle's Development Plan emphasises hinterland connectivity and regional supply chain integration, directly supporting logistics hub development at the Site.[30]

Market and employment opportunities

The key opportunities in the Freight Logistics, Distribution and Supply Chain sectors for the Site are detailed in Table 2-6. Estimated employment per freight logistics, distribution & supply chain facility is 90-400 jobs.

Table 2-6: Freight Logistics, Distribution & Supply Chain market opportunities for the Site

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Cold chain & perishables logistics	The Hunter region's agricultural production and proximity to Sydney markets create demand for temperature-controlled logistics infrastructure including: - Agricultural product aggregation and distribution - Food manufacturing inputs and finished goods logistics	Supported by regional agricultural production and population growth, it requires land, buffers and road access rather than port adjacency.	Conditional	~90
Regional distribution/ e-commerce hub	A 50,000-100,000 m^2 distribution facility could serve the following supply chains across the Lower Hunter and northern NSW: - Major retailers seeking regional distribution capacity - Construction materials distributors serving infrastructure projects - Manufacturing including automotive parts and components distributors	Strong alignment with motorway access and large land parcels; supports decentralisation of Sydney-based logistics Directly aligned with projected construction growth and proximity to Hunter infrastructure pipeline	Long-term optionality	~400

[28] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Economic Outlook for Hunter.

[29] Atlas Economics (2026) Lake Macquarie Development Feasibility Study.

[30] Port of Newcastle (2023) Port of Newcastle Development Plan 2023-2028.

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
	- Fast-moving consumer goods (FMCG) operators			

2.5.3 Community, Tourism and Agriculture

Overview of land use

The Hunter Regional Plan 2041 emphasises sustainable growth, enhanced liveability, and diversified economic opportunities, supporting a balanced approach to land use across community, tourism and agriculture sectors. In the context of post-mining land use, delivering social, recreational and economic benefit to local communities is a key consideration.

An example is the $95 million Black Rock Motor Resort, currently under construction, demonstrating how rehabilitated mining land can be used to support recreation and tourism within the Lake Macquarie region.

Strategic rationale

Community, tourism and agriculture uses at the Site are justified where they support post-mining transition outcomes, including workforce transition, land rehabilitation and activation of constrained areas. These uses align with local policy direction to repurpose former mining land, support emerging sectors and deliver environmental and social outcomes, but should be applied selectively where they complement higher-value employment uses rather than compete with them.

At the Site, these uses are best positioned as supporting and enabling land uses rather than the primary economic driver. Community, tourism and agriculture uses can strengthen the overall Site by activating rehabilitated land, improving amenity, managing buffers between land uses and delivering long-term environmental outcomes.

The market rationale is strongest where these uses align with existing regional demand and site-specific constraints. Lake Macquarie has an established visitor economy, with the Destination Management Plan identifying more than 2.5 million visitors annually,[31] tourism contributes approximately $338.5 million in value added to the local economy.[32]

Tourism, open spaces and recreational facilities leverage the region's natural assets and growing visitor economy. In addition, Lake Macquarie economy has increasingly relied on construction, services and visitor-related sectors to offset industrial decline,[33] reinforcing the role of amenity-led land uses in supporting economic activity. The West Lake sub-precinct and the adjoining lake present potential opportunities for tourism and recreation.

Controlled environment agriculture including greenhouse horticulture and nurseries could also be supported if sufficient water, energy and transport infrastructure are available to enable year-round production and distribution. The Central Coast Primary Industries Centre located in the University of Newcastle and Somersby Field station is a specialist greenhouse and research and development complex to support horticulture and agriculture within the region. Examples of controlled environment agriculture within the Newcastle region include hydroponics farms such as Hydro Produce and Newcastle Greens, and indoor nurseries such as Glendale Select Plants. The Newcastle region already supports a range of commercial horticultural and agricultural activities, including:

- Wholesale nursery production
- Native plant propagation
- Landscape construction and maintenance
- Turf production

[31] Lake Macquarie City Council (2022) Lake Macquarie City Destination Management Plan 2022–2026.

[32] REMPLAN (2025) Lake Macquarie Tourism Profile.

[33] The University of Newcastle (2023) Responding to Structural Change in Lake Macquarie - Briefing Paper for Policy Makers.

- Commercial greenhouse operations
- Environmental rehabilitation and revegetation projects.

Alignment with government policy

The Hunter Regional Plan 2041 supports sustainable growth, lifestyle opportunities and economic diversification, while recognising the Hunter's natural assets and the transition of traditional mining, energy and manufacturing sectors.[34] Tourism and recreation uses align with the NSW Visitor Economy Strategy 2035, which targets visitor economy growth and additional tourism-related jobs.[35] Open space, rehabilitation and offset uses also align with NSW Greener Places policy by delivering green infrastructure with social, environmental and economic benefits.[36]

Agriculture-related uses, including controlled environment agriculture or intensive horticulture, align with NSW primary industries workforce strategy[37] where they support sustainable, innovative and regionally resilient production.[38] Collectively, these policy settings support community, tourism and agriculture uses at the Site as conditional land uses that can activate constrained land, support transition outcomes and improve amenity.

Market and employment opportunities

Key opportunities for community, tourism and agriculture uses are detailed in Table 2-7. Retail uses are not identified as a high-value opportunity due to their lower job density and economic output. However, complementary activities such as shops, restaurants, and retail are expected across identified land uses, supporting functionality and community amenity.

Community, tourism and agriculture land uses are expected to generate relatively limited direct ongoing employment; therefore, job numbers have not been estimated for most of this category. Overall employment opportunities are likely to be modest, with some additional roles potentially supported through the visitor economy and intensive horticulture and controlled environment agriculture activities. Estimated employment per controlled environment agriculture facility is 130 jobs.

Table 2-7: Community, Tourism and Agriculture opportunities for the Site

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Regional open space and recreation	Social infrastructure, open space, trails, entertainment parks, sports and environmental restoration supporting community use and regional visitation.	Strong regional recreation demand; complements Lake Macquarie's outdoor profile; avoids oversupply risks associated with additional golf courses.	Conditional	Not assessed
Visitor economy and eco-tourism experiences	Nature-based tourism, environmental education, events aligned with growing visitor economy.[39]	Builds on established tourism strengths; activates non-industrial land; supports local spending and employment, with tourism currently generating approximately $338 million in value-add[40] (2.5 million visits annually).	Conditional	Not assessed

[34] NSW Department of Planning and Environment (2022) Hunter Regional Plan 2041.

[35] NSW Government and Destination NSW (n.d.) NSW Visitor Economy Strategy 2035.

[36] Government Architect New South Wales (2020) Draft Greener Places Design Guide.

[37] NSW Government (2025) NSW Primary Industries Workforce Strategy 2025 – 2030.

[38] Ibid.

[39] Ibid.

[40] ABS (2025) Tourism Satellite Account.

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Greenhouse horticulture and nurseries	Greenhouse horticulture and nurseries	Large land parcels, access to proximity to Sydney markets and established Hunter supply chains support viability.	Conditional	~130

2.6 Opportunities for complementary land uses

Complementary land uses help improve the area, make it more liveable, and enhance local amenities. While they are not primary economic drivers for the Site, they play an important role in creating a balanced and sustainable precinct and contribute to social value and place-making.

Residential land uses have been identified as complementary.

2.6.1 Residential

Overview of land use

Lake Macquarie's housing infrastructure context indicates demand for more diverse residential land uses. Council's Housing Strategy supports housing in well-located existing urban areas, greater housing diversity, and improved access to jobs, services and transport.[41] The Housing Diversity Planning Proposal also responds to an ageing population and smaller households by supporting infill housing and a broader mix of housing types.[42]

For the Site, residential use should therefore be considered selectively where they support housing diversity, ageing-in-place, access to services and connectivity, while being tested against servicing, contamination, mine subsidence and interface risks with employment-generating uses.

Strategic rationale

Residential and senior living uses at the Site are supporting and enabling land uses, rather than primary economic drivers. Edge-of-site locations with strong accessibility could accommodate residential development that serves as an appropriate interface with industrial uses and helps build community acceptance of broader industrial expansion. Demand for senior living residences is expected to increase due to growth in ageing population.[43]

Alignment with government policy

The Hunter Regional Plan 2041 and Lake Macquarie City Council Local Strategic Planning Statement 2025-2045 promote renewal of former mining land through integrated housing, employment, tourism, recreation and open space. Together, these policies support a balanced approach in which amenity-led uses enhance resilience and adaptive reuse, while residential development remains staged and infrastructure-led, complementing the Site's primary employment function.

Market and employment opportunities

Figure 2-3 indicates approximately seven hectares of land in Teralba Southgate, south-east of the Site is likely to be available for residential and senior living due to access to existing rail and transport-oriented development opportunities. Key opportunities for residential uses are detailed in Table 2-8. While senior living uses may support approximately 50 ongoing jobs per facility, residential development is anticipated to primarily contribute through construction-related employment.

[41] Lake Macquarie City Council (2026) Draft Housing Strategy.

[42] Lake Macquarie City Council (2025) Planning Proposal- Housing Diversity.

[43] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

Table 2-8: Residential opportunities for the Site

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility	Estimated jobs per facility
Senior living	Aged care and senior living responding to population ageing and rising service demand.	Provides stable, non-cyclical employment; supports service gaps.	Complementary	~50
Limited edge-of-site residential	Small-scale residential development on remediated, serviced land.	Strategic edge location.[44]	Complementary	Not applicable

2.7 Opportunities for optional land uses

Optional land uses represent opportunities that are not currently viable but may deliver future regional value under changing market, policy, or infrastructure conditions. While these uses are not expected to form part of near- or medium-term development, they should not be sterilised to preserve flexibility and potential for long-term transformation.

Education, Research & Innovation land use has been designated as optional due to limited market demand in the short term.

2.7.1 Education, Research and Innovation Precinct

Overview of land use

The Site is located in the proximity of the University of Newcastle, Avondale University and TAFE campuses located in Belmont and Glendale. Lake Macquarie Private Hospital plays supports clinical trials and research across Australia, being driven by the Ramsay Research and Development network,[45] providing several avenues of support for the transition of mine workers into other industries.

Strategic rationale

The Site presents a long-term opportunity to establish a regional education, research and innovation precinct. This could be located in the Killingworth, Cockle Creek or potentially Barnsley West sub-precincts. This aligns with regional transition priorities, where education, industry and research collaboration is critical to enabling economic diversification and workforce redeployment. The need for such precincts is further reinforced by workforce dynamics, with up to 76% of fossil fuel workers requiring reskilling or transition pathways.[46]

Co-located precinct models (e.g. Newcastle Institute for Energy and Resources Precinct) show that bringing together universities, industry and government in a single location accelerates skills development, technology adoption and commercialisation outcomes. An education, research and innovation precinct at the Site would leverage proximity to emerging renewable energy and manufacturing activities to provide authentic, industry-integrated learning environments, strengthening pathways between training, research and employment.

Importantly, the Hunter is already investing in this model through initiatives such as the TAFE NSW Manufacturing Centre of Excellence, which aims to support renewable manufacturing and advanced industry capability through industry-aligned training, apprenticeships and applied research.[47] Co-locating education and research functions in proximity to an operational industrial precinct would build on this momentum, creating a connected innovation ecosystem that supports workforce transition, attracts investment and underpins long-term regional competitiveness.

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[44] Partially compatible with edge-of-site development, although proximity to existing industrial uses such as the Holcim concrete batching plant and Structural Concrete Industries facility, may create amenity and land use conflict constraints for residential uses.

[45] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region

[46] Ibid.

[47] NSW Government (2024) Net zero manufacturing TAFE Centre of Excellence in the Hunter.

Alignment with government policy

The NSW Skills Plan 2024-28 identifies Net Zero and energy transition, agriculture and agrifood, and advanced manufacturing as Critical Skills Areas requiring targeted industry, workforce and place-based responses.[48] This aligns with the NSW Industry Policy, which prioritises Net Zero and Energy Transition and Local Manufacturing, and the Net Zero Industry and Innovation Investment Plan, which supports low-carbon manufacturing, mining sector decarbonisation and clean technology development.[49,50]

This policy direction is already being delivered in the Hunter through the $56.2 million Hunter Net Zero Manufacturing Centre of Excellence at TAFE NSW Tighes Hill, focused on renewable manufacturing, clean energy and advanced manufacturing skills.[51] The Future Jobs and Investment Authority also provides a direct policy link to the Site, with a remit covering post-mining land-use planning, skills mapping, training programs and new employment-generating industries in coal regions. Together, these policies position the Site as a credible location for applied training, research and industry partnerships linked to energy transition, circular economy, advanced manufacturing, agrifood and land rehabilitation.[52]

Market opportunities

Key opportunities for education, research & innovation use at the Site are detailed in Table 2-9. [53]

Table 2-9: Education, Research & Innovation market opportunities

Opportunity theme	Core activities & programs	Why it suits the Site	Feasibility
Circular economy & clean technology R&D	Recycling and resource recovery systems; circular manufacturing research	Leverages industrial land, waste streams and proximity to manufacturing uses, supporting regional circular economy priorities.	Long-term optionality
Energy transition training & workforce development	Renewable energy technician and engineer programs; grid operations and microgrid management; energy efficiency certification; apprenticeship pathways for transitioning mining workers	Directly supports reskilling of the mining workforce and addresses regional labour shortages. Co-location with industrial and renewable energy uses enables hands-on training and strong employment pathways.	Long-term optionality
Advanced manufacturing & innovation hub	Composite materials research and prototyping; advanced manufacturing process development; industry-sponsored research projects	Builds on the Hunter's manufacturing capability, supports clustering with industry, attracts funding and enables commercialisation and spin-off activity.	Long-term optionality
Clean energy & industrial innovation precinct	Integration of training, R&D and industry pilots within a clean energy industrial ecosystem	Scale, infrastructure and proximity to REZs position the Site as an ideal location for integrated industry-education clustering.	Long-term optionality
Environmental remediation & land rehabilitation research	Mine rehabilitation techniques; soil remediation; biodiversity restoration; water management and monitoring	The Site could act as a living laboratory, enabling applied research with transferable outcomes across post-mining contexts.	Long-term optionality

[48] NSW Department of Education (2024) NSW Skills Plan 2024-28 – Building skills and shaping success.

[49] NSW Government and Investment NSW (2025) NSW Industry Policy – A forward-looking agenda for a resilient and productive NSW economy.

[50] Office of Energy and Climate Change (2022) Net Zero Industry and Innovation Investment Plan.

[51] Department of Employment and Workplace Relations (2024) Net Zero Manufacturing TAFE Centre of Excellence for the Hunter

[52] NSW Government (n.d.) Future Jobs and Investment Authority.

[53] Due to long-term uncertainty surrounding the development of an education, research and innovation precinct at the Site, potential employment generation has not been assessed.

2.8 Summary of employment opportunities

Approximately **1,130 jobs could be accommodated across the Site**. This estimate was derived based on the following assumptions:

- Aurecon assessed the range of facilities that could be accommodated across the Site, considering highly feasible, conditional and complementary land use opportunities.
- The number of facilities for each opportunity was assigned on a scale of one to three based on market demand drivers.
- Employment generation for each facility type was estimated using the NSW Common Planning Assumptions Group (CPAG) workspace ratio guidance referenced in Section 2.3.1.
- Employment densities were derived using the high-scenario assumptions for comparable facilities presented in Appendix A.
- Employment estimates represent ongoing jobs and do not include construction-related employment.

3 Post-mining impediments, reform consideration and social implications

This Section examines post-mining impediments and the need for reform, focusing on regulatory and delivery framework assessments, structural constraints, and key transition risks. It also outlines overarching mitigation strategies and highlights land use dependencies that influence successful transition outcomes.

3.1 Regulatory and delivery framework assessment

This assessment examines how current regulatory, policy and institutional settings influence the feasibility, timing and delivery of indicative post-mining land uses at the Site. It has been prepared having regard to the NSW Parliamentary Inquiry into Beneficial and Productive Post-Mining Land Use, with a focus on identifying where existing frameworks enable, constrain or distort economically viable redevelopment outcomes.

The Site represents a large-scale, long-life mining asset with a complex transition pathway. Transition from a mine site undergoing active rehabilitation to future industrial, employment, infrastructure and community uses will occur progressively and is dependent on the alignment of rehabilitation, relinquishment, planning approvals and infrastructure provision.

The redevelopment of the Site differs from typical greenfield or infill development due to:

- the scale and staging of rehabilitation across a large and complex Site
- the need to transition land through multiple regulatory states (closed mining > rehabilitation > relinquishment > developable land), noting that the Site is currently undergoing substantial rehabilitation
- the fragmented timing of land availability, rather than a single release point
- limited precedent for coordinated large-scale post-mining transformation of this nature in NSW

As a result, the development pathway is not solely driven by market demand, but by the interaction of regulatory processes and infrastructure readiness. A fast-track pathway approval process will benefit economic development and land use employment development.

3.1.1 Assessment of key frameworks

The proposed land use mix for the Site identified in Section 2 differs significantly in terms of regulatory requirements, infrastructure dependency and delivery risk. Current frameworks do not affect them uniformly.

Mine closure and rehabilitation

The progressive nature of mine closure and rehabilitation across a large, complex mine site is the key determinant for delivering development-ready parcels of land rather than a whole-of-site release.

Closure conditions differ significantly across individual operational areas of the Site. Certain areas have already undergone substantial rehabilitation, including infrastructure removal, contamination remediation and mine sealing works, while others retain fixed infrastructure, rail assets or ongoing closure obligations. Examples include:

- completed contamination remediation and rehabilitation works at Westside Opencut
- retained fixed infrastructure and remaining rail infrastructure at Teralba Southgate and Macquarie Coal Preparation Plant (MCPP)
- sealed shafts and drifts at West Wallsend Under Ground and Stockton borehole areas at MCPP
- progressive removal of coal handling and operational infrastructure across several sites

Closure activities scheduled through 2026-2027 include demolition of certain infrastructure, mine sealing works, tailings storage facility capping and rehabilitation signoff activities, all of which will continue to influence redevelopment sequencing and land availability.

Regulatory certification requirements and final landform outcomes may materially influence redevelopment timing and land use suitability, particularly for residential and other sensitive uses.

Implications by land use:

- Industrial, logistics, energy and circular economy uses are generally more adaptable to staged and partially rehabilitated land
- Residential and community uses require higher levels of land certainty and servicing, making them more dependent on full certification
- Innovation precinct uses require large, coordinated land parcels, which may not align with rehabilitation sequencing

Rehabilitation processes are more conducive to lower-sensitivity uses, while higher-sensitivity or more complex uses are likely to face longer delivery timeframes.

Tenement relinquishment and land transfer

Following rehabilitation certification, the relinquishment of mining lease at the Site would require a clear pathway to avoid uncertainty and fragmentation, which can delay coordinated development and reduce investor confidence. Residual liability considerations may also affect the transfer of land and associated risks to future proponents.

Implications by land use:

- Capital-intensive uses (data centres, advanced manufacturing) require clear land tenure and risk allocation before investment decisions can be made
- Residential and institutional uses are particularly sensitive to legacy risk and require a higher degree of certainty
- Lower commitment uses (e.g. agriculture, some circular economy activities) can proceed with lower levels of tenure complexity

Land use planning and rezoning

The existing planning framework is not currently aligned with the full range of indicative future uses for the Site. Planning framework amendments, including rezoning, are likely to be required to facilitate redevelopment.

Implications by land use:

- Industrial, logistics and energy uses are generally more consistent with the Site's industrial legacy and regional strategic planning objectives, potentially reducing planning complexity relative to other land use outcomes.
- Data centres generate relatively low ongoing employment, which may limit alignment with employment-generating land use objectives.
- Waste and resource recovery uses may be subject to approval scrutiny due to community and environmental considerations, even where planning controls support such uses.
- Residential uses may align with broader regional housing strategies, subject to infrastructure capacity and servicing requirements.
- Research and innovation uses are likely to require strategic anchor institution or industry partnership to justify rezoning and support long-term viability.

The planning system has time, cost and approval risk, particularly for uses that are either non-traditional or require a coordinated precinct-scale approach.

Infrastructure provision and sequencing

Infrastructure is a critical enabler of redevelopment at the Site and a major constraint, despite its proximity to regional transport and energy networks. Key challenges include delivering internal roads and services, upgrading power, water and digital infrastructure, and addressing uncertainty around funding responsibilities and sequencing.

Implications by land use:

- Data centres and advanced manufacturing are highly dependent on reliable, high-capacity power and water infrastructure
- Logistics uses require efficient freight access and road upgrades
- Energy uses may leverage existing connections but still require integration and investment
- Residential and community uses depend on full-service provision and social infrastructure
- Circular economy uses may operate with lower initial infrastructure but still require access and approvals

Infrastructure readiness is likely to be a critical determinant of whether higher-value employment uses can proceed within commercially viable timeframes.

3.1.2 Structural constraint typologies

The structural constraints identified at the Site can be grouped into site-specific, system, legislative and policy, shown in Figure 3-1. Notably, many of the most significant barriers are systemic rather than site-specific, indicating broader structural challenges in the current framework.

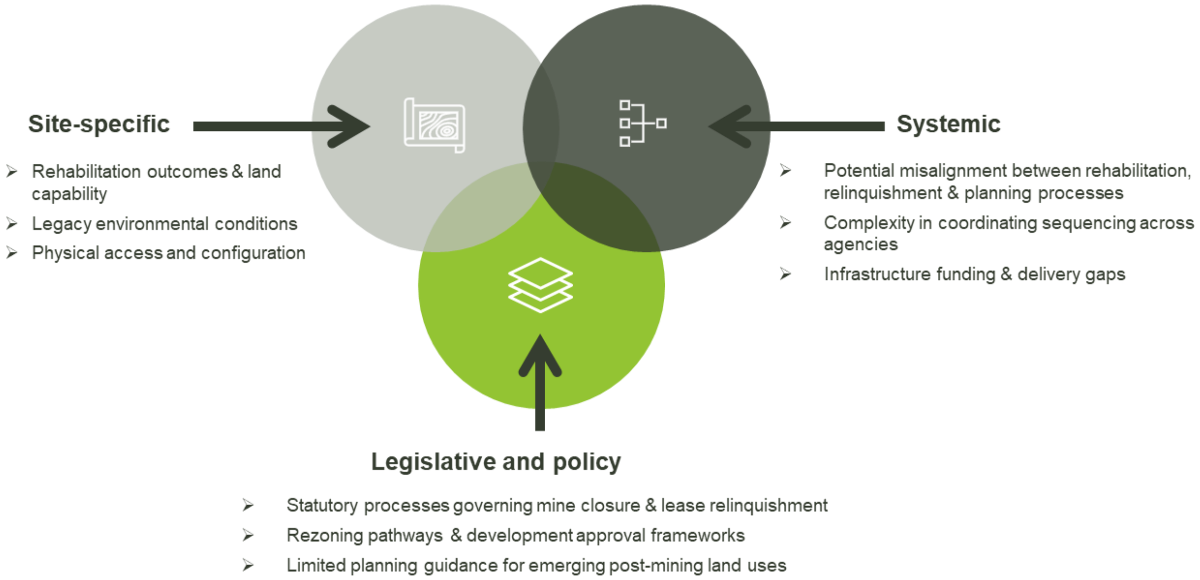

Figure 3-1: The Site structural constraint typologies

The implications of these constraints for the delivery of the Site's indicative land use mix include:

- lower-intensity and transitional uses (e.g. agriculture, recreation, some circular economy activities) are more readily enabled but generate limited economic value
- higher value uses (e.g. advanced manufacturing, data centres, logistics) are viable but dependent on infrastructure and planning certainty
- transformational uses (e.g. innovation precincts) face the greatest barriers due to long lead times and lack of enabling frameworks

This creates a structural risk that current regulatory and delivery settings favour near-term, lower value uses, while delaying or constraining more productive long-term outcomes.

Enablement and reform considerations

Addressing PMLU structural constraints will be critical to unlocking the Site's potential and aligning outcomes with the intent of the Parliamentary Inquiry. Enablement and reform considerations for the Site are outlined in Table 3-1.

Table 3-1: Post-mining site redevelopment enablement and reform considerations

Site-specific enablement	Systemic improvements	Legislative and policy reform
▪ Progress rehabilitation and closure activities in a manner that supports timely post-mining land activation ▪ Consider the future role of retained strategic infrastructure assets, including rail and servicing infrastructure, in supporting industrial, logistics or energy uses ▪ Develop a coordinated infrastructure servicing, funding and sequencing plan ▪ Streamline PMLU compatible with final landform, rehabilitation outcomes and residual risk constraints	▪ Improve coordination between mining, planning, environmental and infrastructure agencies ▪ Improve alignment between mine closure planning, rehabilitation outcomes and future land use objectives ▪ Reduce timing gaps between rehabilitation completion, regulatory certification, tenement relinquishment and redevelopment approvals ▪ Improve alignment between rehabilitation outcomes, infrastructure delivery and rezoning processes ▪ Improve certainty regarding management of residual environmental and geotechnical risks following closure	▪ Streamline rezoning and approval pathways for large-scale post-mining transition sites ▪ Review barriers within the Environmental Planning and Assessment, Protection of the Environment Operations and Mining Acts affecting timely transition to alternative land uses ▪ Establish clearer planning policy pathways for transition from mining land to developable employment land ▪ Support regulatory frameworks that enable productive long-term post-mining land outcomes while maintaining rehabilitation and environmental performance standards

3.2 Transition risks and mitigation strategies, and social considerations

This subsection outlines transition risks including overarching risks and strategies, and those arising from land use dependencies that may influence delivery outcomes.

3.2.1 Land use dependencies and transition risks

This subsection outlines the key dependencies, associated transition risks and social considerations for each land use identified in the assessment, highlighting factors that may influence feasibility, timing, and delivery outcomes.

Circular Economy, Industrial and Advanced Manufacturing

Industrial and advanced manufacturing opportunities offer favourable workforce transition pathways, leveraging the region's strengths in trades, machinery operation, fabrication, and technical supervision. This alignment supports redeployment of mining-dependent workers into adjacent industries with minimal retraining requirements, reducing transition risk relative to more specialised or knowledge-intensive sectors.

However, feasibility across the broader manufacturing mix is subject to several key dependencies:

- **Market scale and demand certainty**: Sustained demand particularly from infrastructure delivery, housing construction, and energy transition is required to support long-term viability and attract operators.
- **Supply chain integration:** Proximity to the Port of Newcastle and REZ provides a strong advantage, but successful outcomes depend on integration with broader manufacturing and export supply chains.

- **Infrastructure readiness:** Reliable power supply, transport connectivity, and industrial servicing will be critical for energy-intensive and logistics-heavy uses.
- **Anchor tenants and clustering effects:** Many advanced manufacturing activities rely on co-location and industrial clustering to achieve efficiency and scale, increasing dependency on early anchor occupiers.
- **Technology maturity and feedstock availability:** Certain sectors, such as critical minerals processing and advanced materials, are more dependent on emerging technologies and market maturity, introducing greater uncertainty.

These factors result in a spectrum of transition risk. Lower-risk opportunities are those with established market demand and proven regional capability, while higher-risk uses are contingent on industry formation, technological advancement, and policy support.

A staged delivery approach is therefore critical. Early phases should prioritise scalable, market-ready industrial uses capable of establishing employment and activity on the Site. Over time, this foundation can support the introduction of more advanced and specialised manufacturing sectors, enabling the gradual development of a diversified industrial ecosystem aligned with long-term economic transformation objectives.

Social considerations and implications

Circular economy, industrial and advanced manufacturing land uses could be located at Cockle Creek and Killingworth (Figure 2-3), adjacent to residential communities including Killingworth, Barnsley and Boolaroo. These land uses align with community aspirations for economic transition, local employment and the retention of established regional industries, while supporting workforce continuity and regional identity.

Key social considerations include:

- **Economic diversification and employment:** Builds on existing regional strengths in steel fabrication, mining equipment and energy infrastructure, supporting local jobs and economic resilience.
- **Housing outcomes:** Activities such as modular construction, prefabrication and construction material manufacturing could directly support housing supply, affordability and housing diversity which is identified as a community priority.[54,55]
- **Community support**: Land uses that contribute to housing outcomes and provide local employment opportunities are likely to receive stronger community support.
- **Social and environmental impacts**: Construction and operational activities may increase traffic, noise, waste generation and land disturbance, affecting local amenity and perceptions of liveability for residents.
- **Managing community acceptance**: Appropriate siting, buffering, operational controls and ongoing community engagement will be important to address both actual and perceived impacts and maintain social licence.[56]
- **Circular economy benefits**: Could deliver broader social value through job creation, resource recovery and reduced waste to landfill.[57]

Waste and Resource Recovery

Waste-related uses are broadly compatible with the existing skills base within the Hunter region, particularly across machinery operation, trades, logistics, and supervisory roles. This alignment reduces transition risk relative to more knowledge-intensive sectors and supports the redeployment of mining-dependent workers into adjacent industries. However, feasibility is contingent on several key dependencies. These include:

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[54] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[55] Lake Macquarie City Council (2026) Draft Housing Strategy.

[56] Berthet, E. et al. (2024) Assessing the social and environmental impacts of critical mineral supply chains for the energy transition in Europe.

[57] NSW Environment Protection Authority (EPA) (2025) Measuring the environmental, economic and social impacts of reuse.

- **Feedstock availability:** Sufficient and consistent waste volumes within the regional catchment are required to sustain operations, particularly for higher-capacity facilities.
- **Infrastructure readiness:** Access to road freight networks, utilities, and (where relevant) grid capacity will influence staging and scalability.
- **Market coordination and anchor tenants:** Larger or integrated precinct models require the presence of initial operators to establish throughput and enable co-located activities. Council and City of Newcastle Circle City Scan initiative could provide assessment of region wide material flow and other datasets to identify circular economy opportunities and initiatives.
- **Regulatory and environmental approvals:** Certain uses, particularly waste-to-energy, face higher approval complexity and community sensitivity.

These dependencies introduce varying degrees of transition risk across the identified opportunities. Lower risk uses are those that can operate as standalone facilities aligned to existing waste streams and logistics networks. In contrast, more integrated or capital-intensive models rely on coordinated delivery, market maturity, and policy support. Energy recovery from residual materials is not a recommended land use in this assessment and would require separate testing if considered in future.

Accordingly, a staged development approach is recommended, prioritising early, market-ready uses that can establish activity on the Site, while preserving flexibility for more complex or integrated opportunities over time. This approach supports both de-risking of initial investment and the gradual development of a broader circular economy ecosystem.

Social considerations and implications

Waste and resource recovery land uses may be suitable within employment-focused areas such as Cockle Creek and Killingworth (Figure 2-3), subject to detailed testing of access, servicing, buffering, amenity, traffic and community acceptance.[58] These uses should be carefully sited to manage interfaces with nearby residential communities, including Killingworth, Barnsley and Boolaroo.

Key social considerations include:

- **Supporting future growth:** Population growth is expected to increase pressure on infrastructure, resources and service systems, reinforcing the need for efficient waste management and resource recovery solutions.[59]
- **Alignment with community priorities:** Waste management and circular economy principles have been identified as important community priorities. Future resource recovery or circular economy uses would need to demonstrate how they support these priorities while managing amenity, traffic, environmental and community acceptance risks.[60]
- **Community perceptions:** Community acceptance varies depending on the type of facility. Recycling expansion and Food Organics and Garden Organics (FOGO) initiatives are viewed positively, with concerns regarding waste importation, waste-to-energy facilities, fairness, safety and trust in decision-making processes.[61,62]
- **Employment and economic benefits**: Waste facilities could improve waste services, enhancing resource recovery outcomes, supporting local employment and generating up to 100 jobs (see Table 2-3).
- **Transport accessibility:** There are varying levels of public transport accessibility, with local communities experiencing limited access to employment opportunities and services (see Section 8.4.1).
- **Construction and operational impacts:** Construction may result in temporary impacts, including increased traffic, noise, dust, reduced accessibility and other amenity effects for neighbouring residential communities. There may also be ongoing operational impacts.[63] These impacts may be acutely experienced by older residents, people

[58] Note: Separated by Cockle Creek.

[59] ABC News (2026) Sydney's landfills are set to reach capacity by 2030 sparking controversial plans to take trash to the regions.

[60] Spectrum Comms (2024) 100 Voices Community Summit - Engagement Summary Report.

[61] Tan, Z. et al. (2024) A Systematic Review on the Social Impact of Construction and Demolition Waste Management in Construction Industry.

[62] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

[63] NSW Department of Planning and Environment (2025) Major Projects Planning Portal document (EXH-74309456).

with long-term health conditions and communities experiencing socio-economic disadvantage. Construction and operational impacts should be investigated further and mitigated during detailed design.

- **Managing community acceptance:** Careful site design, operational controls and ongoing community engagement is necessary to minimise impacts and maintain social licence.

Energy Transition & Renewable Infrastructure

While the Site benefits from large, rehabilitated land parcels and strong high-voltage transmission infrastructure, renewable energy uses such as utility-scale battery storage are subject to intense regional competition, particularly from designated REZs and sites with lower land opportunity costs. In addition, these uses typically deliver lower long-term employment intensity, limiting their contribution to workforce transition objectives.

From a workforce perspective, energy storage and grid support functions offer some alignment with technical and operational skill sets; however, overall employment outcomes are modest compared to industrial or manufacturing uses. As a result, these opportunities are less effective as primary drivers of large-scale workforce redeployment.

Feasibility is further constrained by several key dependencies:

- **Market competitiveness:** Renewable generation and storage projects must compete with alternative sites that offer stronger policy alignment, incentives, or connection advantages.
- **Revenue certainty:** Viability is dependent on long-term energy pricing, market mechanisms, and contracting arrangements for grid services.
- **Grid integration and connection capacity:** While transmission infrastructure is present, proximity to switchyards is critical, and network capacity, queueing, and approval processes remain key constraints.
- **Site suitability and operational requirements:** Battery energy storage facilities require secure, cleared sites with appropriate separation, emergency access and fire safety management. Operational noise from associated equipment should also be considered, particularly where facilities are located near sensitive receivers.
- **Scale and approvals:** Utility-scale battery projects are typically greater than 30 MW, may trigger SSD pathways, and generally require 2-5 years to progress from planning to operation, increasing delivery complexity.
- **Land requirements and staging:** Battery facilities can be developed in modular form, typically commencing at around 2 hectares and expanding up to 15 hectares, enabling staged deployment but requiring sufficient land allocation and long-term planning flexibility.
- **Co-location opportunities:** Hybrid energy and industrial models rely on coordination with energy-intensive users and broader precinct development, introducing additional interdependencies.

These factors result in higher transition and investment risk, particularly for standalone renewable generation. Battery storage and grid-support infrastructure may present more targeted opportunities but are still contingent on broader energy market conditions and system needs.

Thus, renewable energy uses are best considered as enabling elements, rather than primary land use drivers. Their role is likely to support broader industrial and circular economy activities over time, rather than lead site activation. A cautious, opportunistic approach is therefore recommended, with development contingent on market signals, policy settings, and alignment with co-located industrial demand.

Social considerations and implications

Energy transition and renewable infrastructure land uses are located at Cockle Creek and Killingworth (Figure 2-3), adjacent to residential communities including Killingworth, Barnsley and Boolaroo. These land uses support Lake Macquarie's transition from coal-based generation to a more diversified, low-carbon economy, while leveraging the region's established energy infrastructure and workforce.

Key social considerations include:

- **Supporting economic transition:** Aligns with regional decarbonisation objectives and transitions to a more diversified energy economy.[64]
- **Local amenity and environmental impacts:** Construction and operations may result in noise, increased traffic, visual impacts and other disturbances for nearby residential communities. Perceived risks and concerns from other projects focus on hazards, emergency response requirements and environmental management, highlighting the importance of robust planning and mitigation measures.[65,66] Community acceptance is dependent on effective risk and impact management and mitigation.[67]
- **Distribution of benefits and impacts:** Broader benefits such as improved grid stability, energy security and emissions reduction are realised at regional and state levels, local communities may experience a greater share of construction disruption and perceived operational risks.[68]
- **Employment accessibility:** The BESS are expected to produce 20 jobs (Table 2-4). Limited public transport accessibility restricts local employment uptake for residents without private vehicles, particularly those experiencing socio-economic disadvantage, potentially reducing the equitable distribution of economic benefits.

Data Centres and Digital Infrastructure

Data centre and digital infrastructure opportunities are less directly aligned with the existing mining-dependent skills base. Data centres typically generate lower employment intensity and require a more specialised workforce in engineering, and technical operations. As a result, their contribution to large-scale workforce transition is more limited and would require targeted upskilling or workforce mobility to support regional labour participation.

Feasibility is highly dependent on several enabling factors:

- **Power availability and reliability:** The Site benefits from strong high-voltage transmission infrastructure (330kV), providing a key advantage for energy-intensive uses such as data centres and advanced computing facilities.
- **Perceived risks:** The availability of power supports the use of air or fan-based cooling systems, reducing reliance on water-intensive cooling approaches and enhancing operational flexibility.
- **Digital connectivity:** Existing fibre connectivity supports digital operations and positions the Site well for regional or edge data centre development, although further scaling may be required for hyper-scale deployment.
- **Market demand and operator interest:** Attracting major data centre operators depends on demonstrating competitive advantages relative to established Sydney and emerging regional hubs, particularly in relation to connectivity, latency, and cost.

These dependencies introduce a higher degree of delivery and investment risk. Smaller-scale regional or edge computing facilities may be more viable in the medium term, while larger hyper-scale or AI-focused facilities are likely to require longer-term infrastructure upgrades and stronger market signals.

Accordingly, data centre development at the Site should be viewed as an emerging use that may complement broader industrial and energy activities, rather than a primary driver of redevelopment. A staged approach is recommended, allowing infrastructure readiness and market conditions to mature over time.

Social considerations and implications

Data centres and digital infrastructure land uses could be located at Cockle Creek and Killingworth (Figure 2-3), adjacent to residential communities including Killingworth, Barnsley and Boolaroo. These land uses represent a

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[64] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[65] Jacobs (2022) Eraring Battery Energy Storage System- Response to submissions.

[66] Australian and New Zealand Council for fire and emergency services (AFAC) (2025) Large-scale battery energy storage system installations.

[67] SLR Consulting (2024) Pillars for building community awareness and trust for a successful Battery Energy Storage System project.

[68] Australian Renewable Energy Agency (ARENA) (2021) Implementing community-scale batteries.

potential post-mining economic opportunity. However, community acceptance is influenced by the balance between localised impacts and the visibility of community benefits.

Key social considerations include:

- **Economic diversification**: May support economic transition and attract significant investment, contributing to the diversification of the regional economy.
- **Community perceptions**: Community sentiment will be mixed, with concerns focused less on the technology and more on amenity impacts, resource use, land-use compatibility and planning transparency.[69,70]
- **Water demand**: This land use will require substantial and reliable water supplies, creating concerns regarding equitable access to water resources, particularly in the context of increasing drought risk and competing demands from households, industry and the environment.[71]
- **Local amenity impacts**: The 24/7 functions of data centres and continuous operations may generate noise from cooling systems, increased traffic, lighting impacts and concerns associated with backup generators, affecting nearby residential communities.[72,73,74]
- **Employment and economic benefits**: While construction generates short-term employment and investment, ongoing operational employment is comparatively limited (indicated in Table 2-5). However, hyperscale facilities may provide up to 50 jobs per facility including remote and on-site roles (see Table 2-5), supporting broader economic activity.
- **Distribution of benefits and burdens**: Community acceptance will likely be dependent on visible economic benefits, locally realised and perceived outweighing operational impacts.
- **Transport accessibility**: Limited public transport availability may constrain access to employment opportunities for residents without private vehicles, potentially reducing the equitable distribution of economic benefits, particularly for disadvantaged communities.

Freight Logistics, Distribution & Supply Chain Hub

Freight logistics and distribution opportunities at the Site present a moderate but conditional pathway, supported by the Site's access to regional road networks and proximity to key population and industrial centres. These uses align with broader trends toward decentralisation of logistics activity from Sydney and increasing demand driven by population growth, e-commerce, and infrastructure delivery across the Hunter and Central Coast.

From a workforce perspective, logistics and distribution activities offer moderate alignment with the existing skill base, particularly in areas such as machinery operation, transport, warehousing, and supervision. These roles provide accessible transition pathways for a portion of the mining-dependent workforce, although overall employment intensity may vary depending on the level of automation and facility scale.

However, feasibility is influenced by several key dependencies:

- **Market demand and operator interest:** Establishment of large-scale logistics facilities is dependent on securing major tenants, including retailers, distributors, and supply chain operators seeking regional expansion opportunities.
- **Competitive positioning:** The Site must compete with established logistics precincts closer to Newcastle, the Central Coast, and Sydney, which benefit from closer access to major demand centres and established integrated logistics networks.

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[69] Infrastructure NSW (2026) NSW Data Centre Consultation Paper.

[70] NSW Legislative Council Public Accountability and Works Committee (2026) Inquiry into data centres in New South Wales.

[71] Sydney Water (2026) Data centres.

[72] Taylor, J. (2026) Sydney councils fear new datacentres could cause blackouts, block housing and affect locals' health.

[73]NSW Environment Protection Authority (EPA) (2026) Noise.

[74] University of Technology Sydney (2026) Grounding the Cloud.

- **Transport efficiency:** While road access is a strength, logistical viability depends on efficient connections to major freight corridors and end markets. The presence of rail initially suggested a potential logistics opportunity; however, engineering advice indicates that significant investment may be required to make rail-related logistics viable.
- **Site positioning relative to ports:** Opportunities requiring direct port integration are more constrained, given the Site's distance from the Port of Newcastle compared to alternative locations.
- **Operational scale and automation:** Increasing automation within logistics facilities may limit long-term employment outcomes relative to land consumption and infrastructure investment.

These dependencies result in a mixed feasibility profile. Certain uses, such as cold chain and regional logistics, may be viable where aligned to identifiable regional supply chains, while larger distribution hubs or port-linked activities face stronger competition and locational disadvantages.

Hence, freight and logistics uses are best considered as selective or supporting opportunities, rather than primary drivers of Site redevelopment. A targeted approach is recommended, focusing on niche or regionally aligned logistics functions that complement broader industrial and manufacturing uses, rather than attempting to compete directly with established logistics precincts at scale.

Social considerations and implications

Freight logistics, distribution and supply chain hub land uses could be located at Cockle Creek and Killingworth (Figure 2-3), adjacent to residential communities including Killingworth, Barnsley and Boolaroo. These land uses have the potential to deliver significant economic and employment benefits, while also generating local amenity, transport and environmental impacts that will require careful management.

Key social considerations include:

- **Employment and economic growth**: Logistics and distribution facilities could generate substantial employment opportunities, support workforce transition, attract investment and increase demand for local businesses and services. Larger facilities, such as regional distribution and e-commerce hubs, could support up to 400 jobs per facility, making this one of the most significant employment-generating land use options (indicated in Table 2-6).
- **Noise, traffic and transport impacts**: Potential noise associated with 24-hour operations, increased heavy vehicle movements and distribution operations may contribute to traffic congestion, road safety risks and additional pressure on local transport infrastructure and residents.[75][76][77]
- **Workforce accessibility**: Limited public transport accessibility resulting in a high reliance on private vehicles for commuting, increasing demand for car parking, active transport infrastructure and end-of-trip facilities (shown in Section 8.4.1). Accessibility is likely to be slightly improved through active transport routes (Package B).
- **Managing community acceptance**: Detailed assessment of traffic, environmental and amenity impacts, together with appropriate mitigation measures is necessary to minimise impacts maintaining community support.

Community, Tourism and Agriculture

Community, tourism and agricultural uses provide diversified but generally lower-intensity economic outcomes relative to industrial and employment-generating land uses. These uses can nevertheless play an important supporting role in the long-term transition of the Site by contributing to regional liveability, landscape activation, workforce transition and land stewardship outcomes.

Community and recreation-based uses may support local amenity and social infrastructure needs, while tourism and visitor economy activities have the potential to leverage landscape rehabilitation outcomes and regional destination

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[75] Transport for NSW (2025) Delivering freight policy reform in New South Wales.

[76] NSW Department of Planning, Housing and Infrastructure (2026) Augusta Street Warehouse and Distribution Centre – community submissions.

[77] NSW Environment Protection Authority (EPA) (2026) Noise.

appeal. Agricultural uses may also provide productive interim or long-term land outcomes on rehabilitated areas where land capability, access and operational conditions are suitable.

Feasibility across these uses is influenced by several key dependencies:

- **Population growth and visitor demand:** Community and tourism uses are closely linked to regional population trends, visitation patterns and broader economic activity across the Hunter region
- **Landform and rehabilitation outcomes:** Tourism, recreation and agricultural uses are dependent on final landform stability, rehabilitation outcomes and long-term environmental suitability
- **Water access and operational viability:** Agricultural uses benefit from access to non-potable water and large land parcels but remain sensitive to operational inputs, commodity markets and long-term commercial viability
- **Integration with surrounding land uses:** Community, tourism and agricultural uses must maintain compatibility with adjacent industrial, logistics or infrastructure activities across the Site
- **Infrastructure and access requirements:** Certain tourism and recreation uses may require improved site access, servicing, utilities and supporting visitor infrastructure to achieve viable activation outcomes
- **Market positioning and oversupply risk:** Recreation and tourism uses must be appropriately scaled and differentiated to complement, rather than duplicate, existing regional offerings

Thus, these land uses are best considered as part of a broader and staged redevelopment strategy, contributing to landscape activation, regional amenity and land utilisation outcomes while supporting the Site's longer-term economic transition.

Social considerations and implications

Community, tourism and agriculture land uses could be located within the West Lake sub-precinct (Figure 2-3), in proximity to residential communities in Wakefield. These land uses could deliver positive social, economic and wellbeing outcomes by supporting community identity, recreation, tourism and local employment, provided they align with community expectations and future growth needs.

Key social considerations include:

- **Alignment with community priorities:** Community support is likely to be strongest for developments enhancing liveability, protecting local character, providing access to open space services and avoiding unmanaged growth and infrastructure pressures. Open space services are valued by the community for their contribution to wellbeing, social connection, recreation and place identity.[78,79,80]
- **Tourism and visitor economy opportunities:** Tourism development supports economic diversification, employment growth and increased visitation, particularly where environmental, commercial and community outcomes are balanced.
- **Integration with existing communities:** Community acceptance is dependent on new developments complementing existing businesses, maintaining amenity and addressing practical considerations such as parking and local infrastructure capacity.[81]
- **Employment and economic benefits:** Community, tourism and agriculture land uses could generate up to 130 jobs per facility (see Table 2-7), supporting local employment, economic diversification, household income growth and opportunities for skills development.
- **Transport accessibility:** Limited public transport accessibility reduces equitable access to employment, recreation and tourism opportunities, particularly for residents without access to private vehicles. Improved

[78] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[79] Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

[80] Lake Macquarie City Council (2026) Draft Operational Plan and Budget 2026-2027.

[81] Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

connectivity and accessibility are necessary to ensure community, tourism and recreation benefits are available to all sections of the community, including socially disadvantaged groups.

- **Managing community acceptance:** The success of these land uses is dependent on balancing economic opportunities with community expectations for open space, amenity, recreation and access to services while maintaining local character and quality of life.

Residential

Senior and assisted living services may provide stable employment opportunities linked to population ageing and increasing demand for regional community services, while residential development could support longer-term population growth objectives where strategically justified.

However, these uses are also among the most constrained and sensitive due to their reliance on high levels of land supply certainty, infrastructure servicing and environmental assurance.

Feasibility across these uses is influenced by several key dependencies:

- **Rehabilitation completion and land certification:** Residential uses require high levels of certainty regarding land stability, contamination outcomes and long-term environmental performance
- **Infrastructure servicing requirements:** Residential uses depend on comprehensive infrastructure provision, including water, sewer, roads, power, digital connectivity and community services
- **Population growth and housing demand:** Long-term viability is linked to broader regional population and housing market trends, including demonstrated demand for additional residential supply
- **Land use compatibility:** Residential and sensitive community uses may face constraints where located adjacent to industrial, freight or energy-related activities due to amenity, environmental and interface considerations
- **Planning and rezoning complexity:** Edge-of-site locations such as the Teralba Southgate may present a strategic case for residential development but must consider adjoining industrial land uses with 24-hour operations
- **Delivery sequencing and market absorption:** Residential development is unlikely to occur in early transition phases and may require gradual delivery aligned with market absorption and supporting amenity outcomes

These dependencies create a comparatively higher-risk feasibility profile relative to industrial and employment-focused land uses, particularly in the near to medium term. As a result, residential uses are more likely to represent medium-term transition outcomes that complement broader employment and economic activity across the Site, rather than acting as initial catalysts for redevelopment.

Social considerations and implications

Residential land uses could be located within the Teralba Southgate precinct (Figure 2-3), in proximity to the existing residential community of Teralba. These land uses have the potential to deliver significant social benefits by supporting population growth, addressing housing needs and enhancing access to services for existing and future residents.

Key social considerations include:

- **Addressing housing demand:** Residential development will support the growing and ageing population by increasing housing supply and improving access to housing options.
- **Supporting ageing population:** Residential development, including senior living precincts, aligns with community and policy objectives for accessible housing, aged care services and inclusive community spaces that enable residents to remain in their local area as they age. [82][83]

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[82] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[83] Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

- **Community concerns:** Concerns include overdevelopment, loss of green space, impacts on local character and increased pressure on infrastructure and services.[84] However, when delivered alongside appropriate community facilities and services, residential development can contribute to social connectivity.
- **Employment and economic benefits:** Residential development generates construction and operational employment, with the potential to create up to 50 jobs per facility (indicated in Table 2-8) while supporting growth in sectors such as health care and social assistance (indicated in Figure 8-11 and throughout Section 8.6.2).
- **Managing community acceptance**: A balanced approach that integrates housing, social infrastructure and environmental considerations will be important to ensure residential development meets community expectations and delivers positive long-term outcomes.

Education, Research & Innovation

Education, research, and innovation opportunities represent a long-term strategic pathway for the Site, aligned with regional objectives to transition toward knowledge-based industries and higher-value economic activity. These uses have strong conceptual alignment with emerging sectors such as circular economy, clean technology, and advanced manufacturing, but are not currently market-led or investment-ready.

From a workforce perspective, these opportunities support longer-term capability development rather than immediate large-scale employment outcomes. They provide important pathways for reskilling, education, and innovation, particularly in supporting the transition of younger and future workforce cohorts. However, their direct role in absorbing the existing mining-dependent workforce is limited without prior upskilling and institutional support.

Feasibility is highly dependent on several enabling factors:

- **Institutional partnerships and anchor tenants:** Successful delivery requires collaboration with universities, research institutions, and industry partners to establish credibility, funding, and demand.
- **Funding and policy alignment:** These uses typically rely on sustained government investment, research funding, and alignment with state and national innovation strategies.
- **Ecosystem development and clustering:** Innovation uses are most effective when integrated within broader industrial and research ecosystems, rather than operating in isolation.
- **Market maturity and demand:** Many proposed focus areas, particularly in clean technology and advanced manufacturing research, depend on the maturity of emerging industries and downstream commercialisation opportunities.
- **Timing and sequencing:** These uses are more viable once a critical mass of industrial and economic activity has been established on Site.

These dependencies introduce a high degree of delivery uncertainty in the near to medium term, positioning these opportunities as long-term optionality rather than immediate development priorities. Hence, education, research, and innovation uses should be preserved as future opportunities, with land use planning retaining flexibility to accommodate them as the Site evolves. Their role is to support long-term economic transformation, enhance regional capability, and embed innovation within the Site, once foundational industrial and employment-generating uses are established.

Social considerations and implications

Education, research and innovation precinct land uses could be located within the Cockle Creek, Killingworth and potentially Barnsley West sub-precincts (see Figure 2-3), in proximity to the residential communities of Killingworth, Teralba and Boolaroo. These land uses strongly align with regional and community aspirations for economic transition, workforce development, innovation and the growth of diversified, knowledge-based economy.

Key social consideration include:

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[84] Lake Macquarie City Council (2021) Draft Housing Strategy.

- **Supporting economic transition:** Education, research and innovation land use aligns with community and Council objectives to transition towards knowledge-based and service industries while supporting long-term economic resilience.[85]
- **Skills development and workforce capability**: Training and education facilities supports workforce development, upskilling and reskilling opportunities, assisting local workers and income-vulnerable groups to adapt to emerging industries and economic change further supporting the mining transition.
- **Innovation and industry growth**: Research and innovation precincts support emerging sectors such as advanced manufacturing, resource recovery, clean technology and digital industries, fostering new investment and employment opportunities.
- **Alignment with community values:** Environmental sustainability, innovation and future-focused industries are reflected in community priorities, indicating likely support for education and research-focused development.
- **Regional resilience:** By supporting industry transition and workforce pipelines, education-related land uses can enhance economic diversification and reduce reliance on traditional industries.[86]
- **Transport accessibility:** There are varying levels of public transport accessibility, with local communities experiencing limited access to employment opportunities and services.
- **Managing community acceptance:** Community support is likely to be strongest where development delivers visible benefits through local employment, education pathways, skills development and innovation outcomes contributing to long-term prosperity.

Environmentally constrained areas

Environmentally constrained areas are located across the Site (see Figure 2-3), providing environmental protection, ecological restoration and the potential for publicly accessible natural areas that align strongly with community values and sustainability objectives.

Key social consideration include:

- **Environmental protection:** Conserves highly valued areas by the community, including biodiversity, wetlands, riparian corridors and threatened ecological communities while maintaining habitat connectivity across the Site.
- **Alignment with community priorities:** Long-term rehabilitation and stewardship improve environmental condition, supports climate resilience and enhances ecosystem health over time. This responds to strong community support for protecting bushland, biodiversity, open space and Lake Macquarie's natural environment.[87][88]
- **Health and recreation benefits:** Public access to environmentally constrained areas supports recreation, physical activity and improved mental wellbeing through connection with nature.
- **Place identity and liveability:** Enhances environmental amenity, strengthens community connection to place and contributes to the lifestyle and liveability values that define Lake Macquarie.

3.3 Overarching risks and strategies

Former mining and legacy industrial Sites often require staged de-risking before market entry. This includes resolving planning pathways, securing biodiversity approvals, sequencing infrastructure, clarifying rehabilitation obligations, and defining developable parcels. Without this groundwork, market interest is typically speculative or discounted due

[85] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[86] Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

[87] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[88] Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

to risks such as planning uncertainty, environmental approvals, infrastructure cost and timing, unclear yields, subdivision complexity, holding costs, contamination constraints, and market conditions.

Investor confidence strengthens with rezoning certainty, a clear biodiversity pathway, defined development capacity, reduced approvals risk, clear infrastructure and access arrangements, manageable holding costs, and a transparent pathway from planning through to delivery.

This assessment identifies key risks for the Site and presents corresponding mitigation strategies for each in Table 3-2.

Table 3-2: Key transition risks and mitigation strategies for the Site

Key risks	Description	Mitigation
Market (transition uncertainty)	Hunter region transitioning away from coal, with ~11,200 fossil fuel jobs projected to decline by 2035, but replacement demand (energy, manufacturing, logistics) remains uneven with competition from Eraring, Tomago and Williamtown[89]	▪ Stage development and prioritise circular economy, waste, recycling, industrial services uses ▪ Secure anchor tenant commitments before major infrastructure investment; develop flexible infrastructure capable of serving multiple uses ▪ Maintain active market engagement and demand monitoring
Infrastructure / servicing	Existing transmission, rail, and port infrastructure may require significant upgrades to support planned operations, and constrained by water, power, internal access and staging requirements	▪ Conduct detailed infrastructure assessment and upgrade planning and deliver staged servicing strategy aligned to demand ▪ Secure funding commitments for critical upgrades ▪ Coordinate with infrastructure operators (Ausgrid, Transgrid, EnergyCo, Hunter Water, Ports of Newcastle, Transport for NSW(Transport))
Environmental / Site constraints	Former mining land with rehabilitation, contamination and geotechnical uncertainty, constraining land use flexibility and increasing development cost	▪ Conduct early comprehensive environmental assessment ▪ Establish contingency funding ▪ Engage environmental specialists early in planning process ▪ Maintain flexible staging and land allocation
Workforce transition	Mining workers may lack skills or interest in transitioning to renewable energy or manufacturing roles. Skills concentrated in trades, construction and industrial services, with limited alignment to higher-skill manufacturing and emerging sectors.[90]	▪ Develop targeted training programs and transition pathways ▪ Engage workers early in transition planning ▪ Provide income support and relocation assistance where necessary
Housing affordability and displacement pressure	Future employment growth, rezoning and infrastructure investment may increase housing cost pressures over time. Lower-income renters, households in housing stress, older residents on fixed incomes and communities with higher socio-economic sensitivity may have less capacity to absorb these changes. This is a screening-level sensitivity issue, not a confirmed displacement impact.	▪ Align land use planning with housing strategy priorities, including diverse and affordable housing in well-located areas. ▪ Monitor housing stress and rental market change through future stages. ▪ Coordinate workforce growth with transport, housing and social infrastructure planning.
Competitive positioning	Competing Hunter Sites (Eraring, Bayswater/Liddell, Tomago, Williamtown) offer existing	▪ Position around scale, buffers and regional access ▪ Focus on land-intensive uses not suited to constrained precincts

[89] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

[90] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

Key risks	Description	Mitigation
	infrastructure, clustering or grid access, reducing the Site's first-mover advantage.	▪ Avoid direct competition with port/airport-dependent uses
Regulatory / Planning and Approvals	Planning, environmental, or other regulatory approvals may be delayed or impose constraints on development. Rezoning and approvals required to enable large-scale employment uses; coordination required across Council and State agencies.	▪ Engage regulatory agencies early and upfront ▪ Align with regional strategies (Employment Land Use Strategy, Hunter Regional Plan 2041) ▪ Develop comprehensive planning documentation and establish clear approval pathways and timelines
Inequitable access to opportunities due to low levels of public transport accessibility	Limited public transport accessibility particularly in western communities and around the Site, may constrain access to employment, services and community infrastructure. This disproportionately affects lower-income households, older residents and those without private vehicles, limiting who can access future opportunities.	▪ Integrate transport planning with land use planning, including coordination with Transport to improve public and active transport connections ▪ Explore employer-led transport solutions (e.g. shuttle services, staggered shifts) ▪ Prioritise land uses aligned with existing transport corridors or flexible workforce models
Construction and operational impacts on local community and residents	Development of large-scale employment uses may generate localised impacts including increased traffic, noise, dust and reduced amenity during construction and operation. Vulnerable populations may be more sensitive to disruption and environmental change. These may be magnified by cumulative impacts	▪ Further investigation into construction and operational impacts is required ▪ Implement Construction Environmental Management Plans and operational mitigation measures ▪ Stage development to manage cumulative impacts and align with infrastructure delivery ▪ Maintain proactive community communication and complaints management processes ▪ Staging of construction
Construction workforce availability	The region is experiencing high levels of concurrent infrastructure development, creating strong competition for skilled labour across construction, energy and industrial projects. This may constrain labour availability, increase costs and impact delivery timelines for development at the Site.	▪ Stage development to align with workforce availability and broader market conditions ▪ Engage early with contractors and supply chains to secure workforce commitments ▪ Consider workforce attraction strategies, including local hiring targets and partnerships with training providers ▪ Align with regional workforce transition programs to leverage workers exiting declining industries
Community perceptions	Community support may be influenced by perceptions of environmental and amenity impacts, fairness of benefit distribution and transparency of decision-making. Concern may increase where cumulative impacts are unclear or approvals are perceived as fast-tracked.	▪ Develop a clear, consistent communication and engagement strategy aligned with community priorities (e.g. transport, housing, sustainable growth) ▪ Provide transparent information on impacts, benefits and infrastructure planning ▪ Engage early and continuously with stakeholders, including First Nations communities
Impacts to social infrastructure availability	Increased employment and population activity may place additional demand on health, community and recreational infrastructure, which is unevenly distributed and often difficult to access without private transport.	▪ Further investigation into the impacts upon housing availability and affordability is required once more detail becomes available ▪ Plan for social infrastructure provision alongside land use development ▪ Coordinate with Council to align growth with infrastructure delivery strategies

Key risks	Description	Mitigation
Inequitable social benefits / impacts	There is a risk that economic benefits are not evenly distributed, particularly where access to employment, housing or infrastructure is constrained. Some communities may experience localised impacts without receiving proportional benefits.	▪ Embed equity principles in planning, including local employment pathways and inclusive access ▪ Support workforce transition aligned to local skills and industries ▪ Monitor and report on distribution of social and economic outcomes across different community groups

4 Social and community implications for delivery

4.1 Other cumulative social impacts

Within the previous Section, potential social impacts associated with each land use option have been identified. Beyond individual land use impacts, the assessment also considers the cumulative social effects arising from concurrent development occurring across the region, including at the Site. These effects are speculative at this stage and should be further investigated during the delivery phase.

Table 4-1: Cumulative social impacts

Theme	Description	Cumulative impacts
Housing	Housing shows moderate baseline stability, with relatively high home ownership, but rising mortgage and rental costs are increasing financial pressure, particularly for lower-income and ageing households. Affordability remains constrained, with much of Lake Macquarie considered unaffordable for vulnerable groups. SEIFA indicators should be used to identify more sensitive communities.	Impacts are most relevant in areas subject to residential or mixed-use change, such as Teralba and West Lake. Population growth and workforce attraction may increase housing demand, financial stress and insecurity for vulnerable groups. This is a sensitivity issue and requires further assessment once development details are confirmed.
Social infrastructure capacity and access	The 5 km catchment includes a moderate but uneven mix of social infrastructure, clustered in centres such as Teralba, Booragul, Glendale and Cardiff. Areas closer to the Site have limited access and rely on surrounding centres for higher-order services.	Additional workers (approximately 1,130) and residents will increase demand for services, potentially increasing travel to access facilities. While growth is moderate, cumulative impacts may affect accessibility and convenience over time if service provision does not keep pace.
Positive indirect business impacts	Increased workforce presence will drive demand for local goods and services, supporting retail, hospitality and service businesses that currently rely on local population demand.	This will support business growth, local employment and increased economic activity. From a community perspective, this can enhance vibrancy and access to services and local employment outcomes.
Cumulative construction and operational impacts	The concurrent redevelopment of the Site will increase construction activity. The new multiple land uses may generate ongoing operational activity, including workforce movements, servicing requirements and changes to how the area is used and experienced by the community.	Overlapping construction phases may increase traffic, noise and disruption, contributing to construction fatigue. Ongoing activity may result in sustained changes to local amenity. While longer-term benefits include employment and economic diversification.
Workforce availability	The Hunter region is experiencing strong demand for skilled labour, particularly in construction and infrastructure. This reflects a constrained labour market with increasing competition across concurrent projects.	High levels of concurrent development across the region are likely to intensify competition for labour, which may limit local access to employment opportunities if roles are filled by non-local workers. Workforce shortages may also delay delivery, extending periods of disruption for nearby communities and reducing the timely realisation of social and economic benefits.
Workforce transition and reskilling	The transition away from mining is likely to affect workers with industry-specific skills, creating a need for reskilling pathways.	Concurrent development across the region are likely to intensify competition for labour, which may limit local access to employment opportunities if roles are filled by non-local workers. Workforce shortages may also delay delivery, extending disruption for

Theme	Description	Cumulative impacts
		nearby communities and reducing the timely realisation of social and economic benefits.

4.2 Summary of social implications and considerations

The following table summarises the key social implications of each proposed land use, highlighting how they interact with existing community conditions, priorities and cumulative regional change to influence housing, employment, accessibility, amenity and wellbeing.

Table 4-2: Social considerations and implications

Section	Social considerations and implications
Circular Economy, Industrial and Advanced Manufacturing	▪ Stronger support for uses linked to housing outcomes (e.g. modular construction) and aligns with economic transition and existing regional strengths and demand for diverse accessible housing, particularly for the ageing population ▪ Generates localised amenity impacts (traffic, noise, industrial character) near residential areas ▪ Lower job density (Up to 40 jobs per facility) raises risk of uneven distribution of benefits along with low transport access limiting equitable employment uptake
Waste and Resource Recovery	▪ Nearby residential areas may experience construction and operational impacts (traffic, noise, amenity) ▪ Provides local social value, including improved services, resource recovery and modest job creation (Up to 100 jobs per facility). Low transport accessibility limits equitable access to employment opportunities
Energy Transition and Renewable Infrastructure	▪ Strong alignment with low-carbon transition and economic diversification ▪ Community acceptance is conditional, with concerns around safety, noise and risk with potential imbalance of local impacts compared to regional benefits ▪ Low ongoing employment (Up to 20 jobs per facility) and transport constraints limit local benefits
Data Centres and Digital Infrastructure	▪ Community perceptions are cautious and mixed, driven by amenity, fairness and transparency concerns including competition for resources, particularly water. ▪ 24/7 operations create persistent impacts on amenity, sleep and wellbeing for local and nearby residents ▪ Limited ongoing jobs (Up to 50 jobs per facility) and provides some employment and skills opportunities with low transport accessibility constraining equitable participation
Freight Logistics, Distribution and Supply Chain Hub	▪ Delivers significant employment and economic benefits (up to 400 jobs) with low to moderate transport access increasing reliance on private vehicles ▪ 24/7 operations create ongoing noise, increased traffic and congestion along with amenity impacts affecting wellbeing for local and nearby residents ▪ Potential increased demand on housing and social infrastructure
Community, Tourism and Agriculture	▪ Represents strongest positive social outcomes, aligned with liveability and wellbeing with strong support and high demand for green, active and recreational spaces increasing place-based identity ▪ Tourism and visitor economy offer diversification benefits but require balance with local needs with integrated planning needed to manage amenity, congestion and access

Section	Social considerations and implications
	▪ Delivers strong economic employment opportunities (Up to 130 jobs per facility) however with risks of inequitable access due to transport limitations
Residential	▪ Demand for diverse, affordable and accessible housing remains relevant to the broader Site context, particularly in relation to Teralba Southgate and nearby established communities. ▪ Residential uses may support workforce attraction, ageing population needs and access to services where located close to existing centres, transport and social infrastructure. ▪ Key considerations include interface with operational and employment uses, amenity, infrastructure capacity, housing affordability and community concerns about overdevelopment or loss of open space. ▪ Residential outcomes should be treated as complementary and site-specific, rather than a primary driver of the employment-led Precinct.
Education, Research and Innovation Precinct	▪ Strong alignment with economic transition, skills and innovation priorities and supports shift to a knowledge-based economy and future workforce needs ▪ Enables workforce development and industry transition pathways ▪ Could support emerging industries such as advanced manufacturing and circular economy, contributing to long-term economic resilience and diversification
Cumulative impacts	▪ Housing demand and affordability pressures ▪ Increased need for social infrastructure (health, services, community facilities) ▪ Cumulative and ongoing construction and operational impacts, including traffic, noise and amenity effects. ▪ Parking and public transport pressures ▪ Flow-on effects to local businesses (cafes, services)

4.3 Enabling options to create social benefits

The Site presents a significant opportunity to generate positive social outcomes through strategic planning, targeted capital investment, and alignment with community needs and priorities. By embedding social considerations into design, delivery, and land use decisions, the redevelopment can support inclusive economic growth, enhance liveability, and strengthen long-term community outcomes. This Section outlines key enabling strategies and the associated benefits that can be realised across the project lifecycle.

Table 4-3: Social benefits and opportunities

Benefit or opportunity	Description	Action
Aboriginal participation and partnerships	Supports meaningful Aboriginal economic participation, cultural recognition and self-determination through employment, procurement, and co-designed land use outcomes.	▪ Develop Aboriginal employment and procurement targets and plans across construction and operations. Options may also include Aboriginal-identified internship and training. ▪ Partner with local Aboriginal organisations and Biraban Local Aboriginal Land Council (e.g. Awabakal) to co-design land uses and cultural outcomes ▪ Integrate Aboriginal cultural values mapping and land stewardship into Site design

Benefit or opportunity	Description	Action
		▪ Support Aboriginal-owned business participation in supply chains ▪ Establish long-term governance or partnership models (e.g. joint ventures, land access agreements)
Local employment and workforce transition pathways	Enhances equitable access to employment by creating local job opportunities and supporting workforce transition.	▪ Set local employment targets for residents for construction. ▪ Establish reskilling and transition programs for former mining and industrial workers, and training for underrepresented groups. ▪ Create entry-level and pathway roles for disadvantaged groups.
Social procurement and local industry development	Strengthens the local economy by prioritising local suppliers and businesses, while embedding social value outcomes through procurement practices.	▪ Implement a Local Industry Participation Plan (LIPP) ▪ Prioritise local suppliers and subject matter experts in contracts ▪ Include social procurement requirements (e.g. employing priority job seekers) ▪ Support capacity building for local businesses to access contracts
Improved transport	Improves accessibility to employment and services, enabling broader community participation, particularly for disadvantaged and low-access groups.	▪ Improve public transport connections to employment areas ▪ Provide active transport links (walking, cycling) connecting nearby communities ▪ Appropriate transport or travel facilities including additional carparking areas or potential inclusion of bike paths for workers to access the Site ▪ Ensure equitable Site access, particularly for lower socio-economic groups
Community benefit sharing mechanisms	Ensures that project benefits are visible and locally realised through investment in community infrastructure, shared spaces, and ongoing engagement.	▪ Identify opportunities for community benefit funds or investment programs. ▪ Explore community-facing improvements such as walking tracks, recreation links, community facilities or cultural interpretation, subject to ownership, access, funding and delivery arrangements. ▪ Consider opportunities for managed community access to selected Site assets where safe, feasible and compatible with future operations. ▪ Implement long-term engagement strategies, not just one-off consultation. ▪ Use co-design approaches for key community-facing outcomes where appropriate.
Education and skills development	Supports long-term workforce participation and economic resilience by building local skills and aligning education pathways with emerging industries.	▪ Partner with education providers to deliver on-site or nearby training such as University of Newcastle and EmpowerHER. ▪ Support apprenticeships, traineeships and internships, including Aboriginal-identified internship and training ▪ Embed training facilities or partnerships with TAFE/universities within the Site

4.4 Social considerations in economic terms

The social and economic findings are not separate tests. Social conditions affect whether economic benefits can be delivered, accessed and locally captured. Table 4-4 translates the key social findings into their economic and delivery implications.

Table 4-4: Social findings in economic terms

Social finding	Economic translation	Conclusion
Workforce skills are concentrated in trades, machinery operation and industrial services.	Labour supply risk varies by use: lower for circular economy, industrial and logistics uses; higher for digital, advanced and innovation uses.	Workforce alignment supports prioritising circular economy and industrial uses. Training partnerships are an economic enabler for later phases.
Low or no public transport accessibility across most development areas.	Smaller effective local labour catchment; reduced local capture of employment benefits; higher occupier attraction and retention risks.	Transport connectivity is an economic dependency to be sequenced with land release and employment growth.
Ageing population and accessible service needs.	Ageing increases the importance of transport access, service proximity and age-friendly housing in surrounding communities.	Employment and transition use should be planned with access, amenity and service connections in mind, not only job creation.
Housing affordability pressure across the LGA.	Raises workforce attraction costs and may constrain large employment-generating uses if not managed.	Higher-employment land uses should be coordinated with housing supply, transport and social infrastructure planning.
Use-specific social licence risks and uneven distribution of benefits and burdens.	Approval timing risk, holding costs, exhibition risk and reduced investor confidence.	Siting, buffering, local employment pathways and benefit-sharing should be treated as de-risking measures.
Environmental protection and biodiversity are strong community values.	Environmental protection can also support amenity, recreation, cultural value and social licence.	Environmentally constrained areas should be treated as part of the Site's social value, not only as constraints.

5 Key findings and recommendations

5.1 Key findings to support a flexible employment precinct

The Site is best positioned for flexible land use framework capable of accommodating a range of employment outcomes over different time horizons, anchored by foundation employment uses and supported by complementary community and liveability outcomes.

The key findings are:

- **The Site is a strategic opportunity for regional renewal**, aligned with the Hunter Regional Plan 2041, Greater Newcastle and NSW Government transition priorities. Its scale, separation buffers, industrial legacy and access to regional infrastructure favour land uses that need large, flexible, buffered sites.
- **It is not immediately market-ready for all uses**. Delivery depends on rehabilitation, infrastructure, servicing, planning pathways, land release, market testing and occupier demand.
- **Circular economy, industrial and advanced manufacturing, waste and resource recovery and energy transition infrastructure are the strongest near-term uses**, aligning with the Site's industrial context, regional transition objectives and transferable workforce skills.
- **Freight and logistics and data centres are conditional,** depending on power, water, fibre, grid capacity, access, market demand and timing. Smaller logistics and construction support uses may come earlier than a large-scale supply chain hub.
- **The challenge is not simply to create jobs, but to create jobs that are accessible, aligned with local and regional skills, and capable of supporting long-term diversification**. Up to 1,130 ongoing jobs are estimated across the identified uses; noting that construction jobs are excluded, and facility-level benchmarks should not be aggregated beyond this without confirmed land use mix, staging and tenant assumptions.
- **Transport access is the key cross-cutting issue**. Low public transport accessibility limits equitable access to future jobs and services, particularly lower-income households, older residents, people with disabilities and those without private vehicles, and intersects with existing socio-economic disadvantage.
- **The highest contribution land uses can carry the highest local impacts.** Freight offers the most jobs but the greatest traffic, amenity, housing and social infrastructure pressures; data centres raise resource-sharing perceptions, particularly water, with limited local employment relative to scale.
- **Complementary uses balance the Site**. Community, residential, and recreation outcomes deliver stronger direct social value and help offset the impacts of employment-led development; environmentally constrained areas also support health and wellbeing.
- **Social and economic findings are not separate tests**. Workforce skills, transport, housing and social licence translate directly into labour supply, local benefit capture, approval timing and investor confidence.

5.1.1 Integrated land use and employment opportunities

Table 5-1 summarises the land use directions from Section 2 Feasibility Assessment, grouped by strategic opportunity and horizon, with indicative employment and the key social considerations for each. The social considerations are descriptive, they identify what needs to be managed, tested or enabled, not a finding of social feasibility, which would require further social impact assessment (SIA) and engagement.

Table 5-1: Integrated land use key findings

Land use	Classification	Indicative ongoing employment	Social considerations
Near term: 0 to 5 years			
Circular economy, Industrial and advanced manufacturing	Highly feasible	Up to 40 jobs per facility	Strong alignment with community transition expectations, regional identity and transferable workforce skills. Local employment pathways and management of industrial and residential interfaces are key considerations.
Waste and resource recovery	Highly feasible	Up to 100 jobs per facility	Strong alignment with sustainability and resource recovery priorities. Key considerations include amenity impacts, waste importation perceptions, transport access and local benefit distribution, particularly for higher sensitivity uses such as energy recovery from waste
Energy transition and renewable infrastructure	Highly feasible	Up to 20 jobs per facility	Acceptance is likely to depend on impact mitigation, risk communication, safety management and clear explanation of local and regional benefits.
Near Medium: 5 to 10 years			
Community, tourism and agriculture	Conditional	Up to 130 jobs per facility, most relevant to intensive horticulture or controlled environment agriculture rather than general community or recreation uses	Strong alignment with liveability, open space and wellbeing. Equitable access, long-term management and compatibility with employment uses are key considerations.
Residential	Complementary	Up to 50 jobs per facility	Suited to limited, well-integrated edge-of-site development. Integration with social infrastructure, access, servicing and surrounding land use compatibility are key considerations.
Medium term: 10 to 15 years			
Data centres and digital infrastructure	Conditional	Up to 50 jobs per facility	Community perceptions are likely to be cautious and mixed. Amenity, resource-sharing, particularly water, limited local employment intensity and visible local benefit are key considerations.
Environmentally constrained areas	Supporting use: all horizons	Not assessed	Supports community identity, amenity, recreation, wellbeing, cultural values and social licence. Should be integrated with access, recreation, cultural interpretation and long-term land management.
Long term: 15+ years			
Freight logistics, distribution and supply chain hub	Conditional	Up to 400 jobs per facility	The scale of workforce and freight movements means transport, amenity, housing, public transport and social infrastructure implications require further assessment before this use can be confirmed.
Education, research and innovation	Optional	Not assessed	Strong alignment with skills and transition objectives and likely positive community perception. Access, delivery model, funding and institutional partnerships are key considerations.

5.2 Key recommendations

The preliminary assessment indicates the Site is best positioned as a flexible employment precinct. This is consistent with the broader North West Lake Macquarie growth area, which identifies regional capacity for up to 18,300 additional residents, 9,950 dwellings, 4,300 jobs and $7.4 billion in annual economic output. Development at the Site could support and trigger other major assessment, subject to land capability, infrastructure delivery and market take-up.[91]

Key recommendations are:

- **Prioritise early-phase employment uses:** Begin with circular economy, industrial, manufacturing and renewable energy uses, as these offer stronger near-term employment potential and clearer market alignment.
- **Use phased development to manage uncertainty:** Consider logistics, data centres and digital infrastructure, tourism and residential as longer-term opportunities, subject to market testing and Site constraints.
- **Support manufacturing relocation and expansion:** Develop targeted incentives to attract advanced manufacturing and renewable energy equipment production.
- **Establish a regional logistics hub:** Explore opportunities to consolidate freight, warehousing and distribution activity to improve regional supply chain efficiency.
- **Address workforce and delivery constraints:** Partner with TAFE NSW and universities to support workforce transition, while recognising constraints such as market uncertainty, infrastructure servicing and delivery sequencing
- **Embed social outcomes as delivery measures and economic enablers**: Improve transport connectivity, local employment pathways, housing alignment, biodiversity outcomes, social infrastructure and benefit-sharing into precinct planning and controls.

With strategic planning, coordinated governance, and targeted investment, the Site could transition from coal mining to a diversified, resilient regional mixed-use economic precinct. The Site's ultimate employment contribution will depend on the further economic modelling, final land use mix, occupier take-up, infrastructure delivery and staging. Table 5-2 sets out an indicative sequencing logic based on current feasibility findings, not a confirmed delivery plan. This will be tested and refined through the future business case.[92]

Table 5-2: Indicative timeframe development scenario and recommended strategy

Phase	Indicative timeframe (Years)	Recommended focus	Priority land uses	Rationale / link to assessment
Phase 1: Planning, de-risking and early activation	1-5	Site capability and planning pathway and early market interest.	Early circular economy and industrial services, materials handling, contractor yards, energy support uses, advanced manufacturing, temporary activation / open space interfaces.	Prioritise uses that require lower upfront investment and can generate early activity while environmental, servicing and approvals risks are resolved.
Phase 2: Infrastructure and anchor tenant activation	5-10	Deliver staged infrastructure and secure anchor tenants.	Residential and tourism uses	Supports transition from mining while avoiding over-reliance on speculative long-term uses.
Phase 3: Development and scaling	10-15	Scale into higher-value employment uses as infrastructure and market demand mature.	Circular economy precinct, data centres, hyper-scale supply chain hub.	Preserves land for medium-to-long term strategic uses aligned with Hunter transition trends.

[91] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

[92] Note: See Section 2 Preliminary land use feasibility assessment, including the benchmark employment methodology and land use classifications.

Phase	Indicative timeframe (Years)	Recommended focus	Priority land uses	Rationale / link to assessment
Phase 4: Long-term optionality	15+	Preserve and test longer-term opportunities as land, infrastructure and market conditions mature	Larger logistics / supply chain functions, research and innovation, education partnerships, tailings storage facility opportunities subject to declassification and technical assessment	Maintains flexibility for uses that may become more viable over time

5.3 Future business case development considerations

A future preliminary business case should focus on enabling conditions, not only preferred land uses, testing what is required to make each land use deliverable, investable and socially acceptable. This includes:

- Commercial feasibility and market demand testing with potential investors through market sounding
- Infrastructure, servicing and access requirements and sequencing; land release based on land use requirements
- Rehabilitation and certification; planning and approval pathways; governance, delivery responsibilities and risk allocation
- Social enablers identified, including workforce transition, transport access and local job capture, housing, social infrastructure and community benefit, confirmed through further SIA and engagement.

Table 5-3 identifies potential opportunities for further business case testing. These should be treated as candidate opportunities for further investigation, not as confirmed projects or investment commitments.

Table 5-3: Potential land use opportunities for future business case consideration

Opportunity theme	Rationale for business case consideration
Circular Economy, Industrial and Advanced Manufacturing	
Modular construction and prefabrication	Supported by large industrial land parcels and proximity to freight and regional growth areas
Structural steel and metal fabrication	Compatible with industrial land capability and existing heavy industry context
Precision engineering and advanced manufacturing	Potential to support regional industry diversification and higher-value employment outcomes
Critical minerals and battery material recovery	Supported by increasing demand for battery and critical mineral processing capability
Advanced materials and composites	Potential long-term opportunity linked to advanced manufacturing and innovation industries
Waste and Resource Recovery	
Construction and demolition waste recovery hub	Aligns with increasing regional infrastructure activity and potential demand for resource recovery capacity
Mixed waste transfer and sorting facility	Compatible with industrial land outcomes and may leverage large land availability and separation distances
Battery and e-waste recycling centre	Aligns with energy transition industries and emerging circular economy supply chains
Integrated circular economy precinct	Potential to co-locate complementary recycling, recovery and industrial activities within a shared servicing environment

Opportunity theme	Rationale for business case consideration
Energy Transition and Renewable Infrastructure	
BESS	Strong alignment with energy transition objectives and existing electricity network proximity
Data Centres and Digital Infrastructure	
Hyper-scale data centre (long-term)	Potentially supported by regional land availability and access to strategic energy infrastructure, subject to significant power and digital capacity upgrades
Regional data centre / edge computing facility	Smaller-scale opportunity with comparatively lower infrastructure requirements and more flexible delivery potential
Digital services / technology support hub	Potential supporting employment use aligned with digital infrastructure development
Freight Logistics, Distribution and Supply Chain	
Cold chain and perishables logistics	Potential to leverage freight connectivity and support regional agricultural and distribution supply chains
Community, Tourism and Agriculture	
Controlled environment agriculture	Potential productive reuse opportunity where land, water access and operational conditions are suitable
Residential	
Residential and senior living	Linked to demographic change and ageing population trends within the broader region

Part B: Detailed methodology

6 Methodology and assessment framework

This Section outlines the methodology and assessment framework used to evaluate the study, including the definition of social and economic study areas and the rationale for applying both to capture distinct impact dimensions. It describes the analytical approach, data sources, and key assumptions underpinning the assessment, and highlights relevant limitations relating to scope, data, and methodology. The Section also clarifies how the findings should be interpreted and applied within the broader context of the study.

6.1 Social and economic study areas

The assessment uses two study areas to reflect the relevant geographic scales:

- **Social Study Area (SSA)** representing communities most likely to experience direct and localised impacts
- **Economic Study Area (ESA)** representing the wider regional economy, labour market and industry context.

The study areas align, where practical, with Australian Bureau of Statistics (ABS) geographies to support consistent analysis using ABS Census 2021 data, REMPLAN data, employment projections and other available datasets.

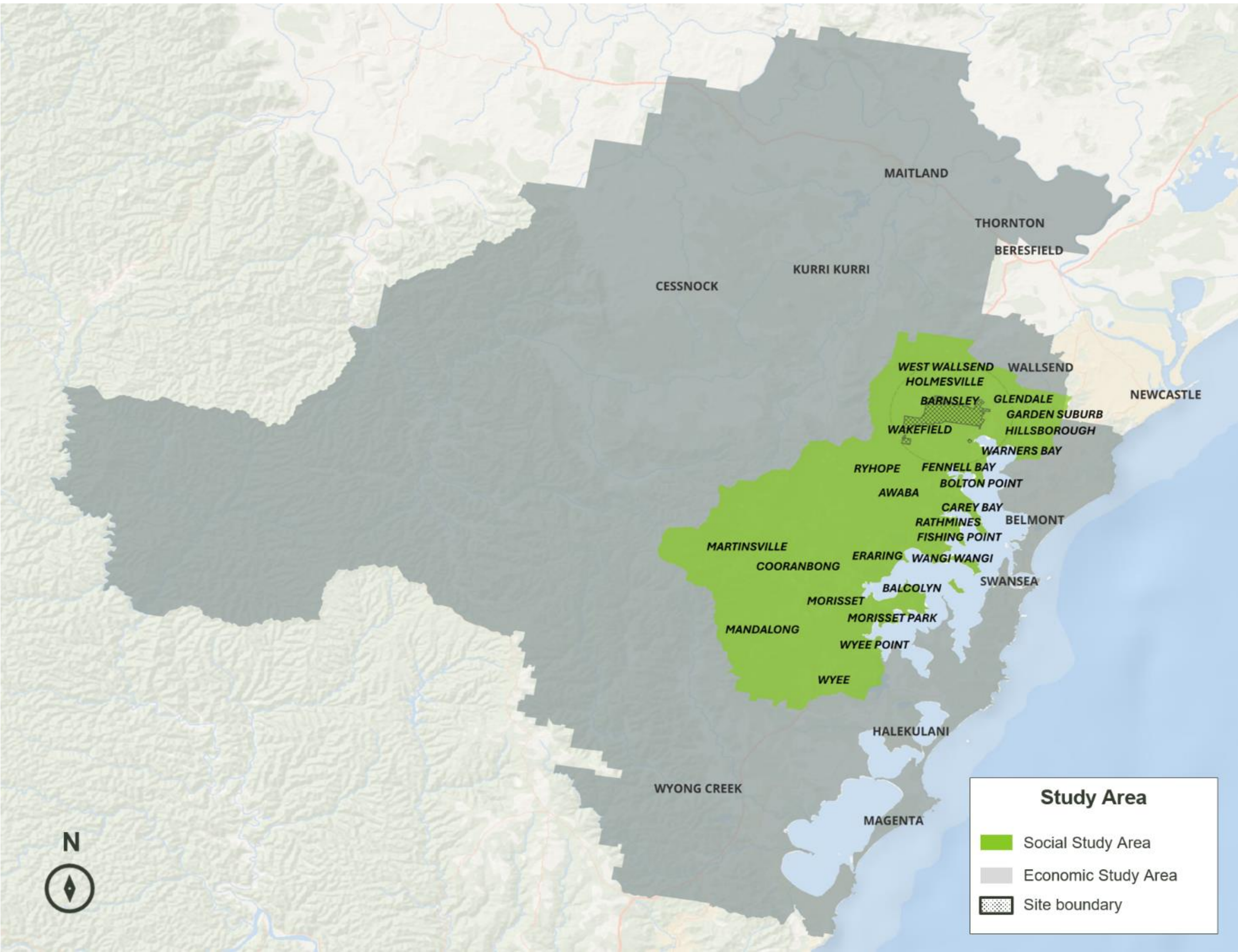

Figure 6-1: Site social and economic study area

Using both study areas allows the assessment to consider local community implications while also recognising the broader regional economic role of the Site. The wider ESA captures the regions of Wallsend – Elermore Vale (SA2) and Maryland - Fletcher – Minmi (SA2) which captures part of the Newcastle LGA. A comparison between both areas is outlined in Table 6-1.

Table 6-1: Social versus economic study area

Study Area	Purpose	Areas included	Key Considerations	
Social	Captures the communities most likely to experience localised social impacts from changes at the Site, including effects on access, amenity, services, infrastructure and perceived change.	▪ Lake Macquarie West (SA3) ▪ Glendale Cardiff – Hillsborough (SA2) ▪ Warners Bay – Boolaroo (SA2)	▪ Local population characteristics ▪ Socio-economic vulnerability ▪ Community values and priorities ▪ Aboriginal and Torres Strait Islander population characteristics	▪ Housing and affordability ▪ Access, transport and mobility ▪ Social infrastructure, services and community facilities ▪ Groups more sensitive to land use change
Economic	Captures the wider functional economy, labour market and industry context in which the Site operates, including broader employment, supply chain, investment and transition impacts.	▪ Lake Macquarie (LGA) ▪ Cessnock LGA ▪ Maitland (LGA) ▪ Wyong (SA3) ▪ Wallsend – Elermore Vale (SA2) ▪ Maryland - Fletcher – Minmi (SA2)	▪ Regional employment and industry structure ▪ Labour market depth and workforce mobility ▪ Skills and occupational characteristics ▪ Economic output and value-added activity	▪ Coal related workforce exposure ▪ Industry transition pressures ▪ Potential sources of workforce, supply chain and investment demand

6.2 Analytical approach

The assessment adopts a staged analytical approach to support early strategic planning and pre-investment decision making as seen in Figure 6-2.

It first establishes the policy, market and technical evidence base, including regional transition priorities, planning directions, mine rehabilitation considerations and comparable investment activity. It then defines the social and economic baseline for the SSA and ESA, focusing on the population, workforce, industry and service conditions most relevant to post-mining land use planning. Potential land use directions emerging from the Enquiry by Design process are then tested against strategic alignment, market plausibility, infrastructure readiness, land suitability, economic contribution, social implications and delivery risk. This provides the basis for identifying key dependencies, evidence gaps and further work required to inform future master planning, business case development and implementation.

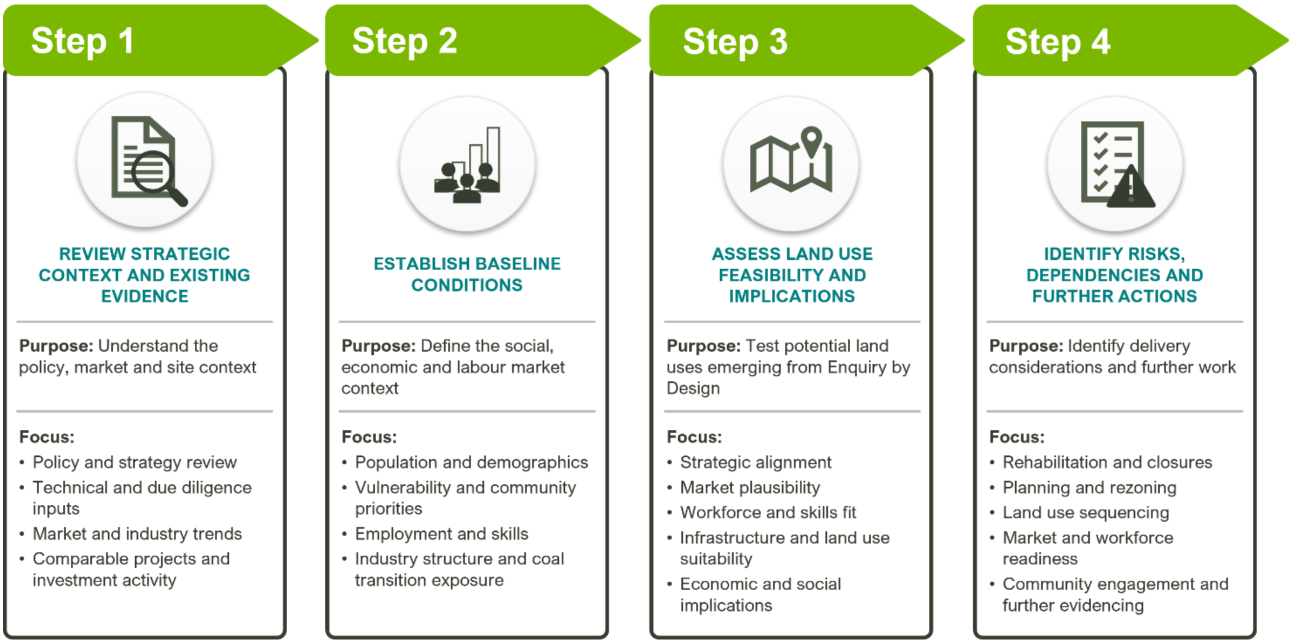

Figure 6-2: Analytical approach for the economic, social and market assessment

6.3 Assumptions, exclusions, and dependencies

This assessment has been prepared using the best available information at the time of writing. It is subject to assumptions, exclusions and limitations that should be considered when interpreting the findings.

6.3.1 General assumptions

The assessment assumes that:

- Council provided information and existing technical studies are suitable for use at this stage of assessment.
- Secondary and desktop data sources provide a sufficient basis for an early strategic assessment, without primary surveys, focus groups or direct community engagement
- ABS Census 2021 data provide a reasonable baseline for demographic, household, workforce and travel behaviour analysis, noting known limitations associated with the COVID period, particularly for journey to work, working from home and population mobility
- REMPLAN and other regional economic datasets provide a reasonable basis for understanding economic structure, industry output and value-added activity
- The emerging land use directions from the Enquiry by Design (EbD) process provide a suitable basis for early feasibility testing
- Future market conditions, investment appetite, infrastructure funding, policy settings and planning pathways may change over time.

6.3.2 Exclusions and dependencies

Table 6-2 summarises the key exclusions and dependencies relevant to the interpretation of this assessment. These include matters outside the scope of Package E, data and evidence constraints, and technical, infrastructure and planning matters that require confirmation at later stages.

Table 6-2: Exclusions and dependencies of assessment

Category	Exclusions and dependencies of the assessment
Scope	▪ Excludes cost-benefit analysis, statutory planning advice, formal market testing, procurement and financial appraisal ▪ Does not provide a detailed socio-economic impact assessment for each land use ▪ Does not determine whether a specific project should proceed to investment ▪ Intended to support strategic land use planning by identifying feasible directions, benefits, risks, dependencies and further work.
Data	▪ Some datasets are only available at LGA, SA3 or other broader geographies and have been used as proxies where required, which limits local precision ▪ The absence of primary surveys, market sounding, tenant negotiations and investor engagement limits certainty around demand, timing and delivery models ▪ Economic contribution estimates are indicative and based on benchmarks and assumptions, not confirmed project design, occupier profiles, capital expenditure or investment commitments
Social Assessment	▪ Provides a high-level strategic social scoping assessment only, identifying baseline social conditions, community values, opportunities, constraints and potential implications ▪ Does not constitute a SIA and does not provide an in-depth social infrastructure or housing analysis ▪ Employment and community outcomes require further testing, particularly given proximity to residential areas ▪ Further SIA and community engagement will be required once options, delivery pathways, built form, timing and operations are better defined ▪ Low historical engagement should not be interpreted as low community interest or low social impact.
Economic Assessment	▪ Uses a high-level benchmark approach suited to master planning ▪ Does not use defined project costs, as detailed costs are not available and cost-based appraisal is outside scope ▪ Employment and economic contribution estimates are order-of-magnitude only, and construction impacts have been excluded ▪ Actual outcomes will depend on land use mix, developable area, floor area, occupier profile, density, infrastructure, market take-up, staging, approvals and investment decisions ▪ More detailed economic modelling, cost-benefit analysis and financial assessment should occur at future business case or investment assessment stages.
Technical and Site constraints	▪ Development potential remains subject to further Site-specific technical testing across environmental, engineering, access, servicing and rehabilitation constraints.
Infrastructure	▪ Infrastructure capacity, connection costs and upgrade requirements are not yet confirmed and will influence which land uses are feasible.
Planning and approvals	▪ Planning and approval pathways require clarification, as existing controls were established for mining and environmental conservation rather than alternative uses ▪ Any transition to new land uses will require coordination between the proponent, Council, DPHI, the Resources Regulator and other relevant agencies.

7 Strategic context and existing evidence

This Section summarises the strategic context and existing evidence relevant to the Site. It sets out what this means for the Site's future role and the land use options being tested. It also carries forward the issues relevant to the feasibility assessment.

7.1 Overview of key studies and inputs

This Section synthesises the key evidence sources informing the assessment, including regional economic transition analysis, local strategic planning documents, and Site-level development feasibility work. Together, these provide a multi-scalar understanding of economic change, land use demand, and delivery constraints across the Hunter region and Lake Macquarie.

Table 7-1: Relevant studies and key findings

Key studies and inputs	Key findings
Hunter region economic transition and outlook[93]	▪ ~11,200 coal related industrial jobs are forecast to be lost by 2035 across Hunter region. ▪ Shift to knowledge-based regional economy could create between 40,000-75,000 jobs over the next decade. ▪ Hunter region carries a comparative advantage in clean energy, hydrogen, defence manufacturing and circular economy industries.
Lake Macquarie housing development feasibility and market considerations[94]	▪ Higher density development is only viable in select locations with strong amenity and market demand: Belmont, Boolaroo, and Cockle Creek. ▪ Detached housing remains the dominant and preferred typology. ▪ Market acceptance of higher density housing is limited, requiring either incentives or premium location attributes. ▪ Development feasibility conditional on market conditions and construction costs.
Macquarie Coal Complex deliverability due diligence findings[95]	▪ The Site is identified as being well suited for energy transition, industrial and logistics and circular economy due to proximity to transmissions infrastructure and freight corridors. ▪ Post-mining assessment identifies approximately 130 hectares with high development potential near rail stations and road corridors. ▪ 44 hectares requires investigation for flooding, biodiversity, and contamination risks. ▪ Tailings storage areas, final voids, and flood-prone land are unsuitable for development and are designated as long-term management zones.
Lake Macquarie City Council Community Strategic Plan 2025-2035[96]	▪ Transition from traditional industries such as mining and manufacturing, towards service-based and knowledge-driven sectors. ▪ Growing demand for housing diversity, transport, health and community infrastructure. ▪ Prioritising the development of connected and inclusive communities with access to open spaces, recreation and improved transport accessibility.

[93] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

[94] Atlas Economics (2026) Lake Macquarie Development Feasibility Study.

[95] Aurecon (2026) Final Deliverability Due Diligence report.

[96] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

7.2 Regional drivers influencing land use feasibility

Local demographic trends and broader structural changes across the Hunter economy shape demand for future land uses at the Site. These drivers operate across different times frames, supporting both near-term activation opportunities and longer-term transition uses.

Population growth, an aging population and ongoing regional migration characterise trends across Lake Macquarie as outlined in Table 7-1. These factors are likely to increase demand for local employment, services, and supporting infrastructure. They also reinforce the need to reduce reliance on out-commuting. Overall, these trends support a diversified mix of employment-generating and population-serving uses, aligned with local community needs.

Table 7-2: Demand drivers of LGA

Demand Driver	Evidence[97]	Opportunity at the LGA level
Population Growth	▪ Lake Macquarie forecast growth 214,000 in 2021 to 271,000 by 2046[98] (26% increase) ▪ North West Lake Macquarie Catalyst Area identified as a regionally significant growth area supporting housing and jobs growth – 3,700 dwellings and 1,800 new jobs by 2036[99,100]	▪ Sustained demand for local jobs, services, social infrastructure, employment land, reinforcing need for diversified employment uses rather than single-industry reliance
Regional Migration	▪ Strong inflows driven by affordability and lifestyle ▪ Growth in working-age cohorts (particularly 25–34 years)[101] ▪ Lake Macquarie ranked in the top five destinations for regional migration nationally	Support demand for: ▪ locally based employment opportunities, reducing out-commuting ▪ regionally connected employment land uses, particularly those leveraging transport access and proximity to Greater Newcastle
Ageing Demographics[102]	▪ Median age increasing (42 to 44 by 2046)[103] ▪ Growth in 65+ cohorts	▪ Supports health, wellbeing, aged care, allied health and accessible recreation uses. ▪ Need to replenish the working cohort with younger workers for sustained employment economy and family friendly activations, including via regional migration
Visitor Economy	▪ Lake Macquarie attracts more than 2.5 million visitors annually[104]	▪ Supports differentiated recreation/tourism, but likely complementary rather than primary.
Workforce Transition	▪ The demand for fossil fuel workers in Hunter is projected to decline, falling by 72% to 2035[105] ▪ Out of the fossil fuel workforce, 77% of workers are likely to undertake a workforce transition in 2035	▪ Shift from goods-producing to service and knowledge economy, with decline in mining and growth in health, education and construction ▪ Creates strong need for replacement employment land uses, particularly industrial diversification, clean energy and service-support industries
Employable Land Scarcity	▪ Limited employment land supply as a city-level constraint.	▪ Supports industrial, business park, logistics and service trade uses if access/servicing resolved

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97 REMPLAN Economy (n.d.) REMPLAN economy profile and data.

98 Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

99 NSW Department of Planning and Environment (2018) Greater Newcastle Metropolitan Plan.

100 Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

101 Australian Bureau of Statistics (ABS) (2021) Age in five year groups (AGE5P).

102 Lake Macquarie City Council (2022) Ageing Population Strategy 2022 - 2026.

103 Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

104 Lake Macquarie City Council (2022) Lake Macquarie City Destination Management Plan 2022–2026

105 Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

Demand Driver	Evidence[97]	Opportunity at the LGA level
	▪ Only ~58 hectares of serviced employment land remaining, with shortfall projected by 2046.[106]	

[106] Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

8 Demographic, social and labour market analysis

This Section provides an overview of the baseline socio-economic conditions within the SSA and ESA. These conditions shape which land uses are feasible, beneficial and aligned with the local context, and they identify the communities most likely to experience change. The social analysis focuses mainly on the SSA, where the most direct effects are expected. The ESA is used for comparison and broader regional context. Data provided in this Section is from the ABS census 2021.[107]

8.1 Demographic profile

Population size, growth and age structure determine the scale of future demand for jobs, housing, services and infrastructure, and the rate at which demand will increase. The SSA is home to approximately 124,304 residents, and the ESA home to 487,624 residents. The SSA broadly mirrors the ESA but has several characteristics relevant to social and land use assessment, including moderate cultural diversity, a relatively high Aboriginal and Torres Strait Islander population compared to NSW average, car-based travel patterns, high home ownership and mixed income profile.

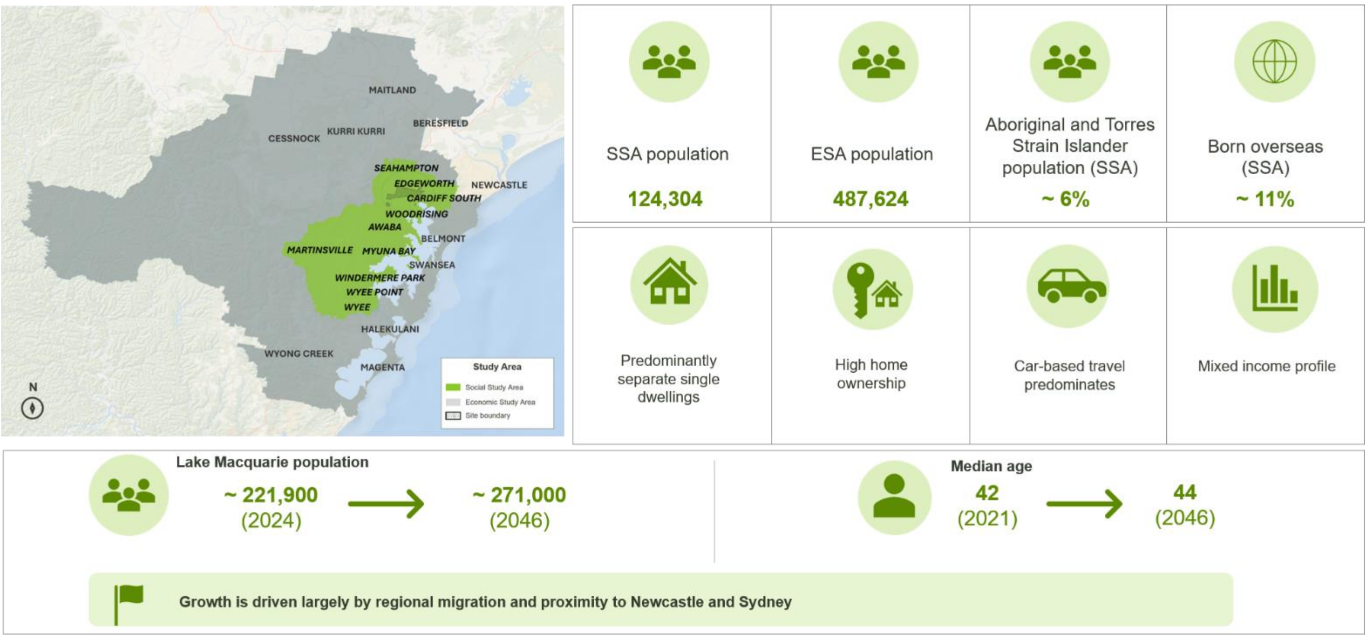

Figure 8-1: Demographic profile of the social and economic study areas[108,109]

8.1.1 Population growth and change

Lake Macquarie population is projected to increase from approximately 221,900 in 2024 to around 271,000 by 2046.[110,111] This growth is largely driven by regional migration, reflecting the city's proximity to Newcastle and Sydney. This reinforces the need for timely planning of supporting infrastructure, including housing, education, health and transport services which supports this growing population.[112] Figure 8-2 sets out this projected population alongside the year-on-year percentage change. Year-on-year growth decreases over time. Lake Macquarie's population is also ageing, with the median age projected to rise from 42 years in 2021 to 44 years by 2046, remaining higher than the NSW average.[113]

[107] Australian Bureau of Statistics (ABS) (2021) 2021 Census

[108] REMPLAN (2025) Lake Macquarie LGA (SEP 2025) Forecast Total Population 2046.

[109] Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

[110] REMPLAN (2025) Lake Macquarie LGA (SEP 2025) Forecast Total Population 2046.

[111] Note: Population growth figures are not available for the SSA and ESA study area data sets. Lake Macquarie has been used as a proxy for the SSA.

[112] Lake Macquarie City Council (2025) Local Strategic Planning Statement 2025-2045.

[113] Lake Macquarie City Council (2026) Draft Employment Land Use Strategy.

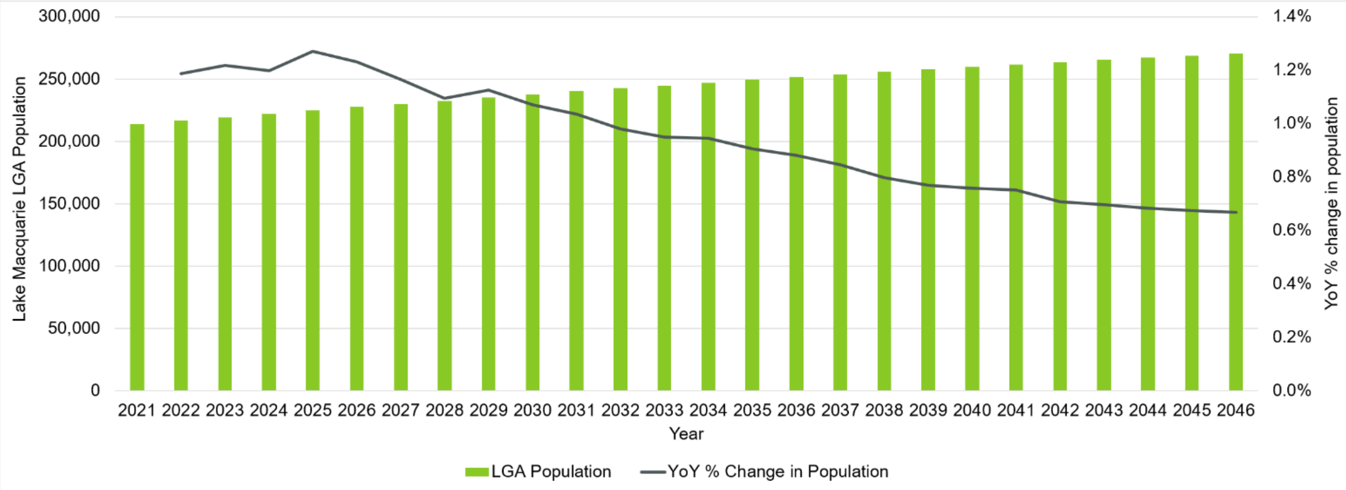

Figure 8-2: Lake Macquarie LGA forecast population and year-on-year growth[114]

8.2 Socio-economic vulnerability indicators

Vulnerability indicators identify which communities have less capacity to absorb the effects of land use change, and where planning should pay closer attention to impacts and equity. Based on the ABS Censes 2021 data, the SSA supports a generally economically stable community with clear indicators of socio-economic vulnerability shaped by the interaction of age structure, health conditions, income distribution, housing costs, transport dependence and cultural and linguistic diversity. A snapshot of socio-economic vulnerability for the SSA and the Site is summarised in Figure 8-3. Table 8-1 outlines vulnerability indicators and the implications for PMLU planning. Further information is available in Appendix H.

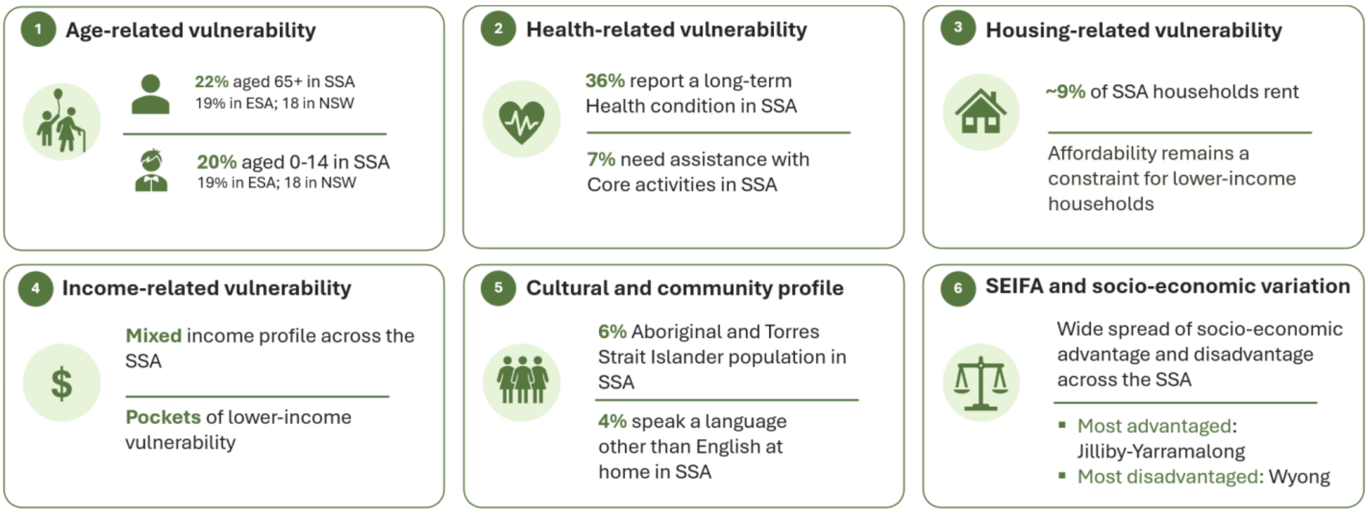

Figure 8-3: Snapshot of Socio-economic vulnerability indicators[115,116,117,118]

Table 8-1: Vulnerability indicators and planning implications

Vulnerability indicator	Implications for the Site
Age related vulnerability	A high proportion of both older people and children, highlighting the need for safe, accessible, low-impact land uses that support mobility, health, and access to local services.
Health related vulnerability	A high proportion of both older people and children, highlighting the need for some PMLU that are safe, accessible, low-impact and that support mobility, health, and access to local services.

[114] REMPLAN (2025) Lake Macquarie LGA (SEP 2025) Forecast Total Population 2046.

[115] Australian Bureau of Statistics (ABS) (2021) Age in five year groups (AGE5P).

[116] Australian Bureau of Statistics (ABS) (2021) Total personal income (weekly) (INCP).

[117] Australian Bureau of Statistics (ABS) (2021) Housing: Census.

[118] Australian Bureau of Statistics (ABS) (2021) Aboriginal and Torres Strait Islander people: Census.

Vulnerability indicator	Implications for the Site
Income related vulnerability	Income vulnerability is observed indirectly through housing stress and employment access, rather than through income figures themselves. To reduce income vulnerability and support equitable transition outcomes, the PMLU options should prioritise stable, accessible employment and incorporate job design, training and procurement approaches that support participation by vulnerable and under-represented cohorts.
Housing related vulnerability	Housing conditions indicate affordability pressures despite high home ownership, reinforcing the need for the PMLU to prioritise land uses that support housing accessibility and reduce cost-of-living stress for vulnerable households.
Cultural and linguistic diversity related vulnerability	There is a relatively high level of people who identify as Aboriginal and Torres Strait Islander within the study area. The PMLU will incorporate Connecting with Country into the design as outlined by the Master Plan.
Socio-economic vulnerability	SEIFA indicates areas that may be more acutely impacted by PMLU changes. Areas with lower SEIFA scores, such as Wyong, may have less capacity to absorb housing cost increases, increased travel costs, disruption to local access or other effects associated with land use change. This does not mean displacement is occurring or will occur. Rather, it indicates where future planning should consider housing affordability, tenure, transport access, service access and the distribution of benefits and impacts. The land use options support new employment, however, the benefits may not be equitably distributed to disadvantaged communities. Lower-income residents without reliable private vehicle access or disabilities may face barriers to accessing new jobs if public transport connections to the Site remain limited. Although the Master Plan considers transport and mobility, further detailed design should not only reflect the number and type of jobs created, but also who can realistically access them via active and public transport.

8.3 Community values and identity

Community identity, values and priorities reveal what people care about most in an area, how they define their sense of place and wellbeing, and the outcomes they want future development and decision-making to support and protect.

Awabakal Country shapes the area's identity. The Awabakal people have maintained strong spiritual, cultural and economic relationships with Lake Macquarie, waterways and surrounding landscapes for thousands of years. The SSA and ESA also have more than a century of non-Indigenous settlement. Key community priorities include caring for Country, protecting cultural heritage, improving health and wellbeing outcomes, and enabling meaningful participation in decision making and employment. These expectations often include practical priorities such as housing affordability, transport access and open space.

Table 8-2 sets out the key themes shaping community values and identity across the SSA, and their planning implications for PMLU. Appendix J gives a high-level description of each non-Indigenous heritage site referenced.

Table 8-2: Community values and identity overview

Theme	Key findings
Aboriginal and Torres Strait Islander	The Site is located on Awabakal Country, with a long and continuing history of Aboriginal occupation, cultural practice and connection to land and water. Key priorities include caring for Country, protecting cultural heritage, improving health and wellbeing, and enabling participation in decision-making and employment. Surrounding waterways, including Cockle Creek and Lake Macquarie, are classified as Sensitive Aboriginal Landscapes under the Lake Macquarie Local Environmental Plan 2014.[119] The Master Plan takes a Country-centred approach to planning, design, delivery and stewardship, including through a Connecting with Country Guideline.

[119] NSW Department of Planning and Environment (n.d.) Lake Macquarie Local Environmental Plan 2014.

Theme	Key findings
Non-Indigenous heritage	The SSA and broader ESA contains 9 non-Indigenous heritage items of local and State significance, spanning education, mining, industrial, defence, healthcare and cultural history (Figure 8-4). These items constrain built form, siting and design in specific locations, and inform place-making across the Site.
Community values and priorities	Community priorities set the standard land use options need to meet, to secure local support and deliver genuine benefit.
Areas of community interest	The Site is well located near a range of recreational, sporting and environmental assets within 5km, providing extensive opportunities for recreation, community activity and access to open space.

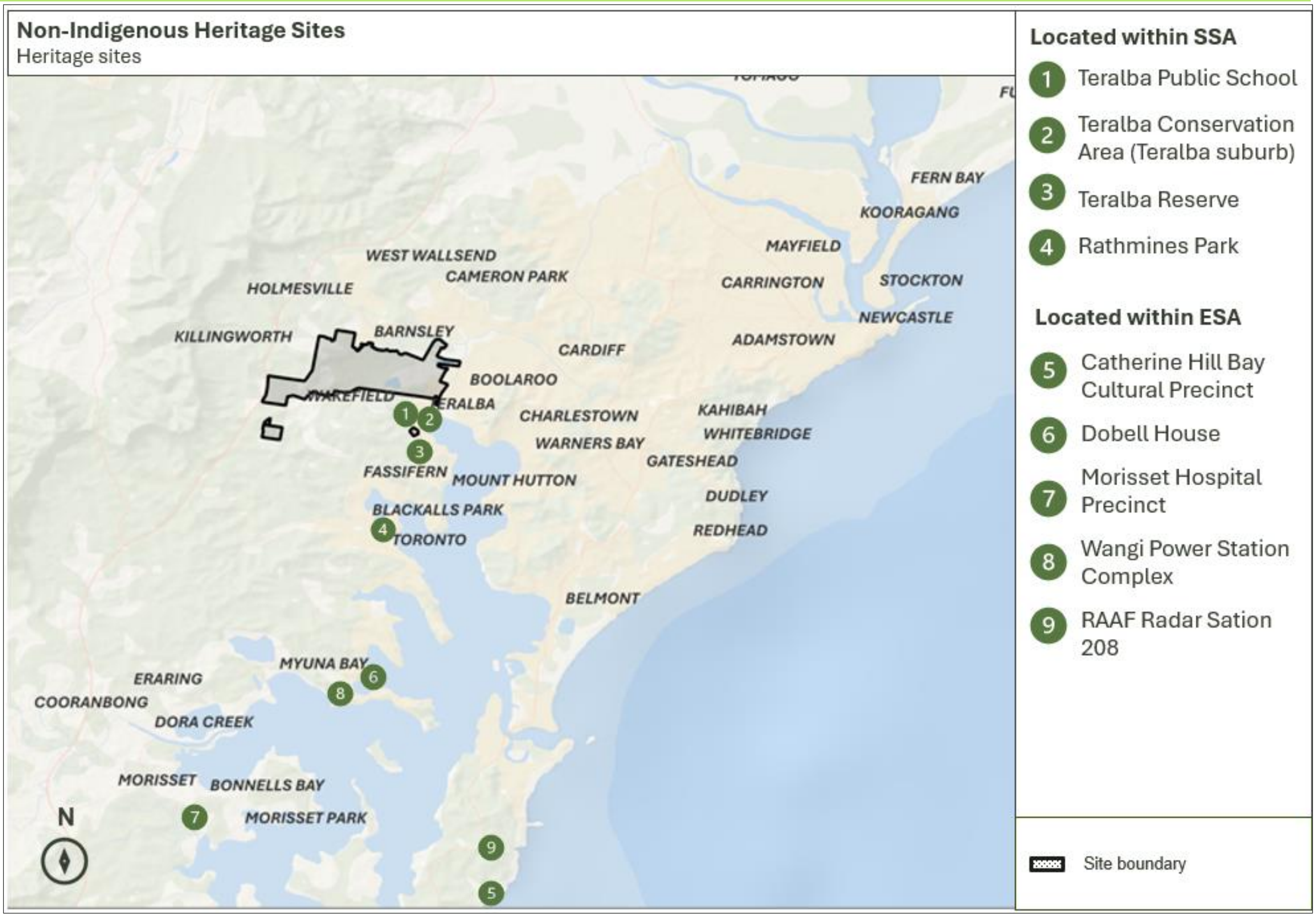

Figure 8-4: Map of all non-Indigenous heritage sites

Table 8-3: Key community values and priorities

Priority		Key findings
Natural environment and climate resilience		▪ Protecting Lake Macquarie's natural environment, biodiversity and climate resilience is central to local identity liveability and wellbeing
Housing affordability and growth		▪ Growth is supported where it improves housing choice, affordability and liveability, and is planned with local input[120] ▪ Concerns regarding rising rental and mortgage stress for low-income households, older and vulnerable residents[121]

[120] Taverner Research Group (2024) Community Satisfaction Survey 2024.

[121] SGS Economics & Planning (2026) Lake Macquarie Housing Affordability and Liveability Study: Interim Housing Demand and Affordability Report.

Priority		Key findings
Transport, access and open space		▪ Prioritise reliable public transport and active travel infrastructure to build healthy, safe, and connected neighbourhoods[122]
Social cohesion and equitable access		▪ Future land use should support social cohesion[123], safety, equitable access to services and local community infrastructure
First Nations, Country and governance		▪ Planning should recognise First Nations connection to Country, protect cultural heritage and support transparent community participation[124]

8.4 Areas of community interest

Table 8-4 provides a list of key locations valued by the community within 5 km of the Site. These locations represent key natural, recreational, and cultural assets ranging from infrastructure and recreational parklands. These locations are shown in relation to the Site in Figure 8-5.

Table 8-4: Key areas of community interest within 5km of the Site

Distance from the Site	Site	Type	Ref
500m	Waratah Golf Club	Recreation and sport	5
1km	Speers Point Park and Jetty	Open space and recreation	2
1km	Speers Point Jetty	Recreational fishing jetty	4
2km	Munibung Hill	Culturally significant site	1
2km	Hunter Sports Centre and Gym	Olympic-standard athletics and community recreation	3
4km	Hunter Ice Skating Stadium	Public skating, training and competitive events	6

[122] Place Score 2025 (2025) Australian Liveability Census: Executive Summary – Lake Macquarie City Council

[123] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[124] Umwelt Environmental Consultants (2011) Sustainable Management of Aboriginal Cultural Heritage in the Lake Macquarie Government Area: Lake Macquarie Aboriginal Heritage Management Strategy.

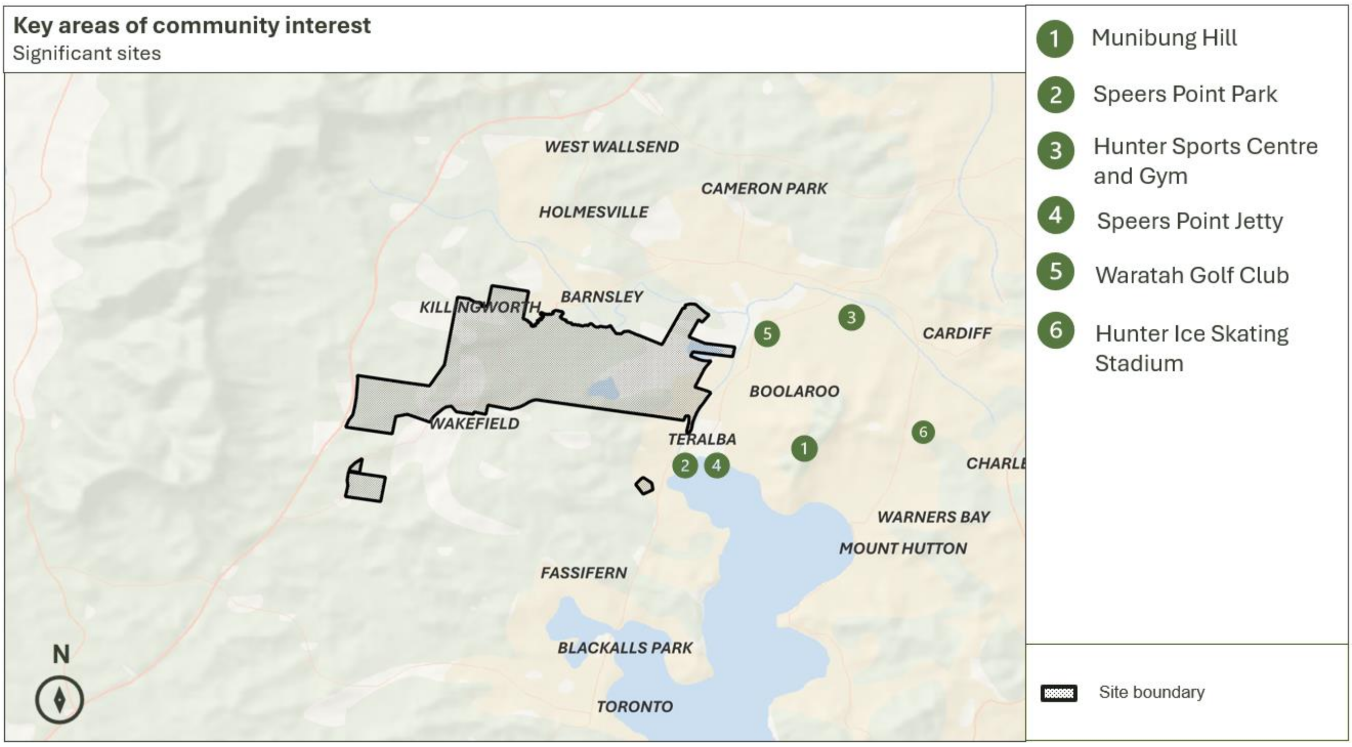

Figure 8-5: Map of key areas of community interest in relation to the Site location.

8.5 Social infrastructure and transport context

Social infrastructure refers to the facilities, structures, and services that support physical, social, cultural, and intellectual development. These can be categorised into healthcare, education, sport and recreation, open spaces and cultural assets. Social infrastructure locations were identified through the NSW social infrastructure and points-of-interest data.[125] The analysis considers two geographies. The SSA provides a strategic view of provision across the wider area. A 5 km radius defines the immediate catchment. The 5 km catchment was selected as a practical screening distance for an early strategic assessment. At typical urban speeds, 5 km equates to roughly 6 to 10 minutes by car.[126] Table 8-5 summarises the baseline social infrastructure and public transport access context.

Table 8-5: key findings of social infrastructure and public transport access

Theme	Key findings
Social infrastructure (SSA and 5 km)	No health and medical facilities are recorded within 5 km of the Site. This indicates reliance on Cardiff, Glendale, Speers Point and other nearby centres for higher-order services.
Public transport	Low to no public transport accessibility (PTAL) around the Site indicates that access to future jobs and services is likely to remain car-dependent unless transport connections are improved.
Combined (Social infrastructure and transport access)	Around 74.4% of SSA social infrastructure assets are in areas with limited public transport access. This means access, not only facility provision, is a relevant consideration for future planning.

Within the SSA, there are 410 facilities, and 127 within 5km of the site. Location of these facilities are shown in Figure 8-6 (the SSA) and Figure 8-7 (5 km). Social infrastructure across the broader SSA and within 5 km of the Site highlights a

[125] Transport for NSW (n.d.) NSW Point of Interest data.

[126] Indicative travel time only. The surrounding road network includes both built-up areas, where the NSW default urban speed limit of 50 km/h applies, and higher-speed rural roads (Transport for NSW, Speed zones and speed management, [link]). At around 50 km/h, 5 km equates to approximately 6 minutes by car.

consistent pattern of uneven geographic distribution, where larger regional centres accommodate a greater diversity of services. Communities closest to the Site rely on these centres to access to higher-order facilities.

Across the SSA, social infrastructure is broad but concentrated in established centres such as Cardiff, Glendale, Boolaroo, Speers Point, Toronto, Morisset and Cooranbong. These locations provide a wider mix of community, cultural, education, recreation, aged care and emergency services, while health infrastructure remains comparatively limited. In contrast, smaller communities including Wakefield, Killingworth, Holmesville and Barnsley have a more constrained local service base. Within 5 km of the Site, infrastructure is more moderate and locally oriented, characterised by schools, community facilities and other community-serving uses rather than higher-order services. Facilities are clustered around Teralba, Booragul, Boolaroo, Edgeworth and West Wallsend.

Implication for the Site

The existing provision of social facilities indicates that communities closest to the Site may be more sensitive to changes in land use, given their limited local access to services and reliance on surrounding centres, particularly the closest suburbs of Teralba, Booragul and Boolaroo.

For PMLU, this highlights the importance of considering how future land uses could either strengthen local service access or maintain connectivity to existing service hubs. This is considered with the Master Plan community facilities and social infrastructure performance criteria.

Table 8-6: Social infrastructure facilities within the SSA and 5 km of the Site

Social infrastructure category	Number of facilities in the SSA	Number of facilities within 5km of the Site
Community[127] and cultural	73	18
Education	64	20
Recreation	40	14
Aged care	38	13
Emergency services	30	11
Early childhood education	21	8
Health and medical	6	0
Other[128]	138	43
Total	**410**	**127**

[127] Note: “Community and cultural” includes source-data records such as places of worship, charities, museums, local government chambers, libraries and art galleries. It is used for high-level screening and is not a detailed service planning category.

[128] Note: “Other” is a mixed source-data category. It includes records not otherwise classified in the source dataset, including selected civic, community / recreation, local service, visitor accommodation, commercial and infrastructure-related assets. It has been retained for transparency in this high-level screening assessment but has not been used as a standalone basis for assessing social infrastructure provision.

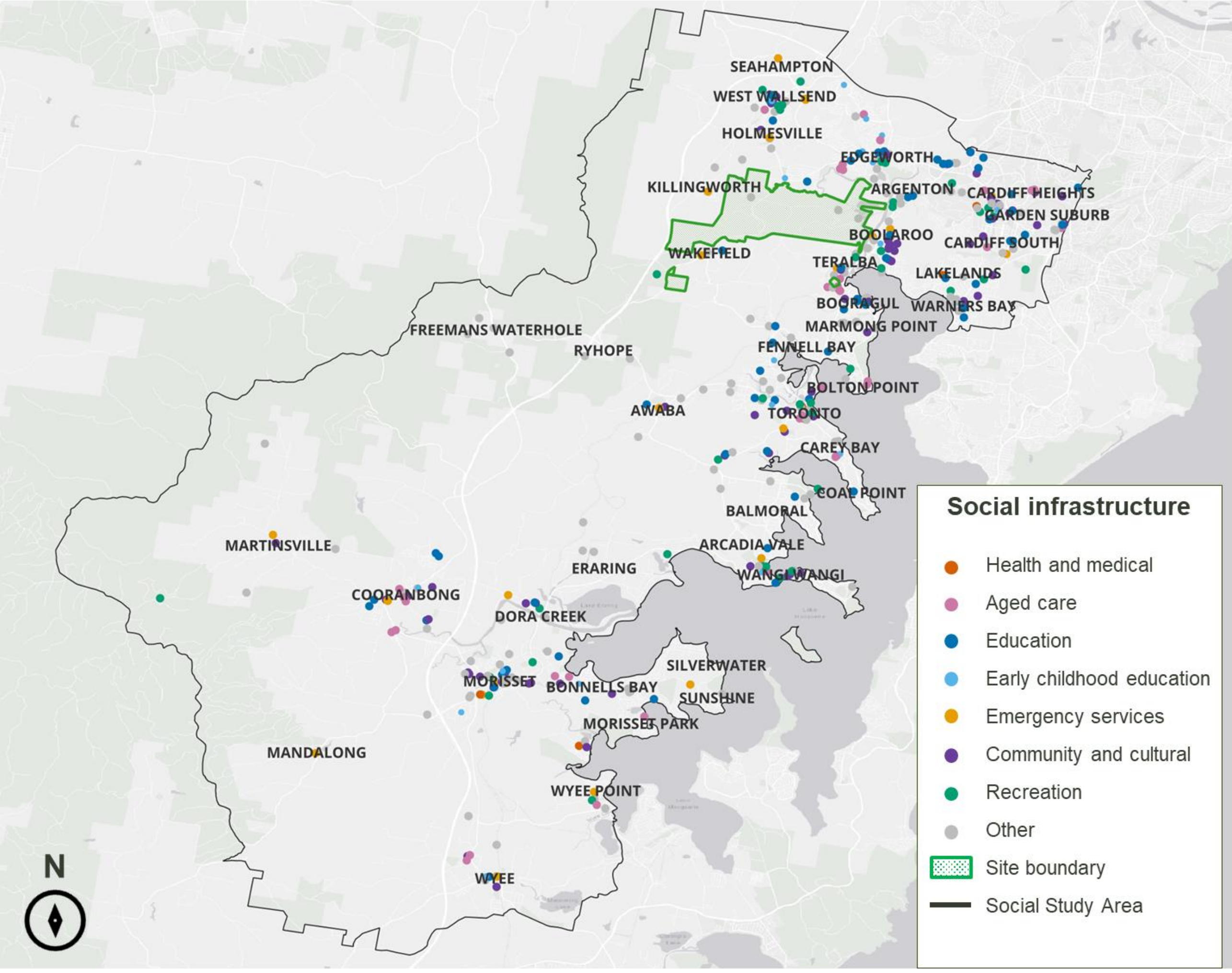

Figure 8-6: Social infrastructure within SSA

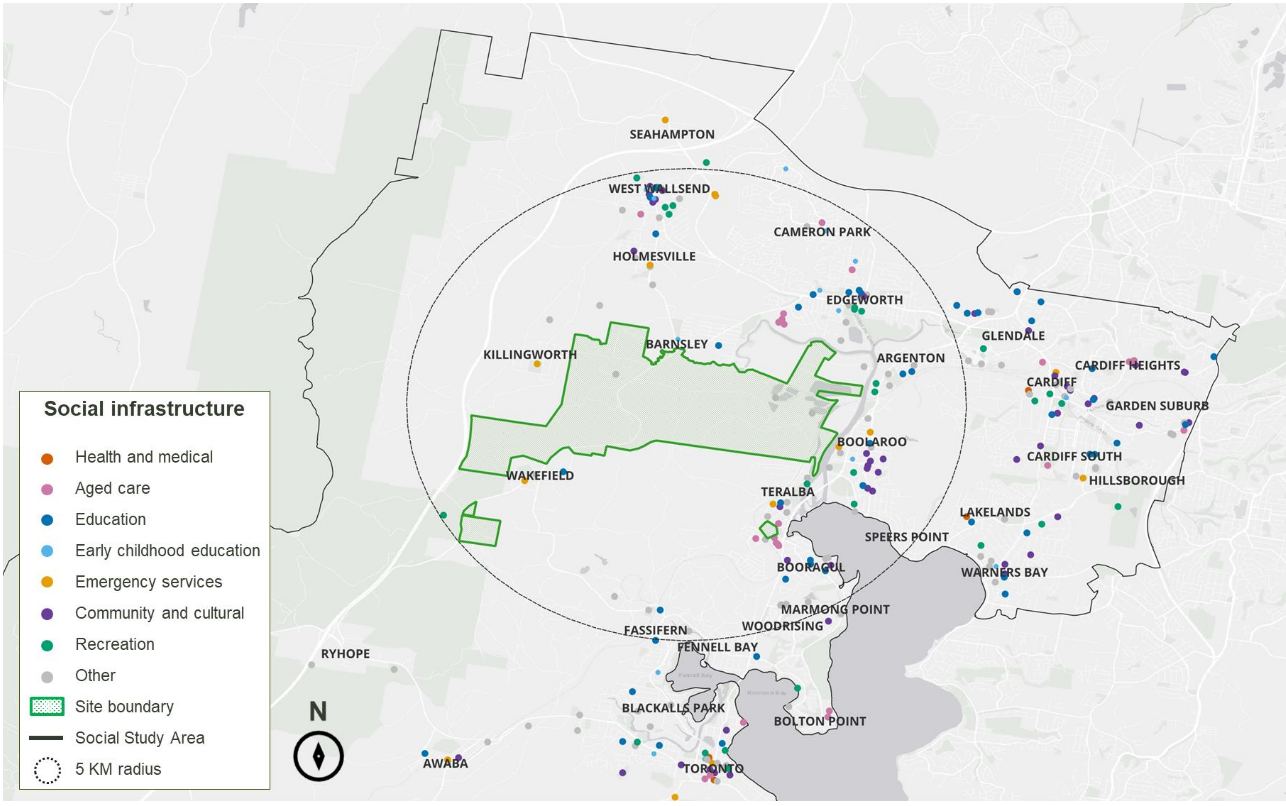

Figure 8-7: Social infrastructure within 5km radius of the Site

8.5.1 Public transport provisions

Public transport accessibility was assessed using the NSW Public Transport Accessibility Level (PTAL) dataset.[129] The detailed methodology is outlined in Appendix E.

Accessibility is generally low across the SSA. Higher levels concentrate around the established urban corridor closer to Cardiff, Glendale, Cardiff Heights, Cardiff South and Warners Bay. The western and central SSA have low to no PTAL. This includes Wakefield, Killingworth, Barnsley, Freemans Waterhole, Awaba, Dora Creek, Morisset, Cooranbong and Wyee. The area around the Site itself is also low to no PTAL (Table 8-7 and Figure 8-8).

Table 8-7: Public transport accessibility by area

Area	Public transport accessibility
Eastern / north-eastern SSA (Cardiff, Glendale, Warners Bay)	Higher
Western / central SSA (Wakefield, Killingworth, Barnsley, Morisset, Cooranbong, Wyee)	Low
Around the Site	Low to no PTAL

Implication for the Site

The Site itself has low to no PTAL. Future employment or community-facing land uses may need to plan for private vehicle access in the near term. Improving public transport connections is also worth considering as part of the Master Plan for future development of the Site.

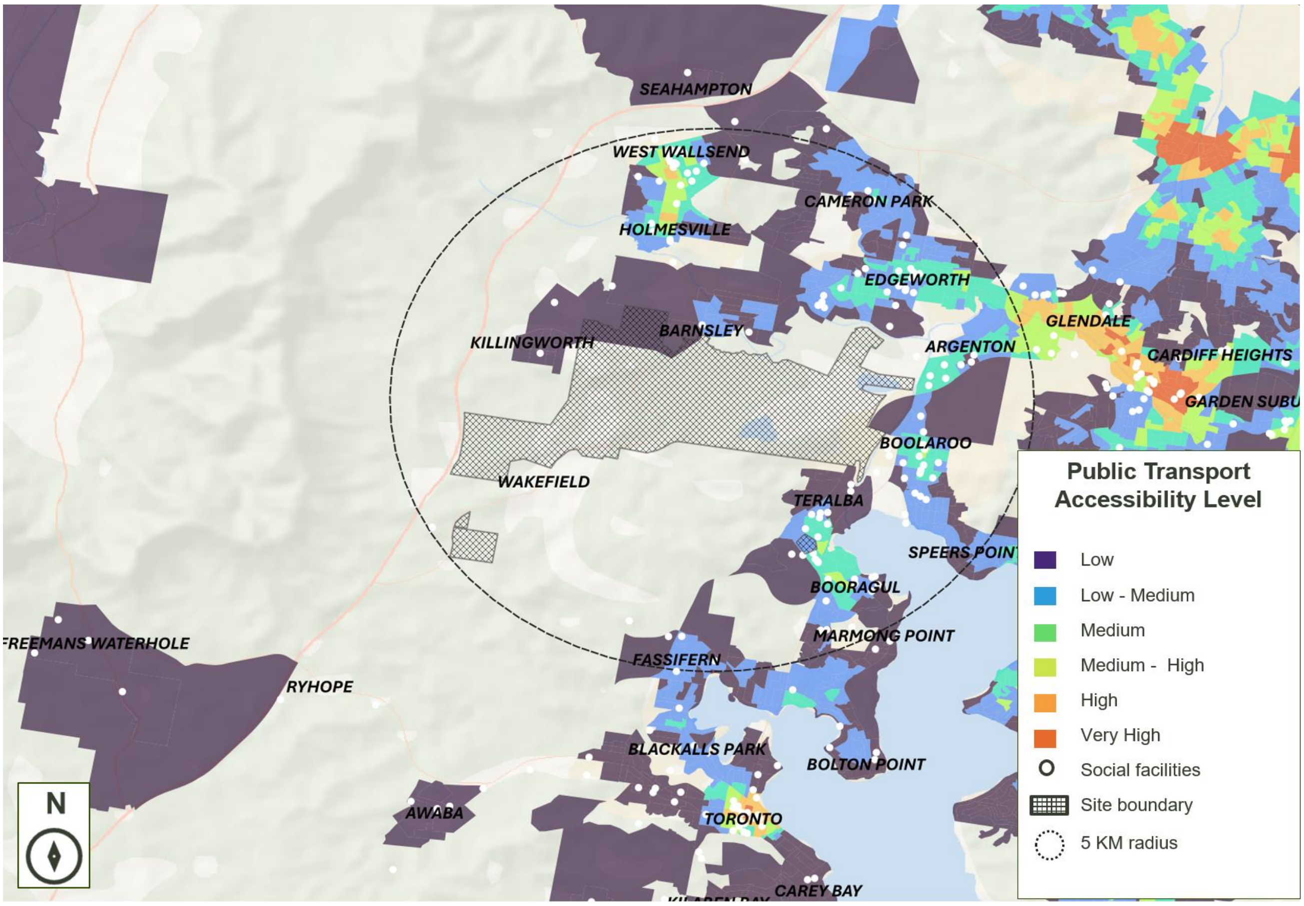

Figure 8-8: PTAL and social infrastructure near the Site

[129] Transport for NSW (n.d.) Public Transport Accessibility Level.

8.5.2 Social infrastructure and public transport accessibility

A spatial analysis combined social infrastructure locations with PTAL scores to assess how accessible existing facilities are for communities around the Site. Facilities are generally more accessible in established urban centres and less accessible in the western, more dispersed parts of the SSA.

Table 8-8: Public transport accessibility of social infrastructure facilities in the SSA[130]

PTAL category	Number of facilities	Share of all SSA facilities	Accessibility interpretation
No PTAL score	95	23%	Not accessible by public transport
Low	138	34%	Poor public transport accessibility
Low Medium	72	18%	Limited public transport accessibility
Medium	53	13%	Moderate public transport accessibility
Medium High	24	6%	Good public transport accessibility
High	18	4%	Strong public transport accessibility
Very High	10	2%	Very strong public transport accessibility
Total	**410**	**100%**	

Implication for the Site

Most social infrastructure in the SSA sits in areas of limited public transport access. Being close to a facility does not always mean it is easy to reach without a car. Future land uses at the Site could increase employment, residential demand or community activity. This points to transport access, not just facility provision, as a relevant equity consideration, particularly for residents who do not drive.

Approximately 305 facilities sit in areas with no, low or low-medium PTAL. This is 74% of all mapped social infrastructure in the SSA (Table 8-8). Access to most of this infrastructure is likely to depend on private vehicle travel outside the main centres. This may be more significant for young people, older residents, people with disability and low-income households.

This indicates that access to social infrastructure across the SSA is likely to be highly dependent on private vehicle travel, particularly outside the main urban centres. Facilities with limited public transport accessibility may be harder to reach for young people, older residents, people with disability, or low-income households.

8.6 Workforce characteristics

This Section outlines workforce characteristics within the SSA and ESA. Workforce characteristics are critical to understanding local skills, employment patterns, vulnerabilities and transition risks that can inform PMLU options. The workforce across both areas reflects a transitioning regional labour market, with relatively strong labour force participation, low unemployment, and a growing working age population that underpins current and future employment capacity. A distinctive feature of the Hunter workforce is the continued concentration of employment in fossil-fuel-related industries, particularly coal mining, which employs a disproportionately male, prime working-age and highly specialised workforce with above-average wages.[131]

At the ESA scale, the Hunter is identified in NSW strategic planning as a major employment region undergoing economic diversification, including the re-use of former mine and power station Sites for new purposes such as

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[130] Note: For this assessment, social infrastructure facilities with no PTAL score are treated as being outside the mapped public transport accessibility catchment. This means they are considered not accessible by mapped public transport for the purpose of this analysis.

[131] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

renewable energy and advanced industry, which directly aligns to PMLU decision-making.[132] Workforce transition planning for the Hunter also highlights the need to manage skills pipelines and ensure new jobs are accessible to impacted workers and under-represented cohorts[133] including First Nations communities.

8.6.1 Population and working age profile

In 2021, the SSA had a working-age population (15–64 years) of 60% (74,841 of 124,294 residents).[134] The largest cohorts were aged 45-64 (25%) and 25-44 years (24%). Residents aged 65 years and over made up 22% of the SSA, slightly above the ESA's 20%. This points to an ageing trend common to both areas, alongside a strong working-age base (Table 8-9).

Table 8-9: Population by broad age group, SSA and ESA[135]

Age group	SSA	ESA
Working age (15-64)	60% (74,841)	61% (349,687)
Aged 25-44	24% (29,734)	25% (141,819)
Aged 45-64	25% (30,896)	25% (140,898)
Aged 65+	22% (27,024)	20% (114,315)

Implication for the Site

The working-age population provides a labour base for future PMLU-related employment. Residents aged 65 and over make up a relatively high share of the SSA. This points to the value of planning for workforce transition over a longer timeframe. Forecast housing growth across the ESA may also increase competition for labour, which is a relevant factor for PMLU delivery timing.

The ESA is forecast to see strong population growth to 2041. Lake Macquarie is one of the LGAs expected to take a large share of new housing. This is likely to increase demand for local jobs, services, construction and infrastructure.

8.6.2 Labour force participation

The SSA labour force made up 48% of the population in 2021 (46% employed, 2% unemployed). The ESA was similar, at 47% labour force, 45% employed and 2% unemployed (Table 8-10). Low unemployment in both areas points to a tight labour market. This is consistent with the broader Hunter region, where fossil-fuel employment has declined over the past decade. Overall workforce growth has remained positive.

Table 8-10: Total SSA and ESA labour force participation and employment (2021)[136]

Area	Labour force	Employed	Unemployed
SSA	48%	46%	2%
ESA	47%	45%	2%

[132] NSW Department of Planning and Environment (2022) Hunter Regional Plan 2041.

[133] Department of Employment and Workplace Relations (2026) Local Jobs Plan: Hunter Employment Region (NSW).

[134] Note: Working age population represents potential pool of people who could participate in employment; however, it also includes residents who are not currently working or seeking work (such as students, carers or people unable to work) and therefore overstates the labour available to support PMLU in the short term.

[135] Australian Bureau of Statistics (ABS) (2021) Age in five year groups (AGE5P).

[136] Australian Bureau of Statistics (ABS) (2021) Employment in the 2021 Census.

Government workforce planning points to challenges in securing skilled workers in growth sectors, such as care, tourism, advanced manufacturing and construction. The ESA also contains an estimated 20,000 employees in carbon-intensive roles, including coal-fired power stations. These are projected to close by 2033, including Eraring in 2027.

Implication for the Site

Unemployment is low in both study areas. New PMLU employment is more likely to draw on workforce mobility and skills transfer than on unemployed residents. This points to the value of attracting and retaining workers locally. The region also has a large carbon-exposed workforce, with coal-fired power stations including Eraring due to close. PMLU options that could absorb or redeploy transferable energy and industrial skills may support this transition.

Government workforce planning similarly identifies challenges in securing adequately skilled workers in growth sectors. These include care, tourism, advanced manufacturing and construction, indicating PMLU proposals should be assessed in terms of their labour requirements such as training needs and a capacity to attract and retain workers locally.[137]

8.6.3 Skills and educational attainment

Educational attainment in the SSA shows a workforce concentrated in certificate-level and vocational qualifications (22% and 9% of residents respectively), with 15% holding a bachelor's degree or higher. This is consistent with a workforce historically based in trades, technical and operational roles, and a high reliance on VET and on-the-job training.[138] The ESA shows a broadly similar profile, with a marginally smaller share of residents holding a bachelor's degree or higher (13%).

Table 8-11: Total SSA and ESA skills and educational attainment (2021)[139]

Area	Certificate	Diploma / Adv. Diploma	Bachelor or higher	Total non-school qualifications
SSA	22%	9%	15%	51%
ESA	22%	7%	13%	49%

Implication for the Site

The SSA's skills profile suggests land uses aligned with trade-based, technical or operational roles may draw well on the existing local workforce, while more specialised roles could require additional training or recruitment from outside the area.

Existing transmission infrastructure and the REZ may also support PMLU options connected to renewable energy or hydrogen production, though this would benefit from a dedicated skills-matching assessment.

NSW strategic planning identifies the Hunter-Central Coast REZ and hydrogen production opportunities, supported by existing transmission infrastructure and a skilled workforce.[140] National workforce projections also point to continued employment growth in service industries and rising demand for digital literacy and higher-skilled roles.

8.6.4 Workforce mobility and travel to work

Travel to work in the SSA and ESA is dominated by car use, at 26% in both areas. Public transport uptake is minimal, at under 1% in both areas. Work-from-home levels (9% SSA, 8% ESA) reflect pandemic-era Census conditions rather than an established long-term pattern.

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[137] Department of Employment and Workplace Relations (February 2024) Local Jobs Plan: Hunter Employment Region (NSW). [link]

[138] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

[139] Australian Bureau of Statistics (ABS) (2021) Education and training: Census

[140] Department of Planning, Housing and Infrastructure (2022) Hunter Regional Plan 2041.

SSA population	By car (Driver)	By public transport	Work from home
124,304	32,414 people (26%)	343 people (>1%)	10,962 people (9%)
ESA population	By car (Driver)	By public transport	Work from home
487,624	126,120 people (26%)	1,452 people (>1%)	41,163 people (8%)

Figure 8-9: SSA and ESA travel to work

Implication for the Site

With car use dominant and public transport uptake minimal, PMLU-related employment may need to plan for road access and parking, particularly for shift-based roles. Transport access has also been identified as a barrier to participation elsewhere in the region, pointing to a possible role for PMLU options that make use of existing road networks or improve transport options in supporting broader workforce accessibility.

NSW regional planning highlights that car dependence contributes to broader equity and cost impacts, reinforcing the importance of integrating transport considerations into PMLU land use selection and staging.[141] Local workforce planning further identifies transport access as a barrier to participation in parts of the region, suggesting PMLU options leveraging existing road networks or incorporate transport solutions could potentially improve job accessibility.

8.7 Economic and industry analysis

8.7.1 Industry output and value-add

Lake Macquarie's mining output, valued at approximately $5.3B in 2023, has remained elevated in the post-COVID period due to high global energy prices driven by supply disruptions and broader geopolitical factors, including the Russia-Ukraine conflict. However, this strength has masked an underlying structural decline in coal, with limited capacity expansion and tightening constraints on long-term growth. A 3% drop in output is forecasted across the mining sector in the Lake Macquarie LGA over the near term, due to closure of Eraring Power Station. It is expected that closure of Eraring will result in further downstream supply chain impacts to mines.[142]

The top 3 industries across Lake Macquarie LGA in terms of industry output are Construction ($6.47B, 19%), Mining ($5.33B, 16%) and Manufacturing ($4.45B, 13%) (Figure 8-10). Across 2011 to 2022 mining output has declined 3%.

[141] NSW Department of Planning and Environment (2022) Hunter Regional Plan 2041.

[142] The University of Newcastle (2023) Responding to Structural Change in Lake Macquarie - Briefing Paper for Policy Makers.

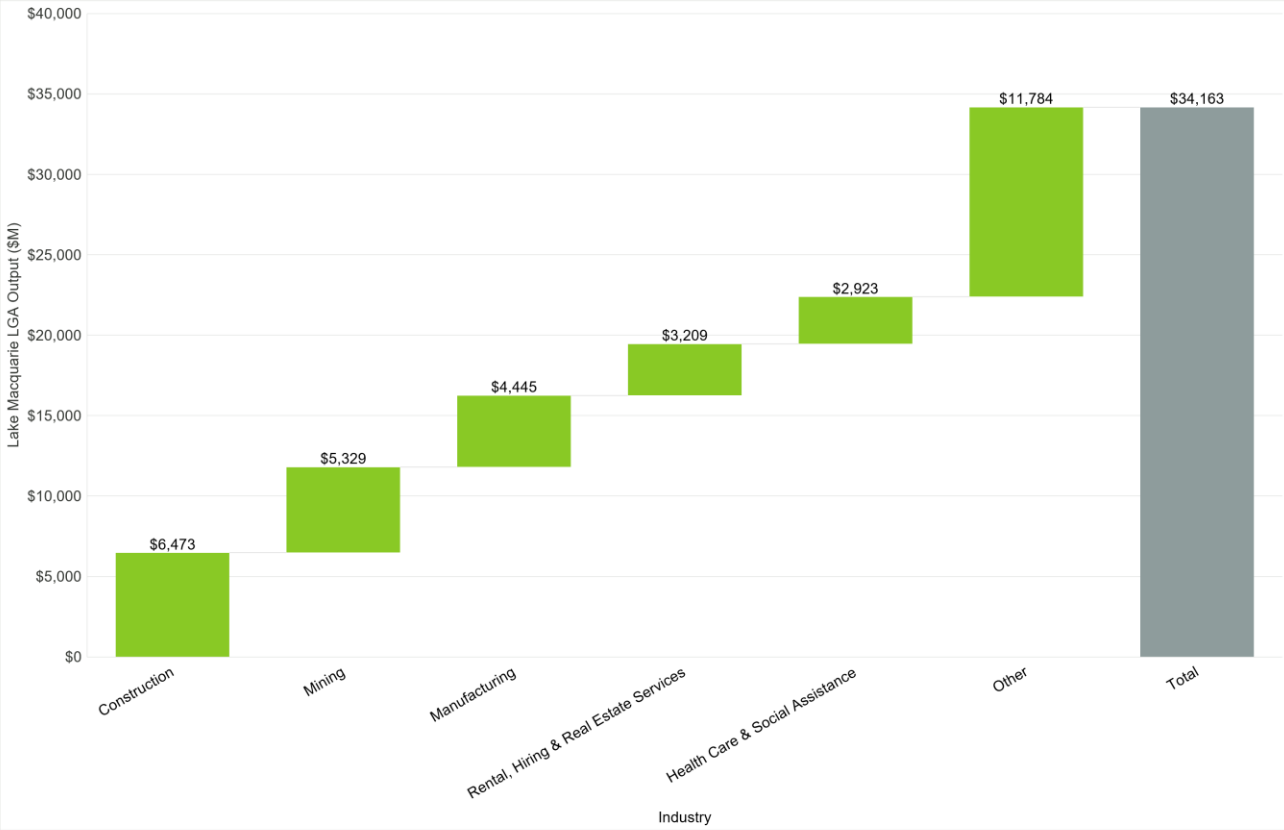

Figure 8-10: Lake Macquarie LGA output by industry ($M)[143]

The ESA contributes modestly to the NSW economy in aggregate terms; however, its role is significant at a regional scale, supporting local employment, supply chains, and sectoral activity. The ESA economy is valued at $71.55B with the SSA economy contributing to 30% to the gross regional product (GRP). The SSA's significant contribution to the region is largely driven by the Mining industry, reflecting the geographic proximity to active coal mining operations. The value added within the Lake Macquarie LGA is $3.90B from the Mining industry alone.

Across industry divisions in the Lake Macquarie LGA, the top 3 industries in terms of value add are Mining $3.9B), Rental, Hiring & Real Estate Services ($2.4B) and Health Care & Social Assistance ($2.1B).

8.7.2 Labour market analysis

Lake Macquarie LGA demonstrates a strong industry specialisation in mining as shown in Figure 8-11, with a location quotient (LQ) of 2.6, indicating that mining employment is approximately 2.6 times more concentrated than the NSW average. This reflects the region's historical role in coal mining and the continued influence of mining activities across the LGA and the broader Lower Hunter. This is followed by Construction (LQ = 1.49) and Health Care and Social Assistance (LQ = 1.37), which also exhibit above-average industry concentrations.

Although mining has the largest industry concentration in the LGA, employment growth has been relatively modest, increasing by 13% between 2016 and 2021 as shown by the size of bubble in Figure 8-11. Industries exhibiting stronger employment growth in the LGA from 2016 to 2021 are Health Care and Social Assistance, Agriculture, Forestry and Fishing, Construction and Professional, Scientific and Technical Services.

[143] REMPLAN (2025) Lake Macquarie LGA (2025 R1) Output by Industry.

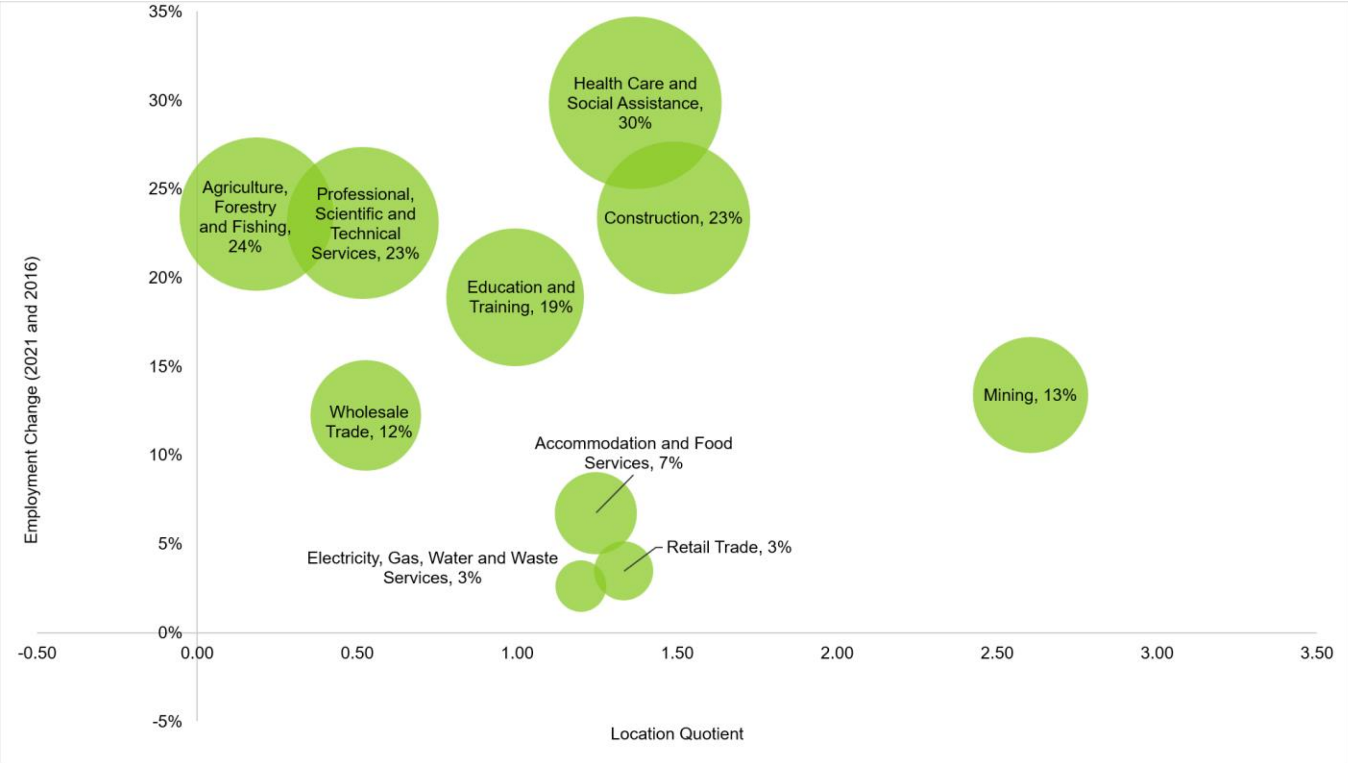

Figure 8-11: Location Quotients of Lake Macquarie LGA and Change in Employment for the ten highest industries[144]

While Health Care and Social Assistance and Construction are among the fastest growing industries, other sectors with the large employment bases in the LGA, ESA and SSA are Retail Trade, Education and Training, and Accommodation and Food Services as shown in Figure 8-12 below.

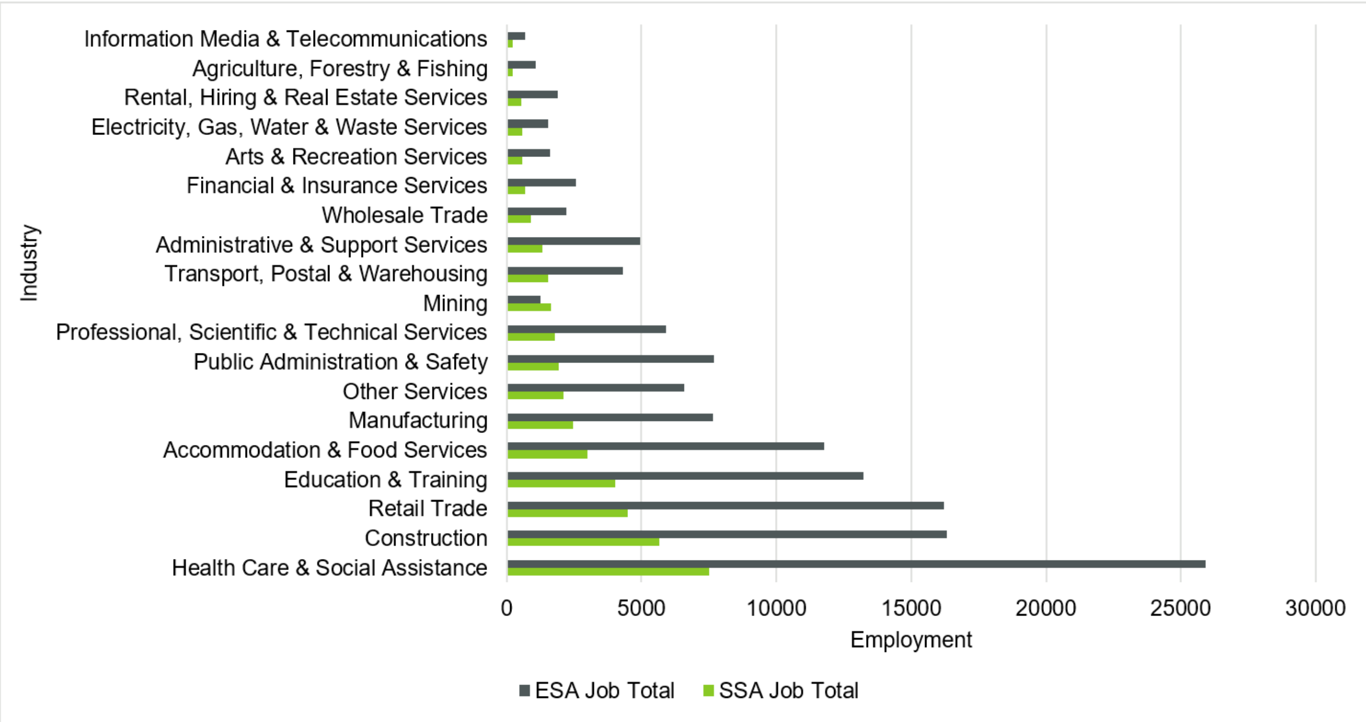

Figure 8-12: Total Employment within the ESA and SSA[145]

[144] REMPLAN (2025) Lake Macquarie LGA (2025 R1) Location Quotients.

[145] Australian Bureau of Statistics (ABS) (2021) Industry of employment (INDP).

8.7.3 Lake Macquarie employment projection

The number of jobs in Lake Macquarie LGA is expected to increase to 102,643 in 2041 from 88,852 in 2026[146], representing a 16% increase. The industries driving the expected increase in employment are Agriculture, Forestry and Fishing (81% increase), Information Media and Telecommunications (34%) and Wholesale Trade (33%). The growth in these sectors signals diversification of the regional economy, presenting new transition pathways, although their capacity to absorb mining workers will depend on skills alignment and scale.

Figure 8-13 also shows an expected 40% decline in mining employment. The decline in mining is influenced by various factors, including mine closures, which result in employees travelling outside the Lake Macquarie LGA to seek similar jobs. This is further explored in Section 8.7.1.[147]

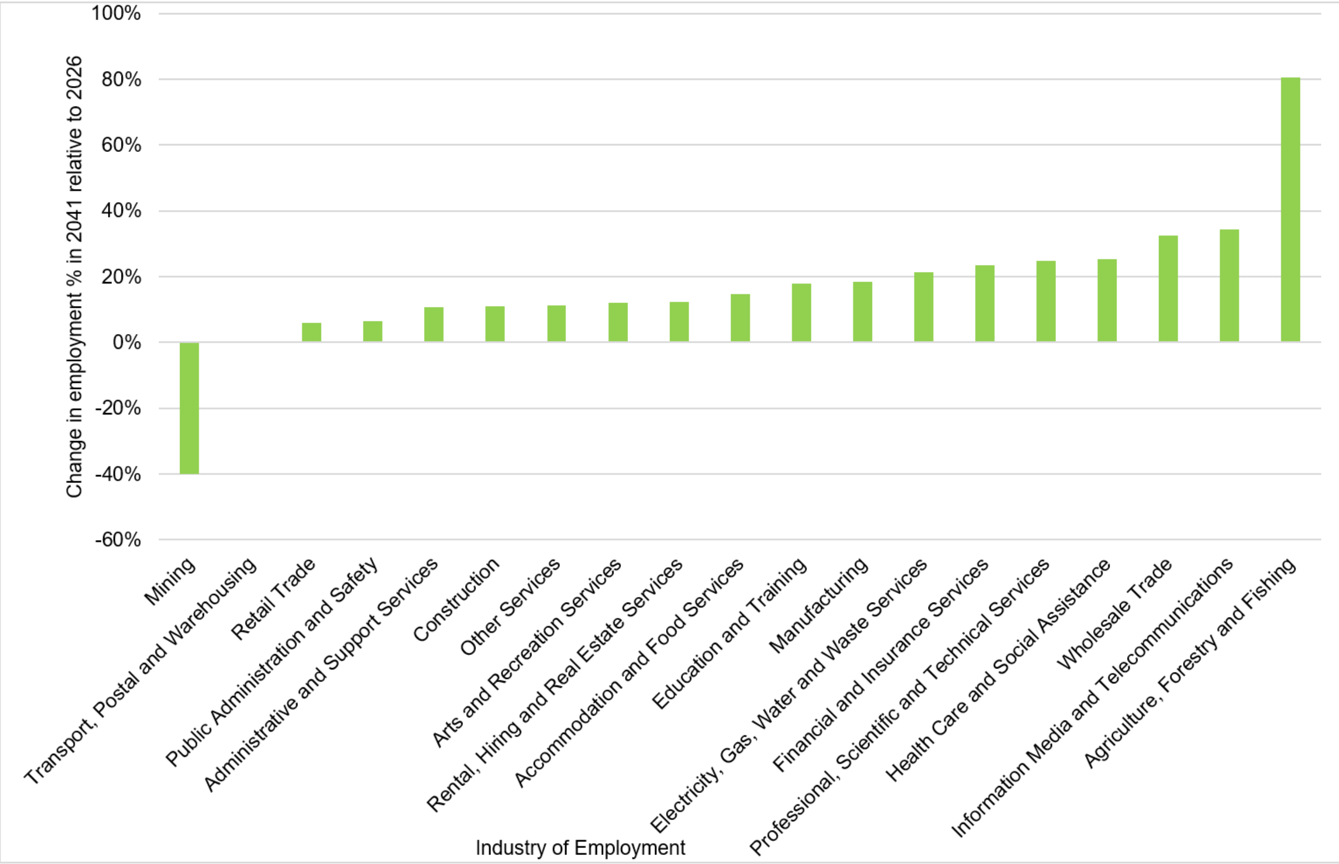

Figure 8-13: Projected change in Industry Employment in 2041 relative to 2026 across Lake Macquarie LGA[148]

Aligning to the expected employment growth in the region, Lake Macquarie City Council's North West Catalyst Area Place Strategy[149] indicates the Cardiff Advanced Industry Precinct is expected to supply 6,337 jobs to the area by 2041, with a mix of industrial, manufacturing, professional and construction-based employment projected. The shift away from mining opportunities in the area will allow Lake Macquarie to grow other industries but will require the necessary and enabling infrastructure to help support the transition, including ICT-related infrastructure, and upgrades to water supply and transport networks.[150]

8.8 Coal mining workforce profile

Mining employment in Lake Macquarie LGA, including Coal Mining, increased (+13%) over the five years to 2021 largely due to the ramp-up of existing operations rather than industry expansion. As these operational effects unwind and no

[146] NSW Government (2025) Employment projections - TZP24 Employment by industry and travel zone 2021-2066.

[147] The University of Newcastle (2023) Responding to Structural Change in Lake Macquarie - Briefing Paper for Policy Makers.

[148] NSW Government (2025) Employment projections - TZP24 Employment by industry and travel zone 2021-2066.

[149] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

[150] Ibid.

material new capacity is added, mining employment is expected to continue its structural decline. Consequently, Mining industry employment share within the Hunter region is forecasted to decline 3% from 2025 to 2050, with mine closures aligning with fossil fuel power plant closures.[151]

As outlined in Figure 8-14, growth in mining employment among Lake Macquarie residents has been sustained by employment in operations located outside the LGA. Approximately 45.96% of mining workers residing in Lake Macquarie travel outside the area for work, while 23.64% of mining workers employed in the region live elsewhere, reflecting strong cross-regional labour linkages.

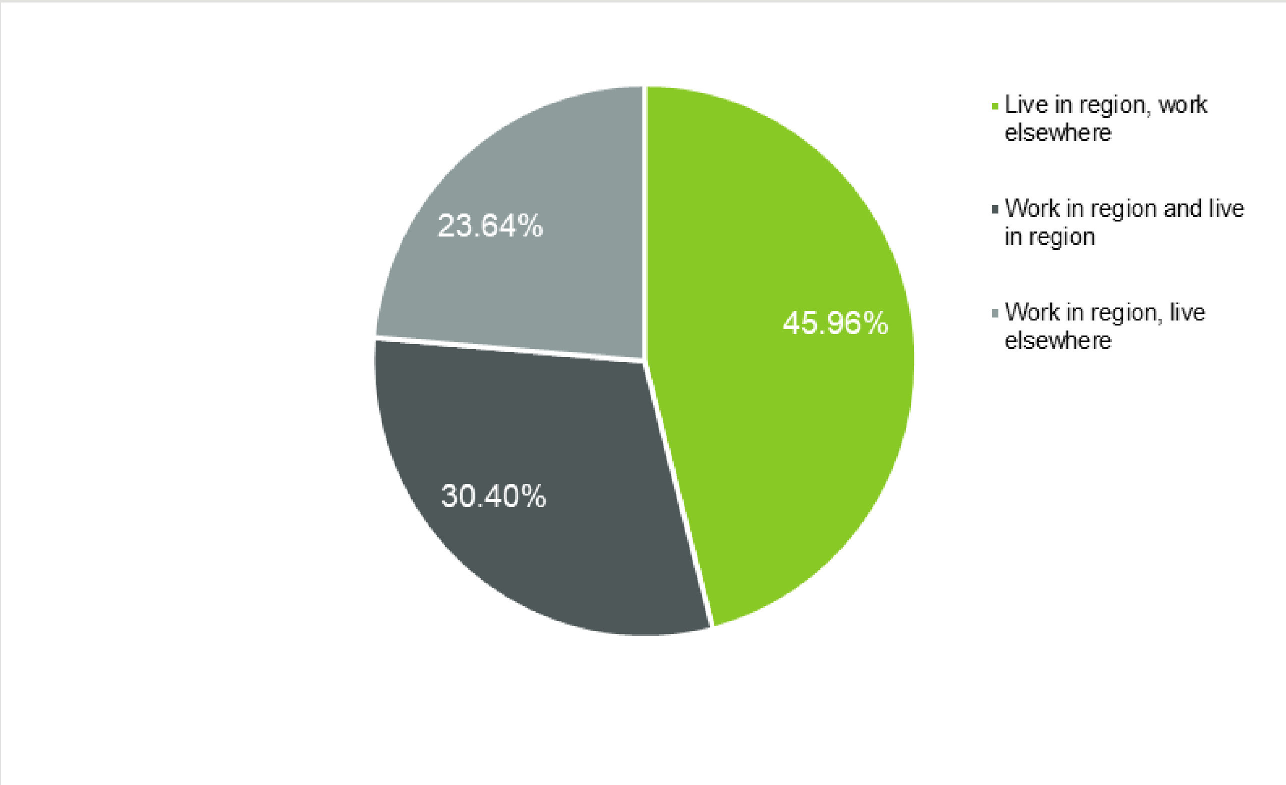

Figure 8-14: Work and residence characteristics of mining workforce in Lake Macquarie LGA[152]

8.8.1 Coal mining dependent employment profile

Figure 8-15 shows that the coal mining workforce is concentrated in the 30-49 age group, consistent with broader mining industry trends. This indicates a substantial cohort of experienced workers who will be transitioning out of the sector over the medium term, alongside an older group approaching retirement. Given the data reflects 2021 conditions, a portion of this workforce is likely already in transition or nearing exit.

This presents a critical window for workforce transition, requiring targeted upskilling and redeployment pathways to retain skills within the region and support emerging industries such as advanced manufacturing, energy, and infrastructure.

[151] Oxford Economics Australia (2025) Regional Economic Transition Analysis - Economic Outlook for Hunter.

[152] REMPLAN (2025) Lake Macquarie LGA (2025 R1) Workforce breakdown – Occupation.

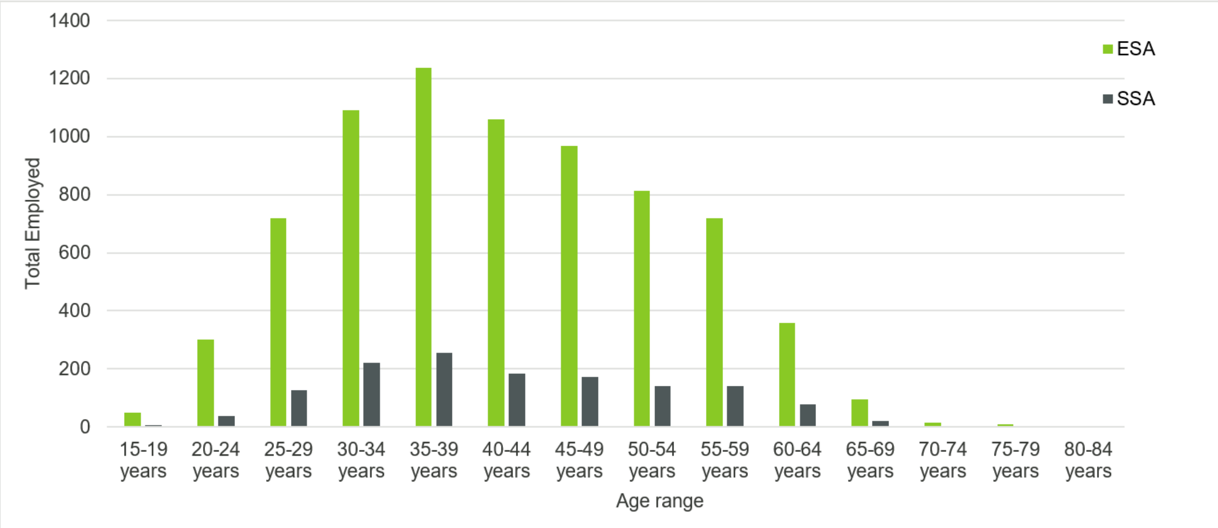

Figure 8-15: Number of coal mining dependent sector workers[153,154,155] distributed by age for ESA and SSA

Table 8-12 shows that Fossil Fuel Electricity Generation experienced a significant decline across both study areas, decreasing by 46% in the SSA and 39% in the ESA between 2016 and 2021, reflecting the broader structural transition away from coal-based energy generation.

While Coal Mining employment remained relatively stable, growing modestly in both the SSA (+10%) and ESA (+8%), several associated industries show mixed trends. Mining and Construction Machinery Manufacturing increased in the SSA (+17%) but declined in the ESA (-7%), likely reflecting a ramp up of activities in Lake Macquarie but reduced demand in the ESA following the mining boom and broader industrial restructuring.

In contrast, Structural Steel Fabricating recorded notable growth in the ESA (+12%) despite a decline in the SSA (-16%), suggesting demand linked to nearby industrial activity, including projects within the Newcastle precinct.

Table 8-12: Number of jobs in the specified industry and change over time for SSA and ESA[156]

Total employment Indicator	SSA Jobs (2021)	ESA Jobs (2021)	SSA (2016-2021) %	ESA (2016-2021) %
Coal Mining	1,223	6,712	10%	8%
Road Freight Transport	600	2,863	-0.33%	1%
Rail Freight Transport	176	621	-39%	-7%
Mining and Construction Machinery Manufacturing	115	366	17%	-7%
Fossil Fuel Electricity Generation	110	414	-46%	-39%
Port and Water Transport Terminal Operations	81	365	-20%	3%
Exploration and Other Mining Support Services	65	342	-34%	-13%
Structural Steel Fabricating	62	331	-16%	12%
Total	**2,432**	**12,014**	**-1.4%**	**4.3%**

Coal mining dependent workforce occupational concentration

The ESA's coal mining-dependent workforce is heavily concentrated in Machinery Operators and Drivers (51%), Technicians and Trades Workers (29%), and Managers (9%) (see Table 8-13 below). The modest growth in Coal Mining

[153] Selected coal dependent sector in this analysis consists of the following industry classes: Coal Mining, Exploration and Other Mining Support Services, Fossil Fuel Electricity Generation.

[154] Australian Bureau of Statistics (2016, 2021) Age in five year groups (AGE5P).

[155] Australian Bureau of Statistics (2016, 2021) Industry of employment (INDP).

[156] Ibid.

employment between 2016 and 2021 is likely associated with post-closure activities, including rehabilitation, care and maintenance, and a higher share of professional and managerial roles. Compared to other coal-dependent sectors, Road Freight Transport demonstrates a relatively higher demand for managerial roles (25%).

This profile indicates a workforce with strong operational and supervisory capabilities, which can be effectively redeployed into aligned sectors such as construction, advanced manufacturing, logistics, and energy infrastructure.

Table 8-13: Occupational concentration within coal mining dependent workforce in the ESA (2021)[157][158]

Occupational concentration	Machinery Operators and Drivers	Technicians and Trades Workers	Managers	Professionals	Laborers	Total
	Jobs					
Coal Mining	2,825	2,196	435	535	157	**6,148**
Road Freight Transport	1,877	108	247	33	101	**2,366**
Rail Freight Transport	399	74	41	32	17	**563**
Exploration and Other Mining Support Services	182	170	53	61	28	**494**
Port and Water Transport Terminal Operations	67	78	57	65	22	**289**
Mining and Construction Machinery Manufacturing	43	165	63	34	18	**323**
Structural Steel Fabricating	16	213	54	40	27	**350**
Fossil Fuel Electricity Generation	29	106	39	10	14	**198**
Total	**5,438**	**3,110**	**989**	**810**	**384**	**10,731**
Share of total jobs	51%	29%	9%	8%	4%	100%

8.8.2 Coal mining dependent workforce transition

This analysis identified approximately 12,000 coal mining dependent employments within the ESA as shown in Table 8-12. Coal mining employment trends across the SSA and ESA highlight a clear structural transition underway, driven by the decline of fossil fuel industries and the gradual diversification of the regional economy. As discussed in Section 8.7.1, Fossil Fuel Electricity Generation jobs declined significantly across both the SSA and ESA in the five years to 2021, reflecting the closure of coal-fired power stations and the broader shift toward renewable energy. While Coal Mining employment has remained relatively stable, this stability masks longer-term transition pressures, particularly given the workforce age profile and exposure to future industry contraction. The demand for fossil fuel workers in the Hunter region is projected to decline by 72% by 2035.[159]

At the same time, employment growth is emerging in sectors such as Health Care and Social Assistance, Agriculture, Forestry and Fishing, Construction, Professional, Scientific and Technical Services. These industries reflect broader economic diversification, with growth concentrated in services, construction, and knowledge-based activities.

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[157] Ibid.

[158] Australian Bureau of Statistics (2021) Occupation (OCCP).

[159] Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

Key transition dynamics

Despite growth in these sectors, several structural challenges remain:

- **Skills mismatch:** The coal mining workforce is heavily skills-based (trade, machinery, operations), which do not directly align with all growth sectors, particularly professional services.
- **Absorptive capacity:** Emerging industries may not be able to absorb the full scale of workers transitioning from mining and energy.
- **Regional labour mobility:** Nearly half of Lake Macquarie residents working in the mining industry travel outside the LGA for employment, reducing the immediate local impact of transition but also limiting opportunities for local workforce retention.
- **Low unemployment baseline:** With unemployment remaining relatively low, workforce transition is less about job scarcity and more about redeployment and skills alignment. However, the planned closure of operations at Eraring and Vales Point power stations, together with the end of the Myuna Colliery coal supply contract, presents a timely opportunity to support targeted workforce transition and reskilling initiatives.

Implications for workforce transition

The data suggests that workforce transition in the region will be gradual rather than abrupt but requires proactive planning to avoid long-term displacement.

Key implications include:

- **Targeted redeployment pathways:** Transition opportunities are strongest into physically aligned sectors such as construction, advanced manufacturing, logistics, and energy infrastructure.
- **Upskilling and reskilling:** There is a need to bridge skill gaps, particularly for higher-value sectors such as professional and technical services.
- **Alignment with emerging industries:** Growth in energy transition infrastructure, advanced manufacturing, and circular economy industries presents a strategic opportunity to retain and redeploy skills locally.
- **Phased transition approach:** Given the ageing workforce and ongoing industry contraction, transition programs should be staged over time rather than treated as a single intervention.

While the region is already diversifying, workforce transition will not occur automatically. The challenge is not only about job creation but ensuring that new employment pathways align with the existing skills base, scale of workforce change, and regional economic direction. Strategic intervention will be required to capture this transition opportunity and support a resilient, future-focused labour market.

Spatial distribution of mining workforce

The spatial distribution of mining workers demonstrates clear concentrations in the northern area (Wallsend-Barnsley-Killingworth SA2) and the western corridor (Morisset-Cooranbong urban centre) of the SSA, indicating that the mining workforce is geographically clustered rather than evenly dispersed. These concentrations represent key residential labour catchments linked to mining employment and highlight that workforce transition will have uneven, place-based impacts across Lake Macquarie.

This reinforces the importance of aligning future employment opportunities, transport connections, and training pathways with these areas to support effective workforce redeployment and minimise localised disruption.

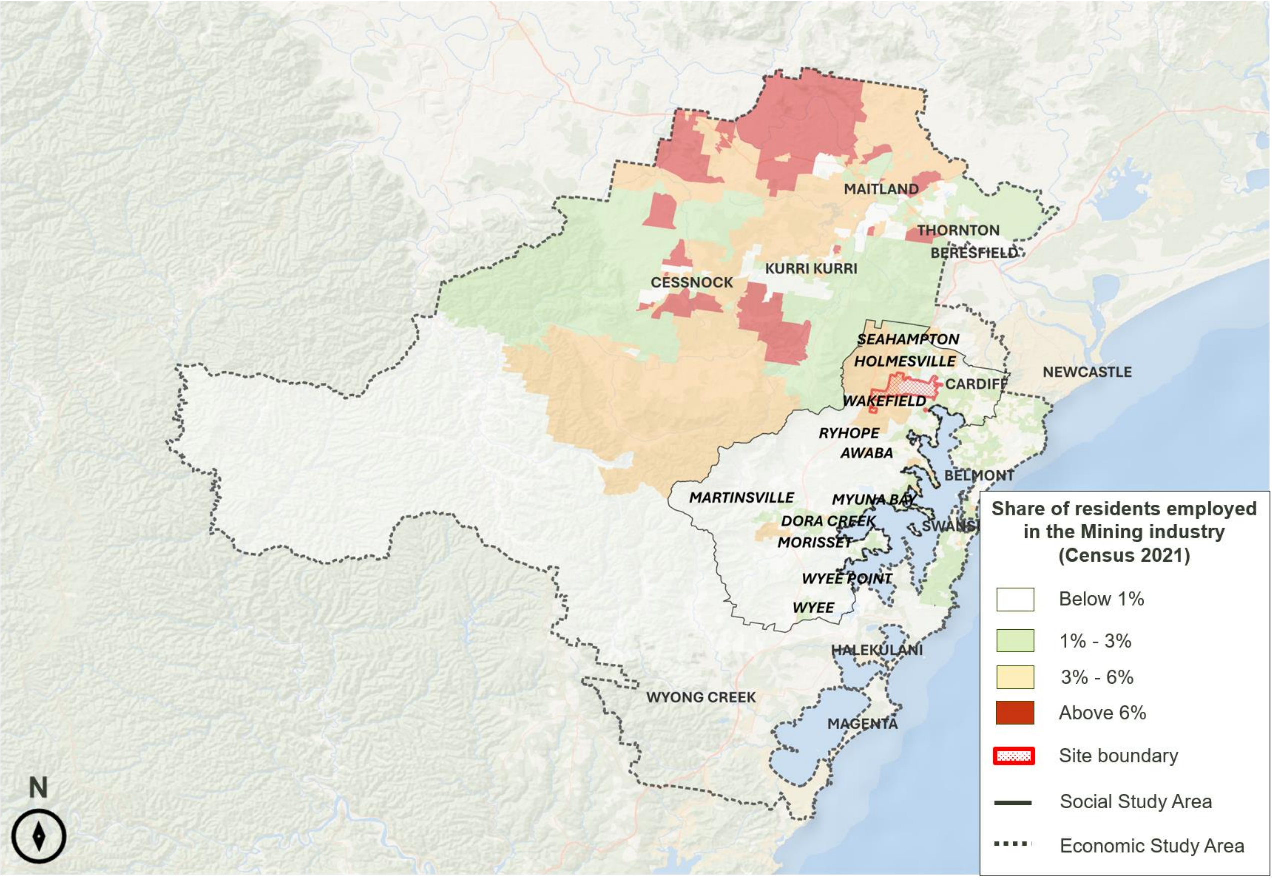

Figure 8-16: Spatial distribution of mining workers in the ESA[160]

Figure 8-16 also shows that the highest mining workforce concentrations extend beyond the immediate Site into the broader Hunter labour catchment. Areas to the north of the ESA, including parts of Cessnock, Kurri Kurri, Abermain, Weston, Branxton and Mulbring, show higher shares of residents employed in mining. This reflects the wider regional geography of mining employment, where workers may live in established mining or mining-adjacent communities while commuting to employment locations across the Hunter.

[160] Australian Bureau of Statistics (2021) Industry of employment (INDP).

9 Appendices

Appendix A: Analysis of economic contributions of land uses

This Appendix outlines the methodology underlying employment generation across the report. Gross Floor Area (GFA) represents the operational or production area, having been determined using satellite imagery. Employment generation is calculated through the NSW Department Planning and Environment's 2023 CPA Workspace ratios. The **low** scenario assumes smaller scale facilities based on Site and GFA, while the **high** scenario assumes larger scale facilities. These scenarios drive labour requirements based on GFA or occupancy. Value-add provides indicative economic productivity for each facility, allowing for comparison and identification of higher value-add themes.

Table 9-1: 2023 CPAG workspace ratios and guidance[161]

Land Use	CPAG Land Classification	CPAG Workplace Ratio (per FTE job position)
Circular Economy, Industrial and Advanced Manufacturing	Traditional Manufacturing	100m^2 GFA
Waste and Resource Recovery	Traditional Manufacturing	100m^2 GFA
Data Centres and Digital Infrastructure	Data Centres	200m^2 GFA[162]
Energy Transition and Renewable Infrastructure	Advanced Manufacturing	200m^2 GFA
Freight Logistics, Distribution and Supply Chain Hub	Warehouse, transport, and storage	150m^2 GFA
Community, Tourism and Agriculture[163]	Advanced Manufacturing	200m^2 GFA
Residential[164]	Aged Care	2.25 residents

Direct operational impacts of Circular Economy, Industrial and Advanced Manufacturing land uses

Table 9-2: Annual economic contribution from circular economy and advanced manufacturing uses

Opportunity theme	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Output ($M)	Value-add ($M)
Modular construction & prefabrication[165]	Low	Hunter Valley Modular Homes	2,100	21	$16	$8
	High	BlueSky Modular Buildings	3,400	34	$26	$14
Structural steel & metal fabrication	Low	Coastal Steel Fabrications	2,840	14	$11	$6
	High	Cullen Steel Fabrications PTY Ltd.	6,600	33	$25	$13

[161] Department of Planning and Environment (2023) 2023 Common Planning Assumptions Workspace Ratios.

[162] Hyperscale data centre jobs determined based on capacity: 0.625 jobs generated per MW of capacity.

[163] Agriculture classified as advanced manufacturing due to absence of specific CPAG land use guidelines for agriculture.

[164] Assisted living job generation calculated based on resident occupancy (2.25 residents).

[165] Modular construction follows a traditional manufacturing CPA Land Classification, other opportunity themes within Industrial and Advanced Manufacturing align with advanced manufacturing classifications.

Opportunity theme	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Output ($M)	Value-add ($M)
Precision engineering & advanced manufacturing	Low	MTECH Engineering – CNC & Wire EDM Machining Services	500	3	$2	$1
	High	Performance Engineering Group Australia Pty Ltd	4,900	25	$19	$10
Renewable energy equipment manufacturing	Low	Precision Oxycut	2,800	14	$11	$6
	High	Vesta Turbine Manufacturing Geelong	3,000	15	$11	$6
Critical minerals and battery material recovery	Low	Ecocycle	1,985	10	$8	$4
	High	Livium Chemicals	7,180	36	$27	$14
Advanced materials & composites	Low	World Class Composites Pty Ltd	2,500	13	$10	$5
	High	Composite Materials Engineering	5,500	28	$21	$11
Integrated circular economy precinct	Low	Cleanaway Kemps Creek Resource Recovery Park	650	7	$5	$3
	High	iQRenew	3,440	34	$26	$14

Direct operational impacts of Waste and Resource Recovery land uses

Table 9-3: Estimated direct economic contribution per annum from waste and resource recovery land use

Opportunity theme	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Output ($M)	Value-add ($M)
Construction and demolition waste recovery hub	Low	Port Phillip Resource Recovery Centre	475	5	$4	$2
	High	Solo Resource Recovery	4,995	50	$38	$20
Mixed waste transfer and sorting facility	Low	Wanless Waste Management	960	10	$7	$4
	High	Veolia Hunter	9,790	98	$74	$39
Battery and e-waste recycling centre	Low	eWaste Sydney	873	7	$7	$4
	High	Beyond eWaste	5,063	51	$38	$20

Direct operational impacts of Energy Transition and Renewable Infrastructure land uses

Table 9-4: Annual economic contribution from energy transition infrastructure

Opportunity theme	Scenario	Comparable facility	MW capacity	Employment generation (jobs)	Economic output ($M)	Value-add ($M)
Battery storage (BESS)	Low	Rangebank BESS	200MW / 400MWh	3	$2.3	$1.2
	Medium	Tomago Battery	500MW / 2,000MWh	7	$5.3	$2.8
	High	Eraring BESS	700MW/ 2,800MWh	11	$8.3	$4.4

Direct operational impacts of Data Centres and Digital Infrastructure land uses

Table 9-5: Direct operational impacts of data centres and digital infrastructure

Opportunity theme	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Output ($M)	Value-add ($M)
Hyperscale/ large-scale data centre (long-term)[166,167]	Low	Equinix Data Centre	11,900	9	$7	$3
	High	Macquarie Data Centres - Macquarie Park Data Centre Campus	31,300	41	$31	$16
Regional data centre/ edge computing facility	Low	Leading Edge Data Centres Newcastle	430	2	$2	$1
	High	EC8 Traralgon Edge Data Centre	565	3	$2	$1
Digital services/ technology support hub	Low	SMIKTECK Business IT Support	310	2	$1	$0.6
	High	DYN I.T. Solutions	650	3	$2	$1

Direct operational impacts of Freight Logistics, Distribution and Supply Chain Hub land uses

Table 9-6: Annual economic contribution from freight, logistics and supply chain uses

Opportunity theme	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Economic output ($M)	Value-add ($M)
Cold chain & perishables logistics	Low	Newcastle Cold Storage	2,800	7	$5	$3
	High	Nationwide Cold Storage & Distribution	8,744	87	$66	$35
Regional distribution/ e-commerce hub	Low	Myer Eastern Creek Distribution	27,737	139	$105	$55
	High	Woolworths Regional Distribution Centre (RDC)	75,000	375	$284	$150

[166] Hyperscale data centres typically incorporate artificial intelligence (AI) and high-performance computing (HPC) facilities within their overall scope.

[167] Hyperscale data centre jobs determined based on capacity: 0.625 jobs generated per MW of capacity.

Direct operational impacts of Community, Tourism and Agriculture land uses

Table 9-7: Estimated direct economic contribution per annum from community, tourism and agriculture land use

Opportunity theme	Category	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Economic output ($M)	Value-add ($M)
Controlled environment agriculture[168]	Intensive horticulture[169]	Low	Newcastle Greens	2,050	21	$16	$8
		High	Pine Crest Orchard	26,125	130	$198	$105
	Controlled environment agriculture[170]	Low	Vertical Patch	2,250	23	$17	$9
		High	InvertiGro	3,250	33	$25	$13

Direct operational impacts of Residential land uses

Table 9-8: Estimated direct economic contribution per annum from residential land use

Opportunity theme	Category	Scenario	Comparable facility	Gross floor area (m^2)	Employment generation (jobs)	Economic output ($M)	Value-add ($M)
Senior living	Aged Care[171]	Low	Hunter Valley Care	2790	27	$20	$11
		High	Amaroo Aged Care Facility	3,250	45	$34	$18

[168] Square meter of greenhouse space.

[169] Greenhouse farming.

[170] Vertical hydroponic farming.

[171] Assisted living job generation calculated based on resident occupancy (2.25 residents).

Appendix B: Consultations with NZEA

Government priorities land use tier system & strategic dependencies

The land use tier system in this Appendix categorises land use opportunities based on their alignment with government priorities. It provides a basis for prioritising investment and policy focus for the Site by highlighting where opportunities are most viable, conditional, or suited for long-term consideration based known government support. This Section was developed based on consultations with NZEA.

Framework principles for the government priorities land use tier system are below:

Tier 1	**Near-term actionable (18 months-5 years)** Government action within 18-24 months could support new projects and investment within 2-3 years. Sectors are investment-ready with minimal preconditions.
Tier 2	**Conditional medium to longer-term (5-10+ years)** Strategic relevance exists but depends on major government, infrastructure, procurement, or industry decisions. Sectors require major infrastructure, policy, or procurement alignment before viability.
Tier 3	**Supporting uses** Possible supporting roles but not proven as near-term anchors. Sectors may support primary uses but require stronger market and occupier evidence.
Tier 4:	**Unlikely** Sectors are deprioritised unless integrated into higher-tier frameworks or a confirmed

Figure 9-1: Framework principles for the government priorities land use tier system

Government priority sectors outlined in Table 9-9 are not automatically suitable for the Site. Each opportunity needs to be tested against site suitability, infrastructure readiness, workforce capability, investment appetite, timing and planning / rehabilitation constraints.

Table 9-9: Land uses tier system based on government priorities

Sector	Strategic positioning	Key dependencies
Tier 1		
Renewable Energy Technology	Strong strategic alignment	Site, infrastructure, and workforce fit
Circular Economy & Recycling	Strong candidate for near-term/medium-term development	Market maturity and operational feasibility
Agri-Food & Food Manufacturing	Potentially relevant	Utilities, servicing, supply chain, and site suitability
Tier 2		
Defence & Aerospace	Strategic priority; anchor-dependent	Confirmed anchor demand; institutional alignment; supply chain evidence
Green Metals	Strategically relevant to Hunter transition	Wider industrial decisions; Tomago capability alignment; offtake agreements
Green Hydrogen	Supported by Commonwealth budget; Port of Newcastle infrastructure alignment	Infrastructure maturity; offtake; water/energy availability; market readiness; policy certainty

Sector	Strategic positioning	Key dependencies
Rail & Rolling Stock	Conditional on procurement	NSW Government and High-Speed Rail procurement decisions
Tier 3		
Freight & Logistics	Possible supporting use; not a near-term anchor	Transport access; rail/road feasibility; Port of Newcastle container/freight outcomes
General Equipment & Machinery	Possible broader industrial use	Stronger market evidence; occupier demand
Health Manufacturing & Other Niche Manufacturing	Niche opportunity	Confirmed proponent; market evidence
Tier 4		
Biofuels & Biorefining	Consider within circular economy, agri-food, or renewable technology frameworks	Confirmed proponent for the Site; integration with broader Hunter region activity

Appendix C: Media scan

Across the articles, there is strong community support for transitioning away from coal, with a clear focus on repurposing mine land for productive uses that create jobs and long-term economic value. However, there is consistent frustration about delays, lack of delivery, and concern that planning and policy ambition are not translating into timely, on-the-ground outcomes. Communities emphasise the need for proactive, locally driven planning that prioritises restoration, future land use, and fair investment in regions that have historically supported the state's energy supply.

Table 9-10: Media scan and key points relevant to PLMU Sites throughout the SSA and ESA.

Publication, Date and Link	Headline	Key Points:
Newy.com **[link]** **20 Nov 2025**	Hunter mine land overhaul; $5m Federal boost articles; Singleton Argus; Hunter Mine Site Rehab Gets $5M Federal Boost	▪ Local councils and community leaders broadly welcome the funding, but stress it is a first step, not delivery. ▪ There is strong sentiment that communities have waited too long for proactive planning and do not want a repeat of "closed Site, idle land, and lost jobs". ▪ Communities emphasise the importance of local involvement in master planning, reflecting distrust of top-down decisions made without regional input.
Mining Weekly **[link]** **20 March 2026**	NSW Shuts door on new greenfield coal mines as transition strategy unveiled	▪ NSW's formal decision to rule out new greenfield coal mines confirms that coal is entering a managed decline, reinforcing the urgency of planning for post-mining land use and alternative industries, particularly in the Hunter. ▪ Communities are less focused on whether coal ends, and more on what replaces it. ▪ There is concern that policy ambition may outpace on-the-ground transition support, especially for land reuse, infrastructure, and skills.
Hunter News **[link]** **16 July 2025**	ERARING – Lake Macquarie Council Calls for Federal Support	▪ Lake Macquarie Council and regional MPs argue that communities hosting major coal assets (Eraring, Myuna, Wangi) have powered NSW for decades and deserve proportionate transition investment, particularly for land redevelopment and economic diversification. ▪ Anxiety about job losses compounding derelict infrastructure, especially if power station and mine Sites remain unused.
ABC News [link] **26 March 2024**	Former Lake Macquarie coal mine Site to become multi-million-dollar racetrack for car enthusiasts	▪ The Rhondda Colliery Site redevelopment into the $95 million Black Rock Motor Resort is highlighted as a rare, tangible example of a former coal mine successfully transitioning into tourism and recreation, creating hundreds of construction and ongoing jobs. ▪ Generally positive and optimistic, as the project proves reuse is possible. ▪ Some concern exists about ensuring such projects are environmentally safe and not merely "development for development's sake". ▪ Pride in seeing former mining land re-enter community and economic life rather than remaining fenced and forgotten.
ABC News [link] **2 November 2015**	Decades after closure, Wangi power station still waiting for redevelopment	▪ Wangi Power Station remains largely derelict decades after closure, symbolising the risks of delayed decision-making around post-industrial land. Multiple proposals have emerged, but none have delivered large-scale redevelopment. Often cited as a warning case of what happens when transition planning stalls.

Publication, Date and Link	Headline	Key Points:
		▪ Deep frustration and fatigue with inaction and desire for redevelopment that balances heritage preservation, public access, and economic use. ▪ Strong attachment to the Site as a heritage and identity symbol.
Mining Digital [link] **15 June 2022**	Repurposing Hunter Valley mines would trigger $3.7bn boost	▪ Reports estimate up to $3.7 billion in economic value could be generated by repurposing mine land for renewables, manufacturing, biodiversity restoration, and regenerative agriculture, especially if buffer lands are also included. ▪ Strong endorsement of "restore the land" narratives. ▪ Communities support renewable energy and clean industries, but stress the need for good-quality, long-term jobs.
The Conversation [link] **8 February 2023**	'We need to restore the land': as coal mines close, here's a community blueprint to sustain the Hunter Valley	▪ This article outlines a community-led vision emphasising large-scale ecological restoration, regenerative agriculture, clean industry, and improved liveability as mines close. ▪ Clear call for healing Country, not just compliance rehabilitation and Strong support for First Nations knowledge and leadership in post-mining land decisions. ▪ Broad consensus that what happens after coal will define the Hunter's identity for generations.
Discovery Alert [link] **21 November 2025**	Federal Government Announces $5M Hunter Mine Site Rehabilitation Funding	▪ Government funding is focused on turning former mine sites into productive land with new economic uses. ▪ Communities are encouraged that government is getting involved earlier in the transition process with strong support for planning ahead, rather than waiting until mines close. ▪ Clear community view that rehabilitation should mean restoring land and giving it a future purpose, not leaving it unused.
Newcastle Herald [link] **3 July 2025**	'We barely are surviving': business hopes councilors vote against recommendation	▪ Councillors were asked to decide whether JET Group could resume a scaled-down green waste processing operation after years of regulatory and environmental issues. Council staff recommended refusal, while the business argued the smaller operation posed minimal risk. ▪ Nearby residents and environmental advocates remain worried about odour, emissions and compliance, citing past breaches. ▪ Small business supporters express sympathy for the operator, pointing to job losses, prolonged uncertainty and perceived regulatory rigidity. ▪ The drawn-out process has fuelled frustration on all sides, with calls for either a clear path forward or a definitive end.

Appendix D: Policy and legislation scan

The policy framework collectively emphasises a coordinated shift from traditional industrial land uses toward diversified, liveable and sustainable urban environments supported by infrastructure, housing diversity, and economic transition. Across state and local strategies, there is strong alignment in directing growth to strategic centres, enhancing connectivity, and promoting new industries such as renewable energy, services, and innovation, alongside high-quality public spaces and inclusive community outcomes. These priorities are highly relevant to post-mining land use, as they support the redevelopment of former mining sites into mixed-use, employment-generating, and liveable precincts that respond to population growth, workforce transition, environmental rehabilitation, and evolving regional economic needs.

Table 9-11: State and local policy and legislation scan

Policy	Summary of key details
State	
Hunter Regional Plan 2041[172]	▪ Sets the regional planning framework for land use, housing and infrastructure. ▪ Promotes 15-minute neighbourhoods and reduced car dependence. ▪ Focuses on economic transition, renewable energy and climate resilience. ▪ Requires high-quality public spaces and better integration of density with amenity.
Greater Newcastle Metropolitan Strategy Economic Prospects to 2036[173]	▪ Shifting from a traditional industrial base (mining and manufacturing) to a diversified economy to knowledge industries, health, education, defence and innovation sectors. ▪ Expected to experience population, employment and economic growth to 2036 ▪ Employment growth will be concentrated in key strategic centres and corridors (e.g. Newcastle CBD, Cardiff–Glendale) to improve access to jobs and support more efficient land use and transport connections ▪ Achieving economic growth depends on investment in infrastructure, transport and workforce skills, ensuring the region is competitive, connected, and able to attract businesses and skilled worker
Local	
Lake Macquarie City Council - Community Strategic Plan 2025-2035[174]	▪ Establishes a 10-year vision guiding all Council decisions, investment, and partnerships across the city. ▪ Built on four pillars: economy, environment, community liveability, and strong governance. ▪ Explicitly acknowledges the economic transition away from coal and need for new industries. ▪ Acts as the top-level framework, aligning with regional/state plans and guiding all other strategies.
Local Strategic Planning Statement 2025-2045[175]	▪ Provides a 20-year land use roadmap responding to population growth, ageing and housing pressures. ▪ Directs growth toward strategic centres to improve access to jobs, services and transport. ▪ Prioritises housing diversity, affordability, and reduced urban sprawl.

[172] NSW Department of Planning and Environment (2022) Hunter Regional Plan 2041.

[173] MacroPlan Holdings Pty Ltd (2017) Greater Newcastle Metropolitan Strategy - Economic Prospects to 2036.

[174] Lake Macquarie City Council (2025) Community Strategic Plan 2025–2035.

[175] Lake Macquarie City Council (2025) Lake Macquarie Local Strategic Planning Statement 2025 – 2045.

Policy	Summary of key details
	▪ Integrates climate resilience, active transport, and infrastructure planning into land use decisions.
Lake Macquarie City Council Aboriginal Community Plan 2025–2029 (Bayikulinan)[176]	▪ A community-led plan addressing social, economic and cultural outcomes for Aboriginal people. ▪ Focuses on reducing inequality, racism, and improving culturally safe services. ▪ Promotes employment, economic participation and stronger community capacity. ▪ Embeds Aboriginal leadership and decision-making in Council planning and governance.
Night-Time Economy Action Plan 2019–2024[177]	▪ Aims to activate Lake Macquarie after dark and support a more vibrant evening economy. ▪ Expands activities across hospitality, arts, retail, health and public spaces. ▪ Focuses on safe, accessible centres with better transport and lighting. ▪ Encourages flexible uses (events, pop-ups) and regulatory reform to support businesses.
Community Engagement Strategy & Participation Plan 2024–2028[178]	▪ Sets out how Council will engage the community in planning and decision-making. ▪ Emphasises early, transparent and inclusive engagement processes. ▪ Targets underrepresented groups to ensure equitable participation. ▪ Aligns with legislation and uses best-practice engagement frameworks (IAP2).
Lake Macquarie City Housing Strategy 2021[179]	▪ Plans for significant population growth and 13,700 new dwellings by 2036. ▪ Focuses on increasing housing diversity beyond detached homes. ▪ Encourages infill housing close to jobs, centres and public transport. ▪ Supports affordability, sustainability, and improved housing design outcomes.
Lake Macquarie City Events & Festivals Strategic Action Plan 2024–2028[180]	▪ Positions events as key drivers of tourism, economic growth and city identity. ▪ Expands night-time economy, large-scale events, and water/adventure offerings. ▪ Builds on strong participation with hundreds of events and large attendance levels. ▪ Targets sustainable delivery, including zero-waste events by 2028.
Lake Macquarie City Arts, Heritage & Cultural Plan 2017–2027[181]	▪ Frames arts and culture as essential infrastructure for identity, wellbeing and economy. ▪ Supports creative industries, local artists, and cultural participation. ▪ Addresses gaps in cultural facilities and infrastructure. ▪ Emphasises Aboriginal culture, storytelling and place-based partnerships.
Lake Macquarie City Parks, Play & Public Amenities Report[182]	▪ Finds parks are valued but need improved quality, maintenance and design. ▪ Key priorities include shade, amenities, inclusive play and safety. ▪ Strong demand for nature-based, unique and locally distinctive parks. ▪ Highlights need for better walking/cycling connections and access to local parks.

[176] Lake Macquarie City Council (2025) Aboriginal Community Plan 2025-2029: Bayikulinan (to act in the future).

[177] Lake Macquarie City Council (2018) Lake Macquarie Night-Time Economy Action Plan 2019–2024.

[178] Lake Macquarie City Council (2024) Community Engagement Strategy- including Community Participation Plan 2024-2028

[179] Lake Macquarie City Council (2021) Draft Housing Strategy.

[180] Lake Macquarie City Council (2024) Events and Festivals Strategic Action Plan 2024-2028.

[181] Lake Macquarie City Council (2017) Arts, Heritage & Cultural Plan 2017–2027.

[182] Otium Planning Group (2021) Lake Macquarie City Council Public Amenities Strategy 2021.

Policy	Summary of key details
Lake Macquarie City Ageing Population Strategy 2022-2026[183]	■ Responds to rapidly ageing population and changing demographic needs. ■ Promotes age-friendly environments (walkability, seating, accessibility). ■ Supports health, social inclusion and active ageing. ■ Encourages community connection and intergenerational programs.
Lake Macquarie City Draft Housing Strategy[184]	■ Responds to housing shortage, affordability crisis and population growth. ■ Promotes greater diversity (apartments, townhouses, secondary dwellings). ■ Focuses on well-located housing near transport and services. ■ Emphasises liveability, public space quality and neighbourhood character.
Studies and surveys within the Lake Macquarie and Hunter Valley area	
100 Voices Community Summit (2024)[185]	■ Captures broad community priorities across infrastructure, housing and environment in Lake Macquarie. ■ Strong demand for better transport, connectivity and local services. ■ Highlights importance of sustainability, liveability and community wellbeing. ■ Confirms alignment between community views and Council strategies.
Lake Macquarie Housing Affordability & Liveability Study (2026)[186]	■ Identifies major housing affordability issues and supply shortfalls. ■ Highlights lack of housing diversity and reliance on detached homes. ■ Projects need for 26,000+ additional dwellings by 2046. ■ Shows growing demand for smaller, more affordable and higher-density housing.
Australian Liveability Census Lake Macquarie City Council Priorities, Values and Performance report [187]	■ Community values environment, public space quality and connectivity most. ■ Strong performance in movement and overall liveability. ■ Identifies gaps in open space quality and active transport links. ■ Calls for better tree canopy, footpaths and transport connections.
Lake Macquarie Community Satisfaction Survey (Taverner Research, June 2024)[188]	■ Shows decline in overall satisfaction, below NSW metro average. ■ High satisfaction with parks, waste services and foreshore management. ■ Low satisfaction with roads, public toilets and engagement in decisions. ■ Residents strongly value environment, liveability and quality of life.
Oxford Economics Australia: Regional Economic Transition Analysis – Worker Transitions in the Hunter - Dec 2025[189]	■ Highlights sharp decline in coal workforce by 2035. ■ Workforce has strong trade, technical and industrial skills. ■ Emphasises need for like-for-like jobs (pay, stability, full-time work). ■ Identifies opportunities in energy, manufacturing, and large-scale land reuse projects.

183 Lake Macquarie City Council (2022) Ageing Population Strategy 2022 - 2026.

184 Lake Macquarie City Council (2026) Draft Housing Strategy.

185 Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

186 SGS Economics & Planning (2026) Lake Macquarie Housing Affordability and Liveability Study: Interim Housing Demand and Affordability Report.

187 Place Score (2023) 2023 Australian Liveability Census Lake Macquarie City Council - Priorities, Values and Performance Report

188 Taverner Research Group (2024) Community Satisfaction Survey 2024.

189 Oxford Economics Australia (2025) Regional Economic Transition Analysis – Hunter Region.

Appendix E: PTAL methodology

PTAL is a spatial measure of how easily people can access public transport from a given location. At a high level, it reflects:

- distance to public transport stops or stations
- frequency of services
- number of available routes
- ease of access to the network
- relative accessibility compared with other locations.[190]

PTAL does not measure whether public transport is good for every possible trip. It measures the availability and intensity of public transport access near a location. A place can have a higher PTAL if it is close to frequent bus, rail or ferry services. A place can have a low PTAL if services are infrequent, distant, or limited in route choice.

For this analysis, PTAL has been assessed using the morning peak period timetable (8AM -9AM), which typically represents the period of highest public transport service frequency. Public transport availability outside peak periods is likely to be lower than shown in the PTAL output, which is particularly relevant for lower-income households, shift workers and residents travelling outside standard commuter hours. As a result, the accessibility identified in this analysis may understate the public transport barriers experienced by some users.

[190] Transport for NSW (2025) Public Transport Accessibility Level (PTAL) Fact sheet.

Appendix F: Detailed PTAL of social infrastructure facilities in the SSA

PTAL category[191]	Number of facilities	Share of all SSA facilities	Accessibility interpretation
No PTAL score	95	23%	Not accessible by public transport
Low	138	34%	Poor public transport accessibility
Low Medium	72	18%	Limited public transport accessibility
Medium	53	13%	Moderate public transport accessibility
Medium High	24	6%	Good public transport accessibility
High	18	4%	Strong public transport accessibility
Very High	10	2%	Very strong public transport accessibility
Total	**410**	**100%**	

[191] Transport for NSW (2025) Public Transport Accessibility Level (PTAL) Fact sheet.

Appendix G: Data sources

The assessment draws on a combination of quantitative datasets, policy documents, existing studies, stakeholder input and desktop research. The analysis has been prepared with reference to technical inputs from other project packages available at the time of writing. Table 9-12 lists key data sources used in this assessment.

Table 9-12: Data sources for the economic, social and market assessment

Source	Purpose
ABS Census 2021	Demographic profile, household characteristics, income, housing, workforce, education, occupation, travel to work and Aboriginal and Torres Strait Islander population indicators
ABS Social Economic Indexes for Areas (SEIFA)	Socio economic advantage and disadvantage indicators
REMPLAN	Economic structure, industry output, value added, employment and regional economic indicators
Transport Travel Zone projections	Employment projections by industry and place of work
NSW Government and regional strategies	Strategic context, regional priorities, economic transition, planning policy and infrastructure context
Lake Macquarie City Council strategies and reports	Local planning context, community priorities, housing, ageing, liveability, employment land and social infrastructure considerations
Net Zero Economy Authority and workforce transition material	Regional transition context, fossil fuel workforce exposure and future industry priorities
Oxford Economics regional transition analysis	Hunter economic outlook, worker transition analysis, regional investment context and industry trends
Aurecon and Patch Deliverability Due Diligence Report	Site context, development potential, constraints and infrastructure related considerations
Enquiry by Design inputs and outputs	Land use directions, emerging master plan ideas and issues raised through the workshop process
Targeted stakeholder discussions	Market appetite, timing, investment conditions, infrastructure dependencies and delivery risks
Desktop market research	Comparable projects, major investments, industry trends and relevant market signals

Appendix H: Socio-economic vulnerability indicators

Age related vulnerability

Age-related vulnerability is an important contextual factor. Approximately 22% of the SSA population (27,033 residents) and 19% of the ESA (94,351 residents) area aged 65 years and over, indicating a higher proportion of older population that is more likely to experience health, mobility and accessibility constraints. This is higher compared to the NSW population totalling 18% (1,424,141 residents), who are aged 65 years and over. With the aging population set to continue growing, the need for age-friendly environment and housing also increases.

There is a slightly higher proportion of the population of children and young people aged 0-14 years in the SSA with 20% (22,431 residents) compared to 19% in the ESA (94,206 residents) and 18% in NSW (1,470,006 residents) reflecting an ongoing reliance on local education, recreation and family support services

Together, these juxtaposing age groups represent around 44% of the SSA population (49,464 residents) and 39% of the ESA population (188,557 residents) underscoring the importance of stable, low-impact land uses supporting community liveability and access to local services. This is reflected in the Master Plan performance criteria for community facilities and social infrastructure and sub-precinct interface considerations.

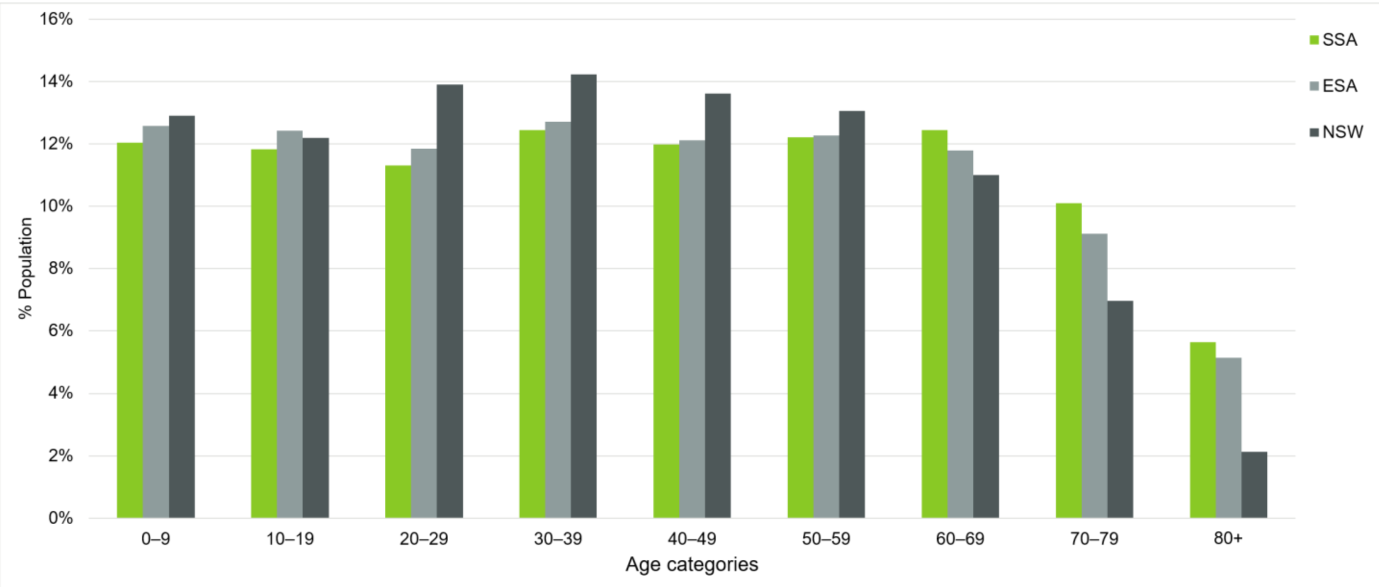

Figure 9-2: Population age-distribution for SSA, ESA and NSW[192]

Health related vulnerability

Health-related vulnerability is a defining characteristic of the SSA, closely linked to aging population statistics. In the SSA approximately 36% of residents (44,970 residents), report a long-term health condition, compared to 31% (170,713 residents) in the ESA, significantly more than NSW (27% of residents). This prevalence increases markedly in older age groups, with a high number of residents aged 65 years and over reporting chronic conditions. Additionally, approximately 7% of residents in the SSA (9,268 residents), 7% in the ESA (36,092 residents) and 6% in NSW (464,712 residents) report a need for assistance with core activities. These is akin to concerns regarding chronic health conditions, aging population and disability intersect with the location of disadvantage within the SSA.[193]

This aligns with wider liveability findings that highlight health and age-based differences as key determinants of community outcomes, with residents within Lake Macquarie reporting positive health overall, but significant variation across cohorts, particularly older populations.[194]

[192] Australian Bureau of Statistics (ABS) (2021) Age in ten year groups (AGE10P).

[193] Spectrum Comms (2024) 100 Voices Community Summit: Engagement Summary Report.

[194] Place Score 2025 (2025) Australian Liveability Census: Executive Summary – Lake Macquarie City Council

Income related vulnerability

Compared to NSW, the SSA and ESA have less lower income earners and more middle-income earners (See Figure 9-3). However, income vulnerability is observed indirectly through housing stress and employment access, rather than through income figures themselves. This vulnerability is heightened by the ESA's exposure to carbon-intensive industries and planned coal-fired power station closures, which may concentrate transition impacts in particular communities and supply chains.[195]

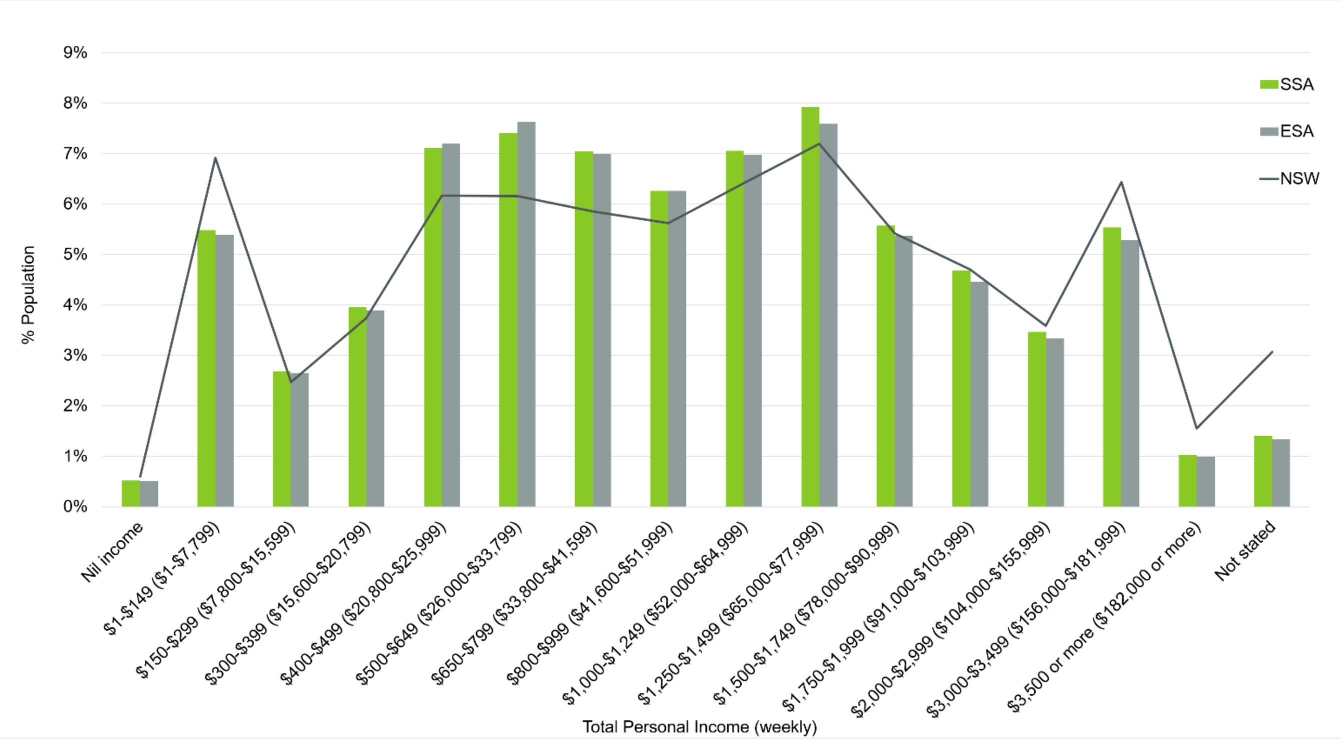

Figure 9-3: Total income weekly across the SSA, ESA and NSW[196]

Housing related vulnerability

Housing characteristics further shape sensitivity. Home ownership outright is common in the SSA (18%) and ESA (17%) however is below the NSW state average (32%). Less people rent in the SSA (12%) and ESA (14%) compared to NSW (33%).

However, large parts of Lake Macquarie are classified as unaffordable, severely unaffordable or extremely unaffordable for lower-income households, including single pensioners and single-parent families.[197] Housing affordability throughout the SSA is considered a major and unresolved challenge with community highlighting issues with stress, rising costs, and structural barriers within the housing system.[198]

Cultural and linguistic diversity related vulnerability

Aboriginal and Torres Strait Islander people comprise around 6% of the SSA population (7,359 residents) and 7% in the ESA (32,733 residents), which is higher than the NSW state residents who make up 3% (278,043 residents) of the population.

The SSA has a largely homogenous language groups with only 4% of the population (4,706 residents) speaking a language other than English at home in the SSA and 4% in the ESA (20,553 residents). There is high English proficiency in the SSA, with 0.4% (466 residents) speaking English not well or not at all and 0.5% in the ESA (2,380 residents). This

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[195] Net Zero Economy Authority (2025) Regional Profiles: Hunter.

[196] Australian Bureau of Statistics (ABS) (2021) Total personal income (weekly) (INCP)

[197] SGS Economics & Planning (2026) Lake Macquarie Housing Affordability and Liveability Study: Interim Housing Demand and Affordability Report.

[198] Lake Macquarie City Council (2025) Lake Macquarie Housing Forum Detailed Outcomes Report.

is lower than the NSW state population with 20% of the population (1,573,216 residents) speaking a language other than English at home and 4% (302,812 residents) speaking English not well or not at all.

Socio-economic profile

The Socio-Economic Indexes for Areas (SEIFA) is a measure used to rank geographic areas in Australia according to levels of socio-economic advantage and disadvantage[199]. SEIFA draws on data relating to income, education, employment, occupation, housing and family structure to describe area-level socio-economic conditions. Figure 9-4 reflects the total SEIFA indicators throughout the SSA in which the higher the vulnerability score the more advantaged the area.

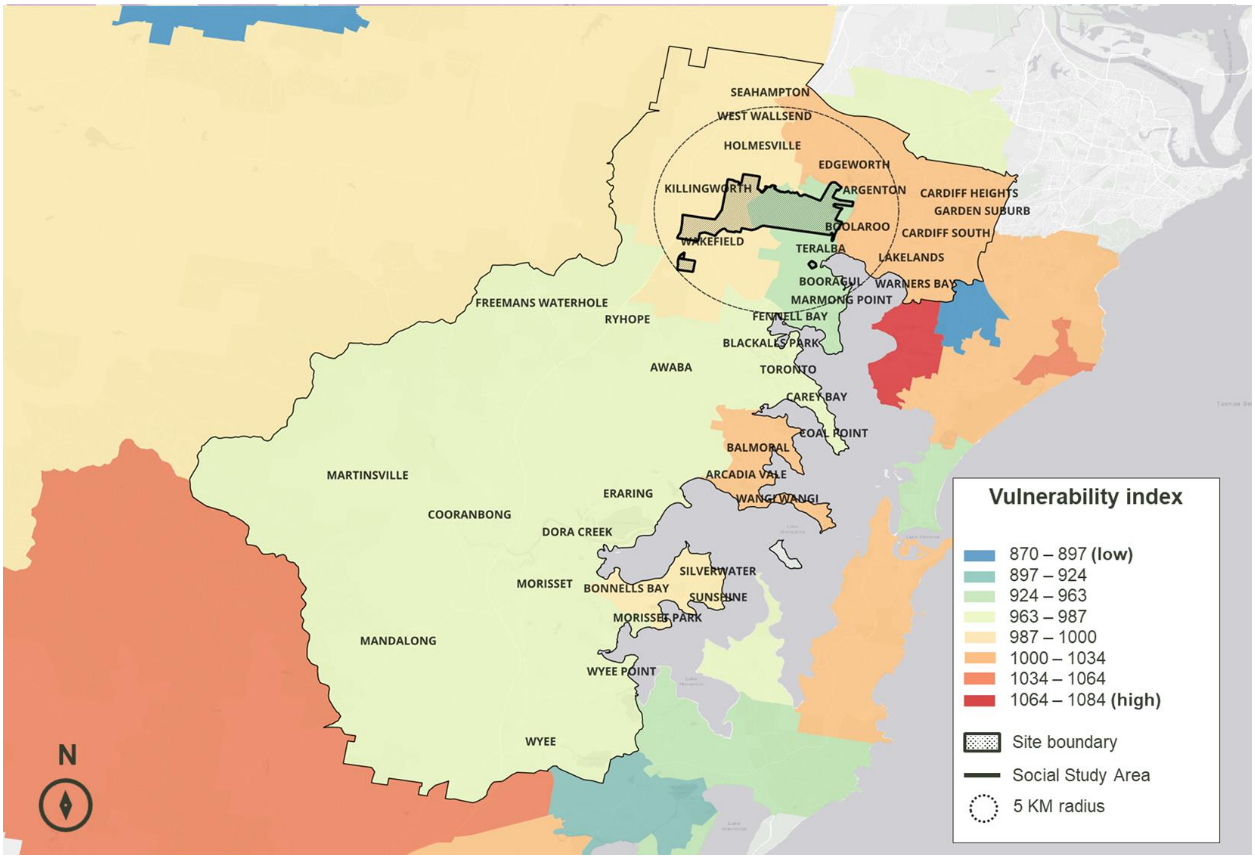

Figure 9-4: SEIFA IRSD index within SSA[200]

Within the SSA, there are highly advantaged rural and coastal suburbs as well as areas experiencing significant disadvantage. Jilliby – Yarramalong is the most advantaged suburb (top 76% of NSW suburbs). In contrast, Wyong is the most disadvantaged suburb (bottom 10% of NSW suburbs). Wide spread of scores indicates pronounced socio-economic inequality within the SSA, with communities experiencing different levels of access to resources, employment opportunities and social advantage. Within the ESA, there are fewer extreme values at either end of the SEIFA distribution. The most advantaged ESA suburbs range from Maryland–Fletcher–Minmi scoring (top 71% of NSW suburbs). The most disadvantaged is Cessnock LGA (bottom 20% of NSW suburbs).

These findings suggest that service planning and resource allocation should be tailored at a local level, particularly within the SSA, to address pronounced inequities that may not be fully visible in broader ESA averages.[201]

The most advantaged and disadvantaged suburbs in the SSA and ESA are listed in Table 9-13.

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[199] Australian Bureau of Statistics (ABS) (2023) Socio-Economic Indexes for Areas (SEIFA).

[200] Ibid.

Table 9-13: Most advantaged and disadvantaged suburbs within the SSA and ESA

Area	Most advantaged in SSA	Most disadvantaged in SSA
SSA	1. Jilliby – Yarramalong - score of 1060 (top 76% in NSW) 2. Warners Bay – Boolaroo – score of 1029 (top 62% in NSW) 3. Glendale – Cardiff – Hillsborough – score of 1017	1. Lake Munmorah – Mannering Park – score of 952 2. Gorokan – Kanwal – Charmhaven – score of 917 (bottom 13% in NSW) 3. Wyong – score of 897 (bottom 10% in NSW)
ESA	1. Maryland – Fletcher – Minmi – score of 1032 (Top 71% in NSW) 2. Lake Macquarie LGA – score of 1004 (top 63% in NSW) 3. Maitland LGA – score of 988	1. Wallsend – Elermore Vale – score of 976 (bottom 32% in NSW) 2. Cessnock LGA – score of 939 (bottom 20% in NSW)

Appendix I: Key areas of community interest

The following areas of community interest are located within 5 km of the Site location and include a range of recreational, sporting and environmental assets.

Table 9-14 Key areas of community interest

Reference number:	Significant Sites	Distance from the Site	
1	**Munibung Hill**	2km	■ A prominent bushland ridge and natural landmark with cultural significance to the Awabakal people, featuring regenerating native vegetation (including threatened species) and managed for conservation, cultural protection, and public access under a 2022–2032 plan.
2	**Speers Point Park**	1km	■ Lake Macquarie's most popular park, offering major recreational facilities including playgrounds, pathways, open spaces, event hire areas, and community amenities. [202]
3	**Hunter Sports Centre and Gym**	2km	■ The region's premier sporting venues, featuring Olympic-standard athletics facilities alongside a wide range of specialised spaces for gymnastics, trampolining, fitness, and community use. It also offers extensive amenities including accessible facilities, event and media support, and catering spaces for community to use.
4	**Speers Point Jetty**	1km	■ An 83-metre recreational jetty within Speers Point Park, popular for fishing and providing easy access to surrounding parklands.
5	**Waratah Golf Club**	500m (across Cockle Creek)	■ A historic Hunter region golf course featuring a traditional layout set within tree-lined fairways and natural landscapes. [203]
6	**Hunter Ice Skating Stadium**	4km	■ The stadium offers public skating sessions, learn-to-skate programs, and ice hockey classes. The stadium also hosts multiple events and nationwide ice hockey tournaments.

[202] Lake Macquarie City Council (2021) Lake Macquarie City Council Parks and Play Strategy 2021.

[203] NSW Government (n.d.) Waratah Golf Club.

Appendix J: Non-Indigenous heritage items

Several non-Indigenous heritage precincts lie within the SSA which include non-Indigenous settlements occurring after colonisation.[204] As indicated in the *Lake Macquarie Local Environmental Plan 2014,* State Heritage Register these sites include:[205]

- **Teralba Public School:** A significant community asset, having served its original educational purpose for more than 90 years. The 1898–99 brick building reflects the architectural influence of the Government Architect's Office during Vernon's tenure. The school evolved from a series of earlier schools established from 1884 with its architectural features reflecting those of Federation Bungalows throughout the Lake Macquarie city.
- **Teralba Conservation Area**: This area has historical significance linked to the railway, coal mining and early settlement. Surviving streets, miners' cottages and community buildings reflect its industrial heritage. Its heritage character, landscape setting and connection to former mining and rail activity warrant protection from inappropriate development.
- **Teralba Reserve:** The reserve is divided into two distinct areas: Teralba Cemetery Park and a newly constructed children's play area. The cemetery contains a variety of grave types, ranging from large formal plots with kerbing and marble headstones to simpler graves marked with bush rock kerbing. Burials within the cemetery date back to the area's earliest occupation in the 1880s, when it functioned as a construction camp. [206] This reserve has also been referred to by the following names including Big Hill Cemetery, Billy Goat Hill Cemetery, Goat Hill Cemetery, Booragul Cemetery and Pioneer Cemetery.[207]
- **Rathmines Park, former RAAF Seaplane Base**: A crucial World War II facility, serving as the largest and longest-operating flying boat base in the Southern Hemisphere and playing a key role in Australia's defence. It supported operations, training, and repairs and remains a historically and socially significant Site with strong ties to the local community.

Within the boarder ESA, there are several non-Indigenous heritage sites as noted in the State Heritage Register and the *Lake Macquarie Local Environmental Plan 2014*[208]*:*

- **Catherine Hill Bay Cultural Precinct:** This heritage area represents a rare and intact example of a 19th-century coal mining "company town," characterised by consistent worker cottages, mining infrastructure, and a distinctive coastal-industrial landscape.
- **Dobell House**: The long-term home, studio, and primary creative setting of artist Sir William Dobell, who produced major Wynn and Archibald Prize works of Wang who landscapes inspired his paintings. The home provides and lens into his artistic life, environment, and legacy as one of the one of the greatest Australian portrait artists of the twentieth century.
- **Morisset Hospital Precinct:** A large early 20th-century psychiatric institution reflecting evolving approaches to mental health care, combining ideals of therapeutic landscapes, patient labour, and institutional isolation within a purpose-built rural setting erected in 1909. Also used as a major Kangaroo spotting Site for locals.
- **Wangi Power Station Complex:** A coalfields-based facility that transformed power generation in NSW, reducing costs and urban pollution becoming the state's largest early power station. Valued for its distinctive industrial architecture, historical significance, and lasting impact on the energy industry, regional development, and community.
- **World War II RAAF Radar Station 208 (former):** A rare remnant of Australia's World War II coastal defence network, representing one of the few surviving radar stations using British ACO technology in NSW. It demonstrates the strategic response to invasion threats, while also holding important historical value for the development of radar technology and its lasting impact on Australian science and research.

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[204] Lake Macquarie City Council (2024) North West Catalyst Area Place Strategy.

[205] Lake Macquarie City Council (2026) State-listed heritage items and precincts.

[206] RPS Australia East Pty Ltd (2025) Teralba Reserve Plan of Management.

[207] Lake Mac Libraries (2026) *Teralba Cemetery: the Big Hill*.

[208] Lake Macquarie City Council (2026) State-listed heritage items and precincts.

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